BSIM-SOI 100.1.1 MOSFET Compact Model

Technical Manual

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1 Introduction

BSIM-SOI 100.x.x (formerly known as symmetric BSIM-SOI) is an international standard model for SOI (Silicon-On-Insulator) circuit design. This model offers two different mode of operations, SOIMOD=1 is dedicated to the dynamic depletion devices, and it has surface potential based core IV and CV formulation, SOIMOD=0 is developed for only partial depletion devices, and it is formulated on top of the BSIM-BULK framework [1, 2, 3]. SOIMOD=0 shares the same basic equations with the bulk model so that the physical nature and smoothness of BSIM-BULK are retained. For both the SOIMODs, the most parameters related to general MOSFET operation (non-SOI specific) are directly imported from BSIM-BULK to ensure parameter compatibility.

SOIMOD=1 is the Dynamic Depletion (DD) mode [4, 5], and SOIMOD=0 is the Partial-Depletion (PD) mode [6] of the BSIM-SOI 100.1.1 model. Many enhanced features, as listed below, are included in BSIM-SOI 100.2.0 through the joint effort of the BSIM Team at UC Berkeley and GlobalFoundries Inc. (GF), Bangalore, India. In particular, the model has been tested extensively within GF on its state-of-the-art SOI technology [7, 8].

SOIMOD=1(DD) has newly developed surface potential-based core IV and CV modules, while SOIMOD=0(PD) uses the core model of BSIM-BULK 107.1.0 for I-V and C-V. Additionally, effects associated with SOI operation have been added (similar to the standard BSIM-SOI 4.x.x). BSIM-SOI 100.x.x has the following features:

- Real floating body simulation in both I-V and C-V. The body potential is determined by the balance of all the body-current components.
- An improved parasitic bipolar current model. This includes enhancements in the various diode leakage components, second-order effects (high-level injection and Early effect), diffusion charge equation, and temperature dependence of the diode junction capacitance.
- An improved impact-ionization current model. The contribution from BJT current is also modeled by the parameter FBJTII.
- An improved GIDL/GISL model.
- Asymmetric current/capacitance model Source/Drain diode and asymmetric Source/Drain resistance.
- Gate tunneling currents with a gate current component in the body contact region.

- Linear and non-linear body resistance models for body-contacted SOI-devices.
- Instance parameters (PDBCP, PSBCP, AGBCP, AGBCP2, AEBCP, NBC) are provided to model the parasitics of devices with various body-contact and isolation structures.
- Improved history dependence of the body charges with two new parameters (FBODY, DLCB).
- Improved models for parasitic capacitances such as opposite-type gate parasitic FET due to ion-implanted body contacts (enabled by MODAGBCP2), source/drain to substrate capacitances, etc.

2 MOS I-V Model

A typical SOI MOSFET structure is shown in Fig. 1. The device is formed on a thin SOI film of thickness T_{Si} on top of a layer of buried oxide with thickness T_{box}. In the floating body configuration, there are four external biases which are gate voltage (V_{g}), drain voltage (V_{d}), source voltage (V_{s}) and substrate bias (V_{e}). The body potential ($V_{\mathrm{b_i}}$) is iterated in circuit simulation. If a body contact is applied, there will be one more external bias, the body contact voltage (V_{b}).

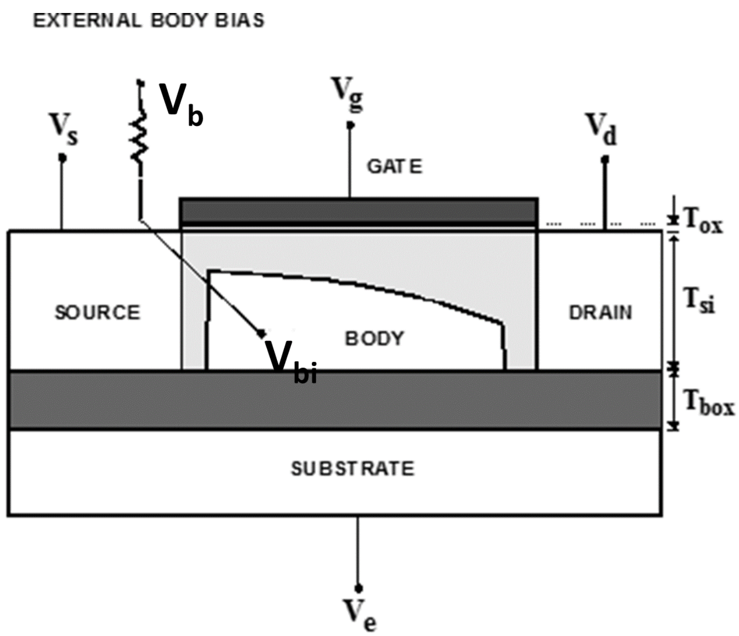

Figure 1: Schematic of a typical SOI MOSFET.

SOIMOD=1 (DD), solves coupled surface potential Poisson's equation to model the dynamic depletion operation. It has different poly depletion, Velocity saturation, DIBL in subthreshold, short channel modules compared to SOIMOD=0. For SOIMOD=0 (PD), Since the backgate (V_{e}) effect is decoupled by the neutral body, PD SOI MOSFETs have similar characteristics as bulk devices. Hence most PD SOI models reported were developed by adding some SOI specific effects onto a BSIM-BULK 107.1.0 model. These effects include parasitic bipolar current, self-heating and body contact resistance, reverse short channel effect, poly depletion, velocity saturation, DIBL in subthreshold and output resistance, short channel effect, mobility degradation, narrow width effect and source/drain series resistance.

2.1 Floating Body Operation

In BSIM-SOI 100.x.x model, the floating body voltage is iterated by the SPICE engine. The result of iteration is determined by the body currents. In the case of DC, body currents include diode current, impact ionization, gate-induced drain leakage (GIDL), oxide tunneling and body contact current. For AC or transient simulations, the displacement currents originated from the capacitive coupling are also contributive.

2.2 Parasitic Series Resistance

The total parasitic resistance at the source/drain terminal consists of two parts: (a) Bias independent and (b) Bias dependent. BSIM-SOI 100.x.x (simillar to the BSIM-BULK 107.1.0) offers three different options to model parasitic resistance with variations on the way the bias dependent and bias independent parts of the parasitic resistance are handled. These options can be exercised by the switch RDSMOD as described below:

(a) RDSMOD = 0, External resistance are bias independent while internal resistance is bias dependent.
(b) RDSMOD = 1, No internal resistance. Both bias dependent and independent resistor are kept externally.
(c) RDSMOD = 2, No external resistance. Both bias dependent and independent resistor are kept internally.

RDSMOD=0:

$$T_0 = 1 + PRWG \cdot q_{ia} \tag{2.1}$$

$$T_1 = PRWB \cdot (\sqrt{\phi_s - V_{bs}} - \sqrt{\phi_s}) \tag{2.2}$$

$$T_2 = \frac{1}{T_0} + T_1 \tag{2.3}$$

$$T_3 = \frac{1}{2}\left[T_2 + \sqrt{T_2^2 + 0.01}\right] \tag{2.4}$$

$$R_{ds}(V) = NF \cdot \left(W_{eff}^{WR} \left[RDSWMIN + RDSW \cdot T_3 \right] \right) \tag{2.5}$$

$$D_r = 1.0 + \frac{\mu_0}{D_{mob}.D_{vsat}} \cdot C_{ox} \cdot \frac{W_{eff}}{L_{eff}} \cdot q_{ia} \cdot R_{ds} \tag{2.6}$$

$$R_{source} = R_{s,geo} \tag{2.7}$$

$$R_{drain} = R_{d,geo} \tag{2.8}$$

$R_{s,geo}$ and $R_{d,geo}$ are the source and drain diffusion resistances, which are described later. And, D_r goes into the denominator of the final I_{ds} expression.

RDSMOD = 1:
The bias-dependent external resistance model is adopted from BSIM4 and can be invoked by setting model selector RDSMOD=1. BSIM4 and BSIM-SOI 100.x.x allow the source extension resistance $R_s(V)$ and the drain extension resistance $R_d(V)$ to be external and asymmetric. The source/drain series resistance is the sum of a bias-independent component and a bias-dependent component.

$$V_{gs,eff} = \frac{1}{2} \left[V_{gs1} - V_{fbsdr} + \sqrt{(V_{gs\_noswap} - V_{fbsdr})^2 + 10^{-2}} \right]$$

$$V_{gd,eff} = \frac{1}{2} \left[V_{gd1} - V_{fbsdr} + \sqrt{(V_{gd\_noswap} - V_{fbsdr})^2 + 10^{-2}} \right] \tag{2.9}$$

$$R_{source} = \frac{1}{W_{eff}^{WR} \cdot NF} \cdot \left(RSWMIN + RSW \cdot \left[-PRWB \cdot V_{sb\_noswap} + \frac{1}{1 + PRWG_i \cdot V_{gs,eff}} \right] \right) + R_{s,geo} \tag{2.10}$$

$$R_{drain} = \frac{1}{W_{eff}^{WR} \cdot NF} \cdot \left(RDWMIN + RDW \cdot \left[-PRWB \cdot V_{db\_noswap} + \frac{1}{1 + PRWG_i \cdot V_{gd,eff}} \right] \right) + R_{d,geo} \tag{2.11}$$

$R_{s,geo}$ and $R_{d,geo}$ are the source and drain diffusion resistances.

RDSMOD=2

$$R_{ds}(V) = R_{s,geo} + NF \cdot \left(W_{eff}^{WR} \left[RDSWMIN + RDSW \cdot T_3 \right] \right) + R_{d,geo} \tag{2.12}$$

$$D_r = 1.0 + \frac{\mu_0}{D_{mob}.D_{vsat}} \cdot C_{ox} \cdot \frac{W_{eff}}{L_{eff}} \cdot q_{ia} \cdot R_{ds} \tag{2.13}$$

where T3 is given by (5.410).

The resistances $R_{s,geo}$ and $R_{d,geo}$ are simply calculated as the sheet resistances ($RSHS$,$RSHD$) times the number of squares (NRS,NRD):

$$\begin{aligned} R_{s,geo} &= NRS \cdot RSHS \\ R_{d,geo} &= NRD \cdot RSHD \end{aligned} \tag{2.14}$$

2.3 Drain Current Equation

In BSIM-SOI 100.x.x, the MOSFET drain current for both the SOIMODs is described later, refer to Sec. (5.14).

3 Body Currents Model

Body currents determine the body potential and therefore the drain current through the body effect. The impact ionization current, diode (bipolar) current, GIDL, oxide tunneling and body contact current are all included in the BSIM-SOI 100.x.x model [see 2] to give an accurate body-potential prediction in the floating body simulation

3.1 Diode and Parasitic BJT Currents

In this section we describe various current components originated from Body-to-Source/Drain (B-S/D) injection, recombination in the B-S/D junction depletion region, Source/Drain-to-Body (S/D-B) injection, recombination current in the neutral body, and diode tunneling current.

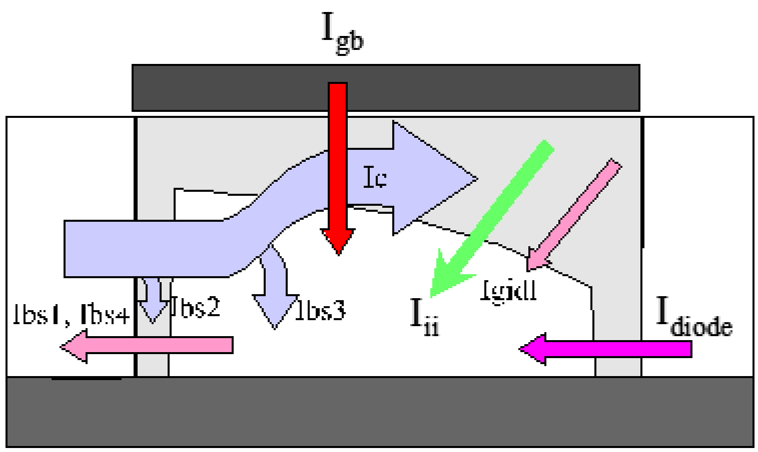

Figure 2: Schematic of a typical PD SOI MOSFET.

The backward injection current:
The backward injection current in the B-S/D diode can be expressed as

$$
\begin{aligned}
I_{bs1} &= W_{dios} T_{si} j_{difs} \left(exp\left(\frac{V_{bs}}{n_{diodes} V_t} \right) - 1 \right) \\
I_{bd1} &= W_{diod} T_{si} j_{difd} \left(exp\left(\frac{V_{bd}}{n_{dioded} V_t} \right) - 1 \right)
\end{aligned} \tag{3.1}
$$

Here n_{diodes},j_{difs},W_{dios},n_{dioded},j_{difd},W_{diod} are the non-ideality factor, the saturation current, the effective B-S diode width and the B-D diode width, respectively.

The carrier recombination and trap-assisted tunneling current:
The carrier recombination and trap-assisted tunneling current in the space-charge region is modeled by

$$
\begin{aligned}
I_{bs2} &= W_{dios} T_{si} j_{recs} \left(exp\left(\frac{V_{bs}}{0.026 n_{recfs}}\right) - exp\left(\frac{V_{sb}}{0.026 n_{recrs}} \frac{V_{rec0s}}{V_{rec0s} + V_{sb}}\right)\right) \\
I_{bd2} &= W_{diod} T_{si} j_{recd} \left(exp\left(\frac{V_{bd}}{0.026 n_{recfd}}\right) - exp\left(\frac{V_{db}}{0.026 n_{recrd}} \frac{V_{rec0d}}{V_{rec0d} + V_{db}}\right)\right)
\end{aligned}
\tag{3.2}
$$

Here n_{recfs}, n_{recrs}, j_{recs}, n_{recfd}, n_{recrd}, j_{recd} are non-ideality factors for forward bias and reverse bias, the saturation current, respectively. Note that the parameters V_{rec0s}, V_{rec0d} are provided to model the current roll-off in the high reverse bias regime.

The reverse bias tunneling current:
The reverse bias tunneling current, which may be significant in junctions with high doping concentration, can be expressed as

$$
\begin{aligned}
I_{bs4} &= W_{dios} T_{si} j_{tuns} \left(1 - exp\left(\frac{V_{sb}}{0.026 n_{tuns}} \frac{V_{tun0s}}{V_{tun0s} + V_{sb}}\right)\right) \\
I_{bd4} &= W_{diod} T_{si} j_{tund} \left(1 - exp\left(\frac{V_{db}}{0.026 n_{tund}} \frac{V_{tun0d}}{V_{tun0d} + V_{db}}\right)\right)
\end{aligned}
\tag{3.3}
$$

where j_{tuns}, j_{tund} are the saturation currents. The parameters n_{tuns}, n_{tund} and V_{tun0s}, V_{tun0d} are provided to better fit the data.

The recombination current in the neutral body:
The recombination current in the neutral body can be described by

$$
\begin{aligned}
I_{bs3} &= (1 - \alpha_{bjt})\, I_{ens} \left[exp\left(\frac{V_{bs}}{n_{diodes} V_t}\right) - 1\right] \frac{1}{\sqrt{E_{hlis} + 1}} \\
I_{bd3} &= (1 - \alpha_{bjt})\, I_{end} \left[exp\left(\frac{V_{bd}}{n_{dioded} V_t}\right) - 1\right] \frac{1}{\sqrt{E_{hlid} + 1}}
\end{aligned}
\tag{3.4}
$$

$$
\begin{aligned}
I_{ens} &= W'_{eff} T_{si} j_{bjts} \left[L_{bjt0} \left(\frac{1}{L_{eff}} + \frac{1}{L_n} \right) \right]^{N_{bjt}} \\
I_{end} &= W'_{eff} T_{si} j_{bjtd} \left[L_{bjt0} \left(\frac{1}{L_{eff}} + \frac{1}{L_n} \right) \right]^{N_{bjt}} \qquad (3.5)
\end{aligned}
$$

$$
\begin{aligned}
E_{hlis} &= A_{hlis_{eff}} \left[exp\left(\frac{V_{bs}}{n_{diodes} V_t} \right) - 1 \right] \\
E_{hlid} &= A_{hlid_{eff}} \left[exp\left(\frac{V_{bd}}{n_{dioded} V_t} \right) - 1 \right] \qquad (3.6) \\
\alpha_{bjt} &= exp\left[-0.5 \left(\frac{L_{eff}}{L_n} \right)^2 \right] \qquad (3.7)
\end{aligned}
$$

Here α_{bjt} is the bipolar transport factor, whose value depends on the ratio of the effective channel length L_{eff} and the minority carrier diffusion length L_n. j_{bjts} and j_{bjtd} are the saturation currents, while the parameters L_{bjt0} and N_{bjt} are provided to better fit the forward injection characteristics. Notice that E_{hlis} and E_{hlid}, determined by the parameter A_{hlis} and A_{hlid}, stand for the high level injection effect in the B-S/D diode, respectively.

The parasitic bipolar transistor current:
The parasitic bipolar transistor current is important in transient body discharge, especially in pass-gate floating body SOI designs. The BJT collector current is modeled as

$$
I_c = \alpha_{bjt} I_{en} \left\{ exp\left[\frac{V_{bs}}{n_{diodes} V_t} \right] - exp\left[\frac{V_{bd}}{n_{dioded} V_t} \right] \right\} \frac{1}{E_{2nd}} \qquad (3.8)
$$

$$
\begin{aligned}
E_{2nd} &= \frac{E_{ely} + \sqrt{{E_{ely}}^2 + 4E_{hli}}}{2} \qquad (3.9) \\
E_{ely} &= 1 + \frac{V_{bs} + V_{bd}}{V_{Abjt} + A_{ely} L_{eff}} \qquad (3.10) \\
E_{hli} &= E_{hlis} + E_{hlid} \qquad (3.11)
\end{aligned}
$$

where E_{2nd} is composed of the Early effect E_{ely} and the high level injection roll-off E_{hli}. Note that $E_{2nd} \rightarrow E_{ely}$ as $E_{ely} >> E_{hli}$. While $E_{2nd} \rightarrow \sqrt{E_{hli}}$ as $E_{hli} >> E_{ely}$, in which case the Early voltage $V_{Abjt} + A_{ely}L_{eff}$ is high.

To sum up, the total B-S current is $I_{bs} = \sum_{i=1}^{4} I_{bsi}$, and the total B-D current is $I_{bd} = \sum_{i=1}^{4} I_{bdi}$. The total drain current including the BJT component can then be expressed as

$$I_{ds,total} = I_{ds,MOSFET} + I_c \tag{3.12}$$

3.2 Impact Ionization Current

IIIMOD = 0

This impact ionization current model is same as BSIM-BULK 107.1.0, and is modeled by

$$I_{ii} = ALPHA0 \ \cdot (V_{ds} - V_{dseff}) \cdot \exp\left(\frac{BETA0}{V_{ds} - V_{dseff}}\right) \cdot \left(\frac{I_{ds}}{M_{SCBE}}\right) \tag{3.13}$$

where parameters ALPHA0 and BETA0 are impact ionization coefficients. ALPHA0L and ALPHA0LEXP are length scaling parameters for ALPHA0.

IIIMOD = 1

An accurate impact ionization current equation is crucial to the SOI model since it may affect the transistor output characteristics through the body effect. Hence in both the SOIMOD we use a more recent expression to formulate the impact ionization current I_{ii} as

$$I_{ii} = ALPHA0 \cdot (I_{ds} + I_{ii,BJT}) \cdot exp\left(\frac{V_{diff}}{BETA2 + BETA1 \cdot V_{diff} + BETA0 \cdot {V_{diff}}^2}\right) \tag{3.14}$$

$I_{ii,BJT}$ is defined as:

$$I_{ii,BJT} = F_{bjtii} I_c \tag{3.15}$$

$$V_{diff} = V_{ds} - V_{dsatii} \tag{3.16}$$

$$V_{dsatii} = VgsStep + \left[V_{dsatii0} \left(1 + T_{ii} \left(\frac{T}{T_{nom}} - 1 \right) \right) - \frac{L_{ii}}{L_{eff}} \right] \tag{3.17}$$

$$VgsStep = \left(\frac{E_{satii} L_{eff}}{1 + E_{satii} L_{eff}} \right) \left(\frac{1}{1 + S_{ii1} V_{gsteff}} + S_{ii2} \right) \left(\frac{S_{ii0} V_{gst}}{1 + S_{iid} V_{ds}} \right) \tag{3.18}$$

Here the $F_{bjtii} I_c$ term represents the contribution from the parasitic bipolar current. The extracted saturation drain voltage V_{dsatii} depends on the gate overdrive voltage V_{gst} and L_{eff}. One can first extract the parameters $(V_{dsatii0}, L_{ii})$ by the $V_{dsatii} - L_{eff}$ characteristics at $V_{gst} = 0$. All the other parameters $(E_{satii}, S_{ii1}, S_{ii2}, S_{ii0}, S_{iid})$ can then be determined by the plot of V_{dsatii} versus V_{gs} for different L_{eff}. Notice that a linear temperature dependence of $V_{dsatii0}$ with the parameter T_{ii} is also included.

IIIMOD = 2

When IIIMod = 1, the two component currents I_{ds} and I_c have a same bias dependence for impact ionization rate. This approximation generally won't cause accuracy problem because the MOSFET drain current is the major contribution on impact ionization current in the interested operation regions. While SOI MOSFET device operates in subthreshold to accumulation regions, parasitic BJT effect starts to dominant nodal drain current at high drain bias. In order to model I_{ii} better, IIIMOD =2 is introduce to treat I_{ds} and I_c separately. It means that these two components have the different impact ionization rate.

Here I_{ds} still uses the old impact ionization model. The BJT contribution is expressed below. The temperature dependence is also improved.

$$I_{ii,BJT} = \frac{CBJTII + EBJTII \cdot L_{eff}}{L_{eff}} \; I_c \; (V_{bci} - V_{bd}) \cdot \exp\left(- ABJTII \cdot (V_{bci} - V_{bd})^{(MBJTII-1)} \right) \tag{3.19}$$

$$V_{bci} = VBCI \left(1 + TVBCI \left(\frac{T}{Tnom} - 1 \right) \right) \tag{3.20}$$

where ABJTII, CBJTII, EBJTII, MBJTII, VBCI and TVBCI are model parameters.

3.3 Gate Induced Source/Drain Leakage Current

GIDLMOD = 0
GISL/GIDL can be important in SOI device because it can affect the body potential in the low V_{gs} and high V_{ds} regime.

The formula for GIDL current is:

$$I_{GIDL} = AGIDL \cdot W_{diod} \cdot Nf \cdot \frac{V_{ds} - V_{gse} - EGIDL + V_{fbsd}}{3 \cdot T_{oxe}} \cdot exp\left(-\frac{3 \cdot T_{oxe} \cdot BGIDL}{V_{ds} - V_{gse} - EGIDL}\right) \cdot \frac{V_{db}^3}{CGIDL + V_{db}^3} \tag{3.21}$$

Where, AGIDL, BGIDL, CGIDL, and EGIDL are model parameters and explained in Appendix A. CGIDL accounts for the body-bias dependence of IGIDL and IGISL. Here Vgse accounts for poly depletion effect.

Following BSIM-BULK 107.1.0, BSIM-SOI 100.x.x also introduces GISL current. In order to model asymmetric source/drain, GISL model has another set of parameters: AGISL, BGISL, CGISL, and EGISL.

$$I_{GISL} = AGISL \cdot W_{dios} \cdot NF \cdot \frac{-V_{ds} - V_{gse} - EGISL + V_{fbsd}}{3 \cdot T_{oxe}} \cdot exp\left(-\frac{3 \cdot T_{oxe} \cdot BGISL}{-V_{ds} - V_{gse} - EGISL}\right) \cdot \frac{V_{sb}^3}{CGISL + V_{sb}^3} \tag{3.22}$$

GIDLMOD = 1
In this new model, the basic idea is to decouple V_{ds} and V_{gs} dependence by introducing an extra parameter RGIDL. The body bias dependence part is also revised. Here, KGIDL and FGILD are V_{bs} dependent parameters.

$$I_{GIDL} = AGIDL \cdot W_{diod} \cdot Nf \cdot \frac{V_{ds} - RGIDL \cdot V_{gse} - EGIDL + V_{fbsd}}{3 \cdot T_{oxe}} \times exp\left(-\frac{3 \cdot T_{oxe} \cdot BGIDL}{V_{ds} - V_{gse} - EGIDL}\right) \cdot exp\left(\frac{KGIDL}{V_{ds} - FGIDL}\right) \tag{3.23}$$

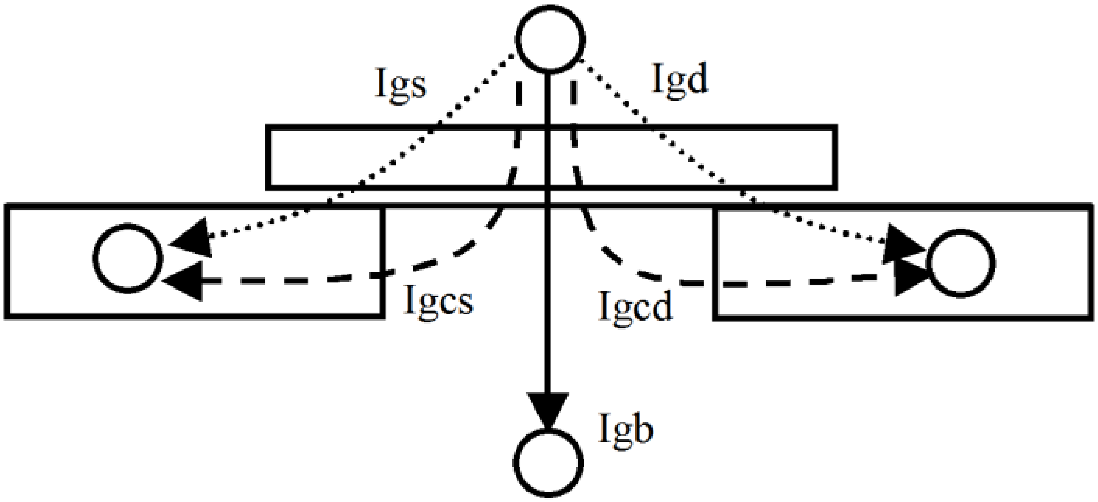

Figure 3: Schematic gate current components flowing between MOSFET terminals.

$$I_{GISL} = AGISL \cdot W_{dios} \cdot NF \cdot \frac{-V_{ds} - RGISL \cdot V_{gse} - EGISL + V_{fbsd}}{3 \cdot T_{oxe}} \times exp\left(-\frac{3 \cdot T_{oxe} \cdot BGISL}{-V_{ds} - V_{gse} - EGISL}\right) \cdot exp\left(\frac{KGISL}{-V_{bs} - FGISL}\right) \tag{3.24}$$

3.4 Gate Tunneling Current Model

In BSIM-SOI 100.x.x, the gate tunneling current components include the tunneling current between gate and substrate (I_{gb}), and the current between gate and channel (I_{gc}), which is partitioned between the source and drain terminals by $I_{gc} = I_{gcs} + I_{gcd}$. The third component happens between gate and source/drain diffusion regions (I_{gs} and I_{gd}). Figure 3 shows the schematic gate tunneling current flows.

3.4.1 Model Selectors

Two global selectors are provided to turn on or off the tunneling components. **IGCMOD** = 1 turns on I_{gc}, I_{gs}, and I_{gd}; **IGBMOD** = 1 turns on I_{gb}. When the selectors are set to zero, no gate tunneling currents are modeled.

$$V_{ox} = nVt \cdot (v_g - v_{fb} - \psi_p + q_s + q_{deff}) \tag{3.25}$$

$$V_{oxacc} = \frac{1}{2}\left(-V_{ox} + \sqrt{V_{ox}^2 + 10^{-4}}\right) \tag{3.26}$$

$$V_{oxdepinv} = \frac{1}{2}\left(V_{ox} + \sqrt{V_{ox}^2 + 10^{-4}}\right) \tag{3.27}$$

Eq. (3.26) and (3.27) are valid and continuous from accumulation through depletion to inversion.

3.4.2 Oxide Tunneling Current

For thin oxide (below 20 Å), oxide tunneling is important in the determination of floating-body potential. In PDSOIMOD the following equations are used to calculate the tunneling current density J_{gb} :

In inversion,

$$J_{gbinv} = A\frac{V_{gb}V_{aux}}{TOXE^2}\left(\frac{T_{\mathrm{OXREF}}}{TOXE}\right)^{N_{\mathrm{tox}}} exp\left(\frac{-B\left(\alpha_{\mathrm{gb1}} - \beta_{\mathrm{gb1}}\left|V_{ox}\right|\right)T_{ox}}{1 - \frac{\left|V_{ox}\right|}{V_{\mathrm{gb1}}}}\right) \tag{3.28}$$

In accumulation,

$$J_{gbacc} = A\frac{V_{gb}V_{aux}}{T_{ox}^2}\left(\frac{T_{\mathrm{OXREF}}}{TOXE}\right)^{N_{\mathrm{tox}}} exp\left(\frac{-B\left(\alpha_{\mathrm{gb2}} - \beta_{\mathrm{gb2}}\left|V_{ox}\right|\right)T_{ox}}{1 - \frac{\left|V_{ox}\right|}{V_{\mathrm{gb2}}}}\right) \tag{3.29}$$

$$J_{gb} = (J_{gbinv} + J_{gbacc}) \times J_{gtemp} \tag{3.30}$$

J_{gb} is evaluated in IGBMOD = 1. IGBMOD = 0 turns it off.

Gate Tunneling Component in the AGBCP2 Region:
In BSIM-SOI 100.x.x, the instance parameter AGBCP2 represents the parasitic opposite type gate-to-body overlap area due to the body contact. This applies to the opposite-type gate, which is shown in Fig. 4.4. To account for the tunneling current in this region, $I_{g-agbcp2}$ is introduced as follows:

$$I_{g-agbcp2} = A \times A_{agbcp2} \times min(V_{gp} - V_{fb2}, 0) \times V_{gp-eff} T_{oxRatio}$$
$$\times\ exp[-B \times TOXE\,(AIGBCP2 - BIGBCP2 \times V_{gp_{eff}})\,(1 + CIGBCP2 \times V_{gp_{eff}})] \tag{3.31}$$

$$V_{gp-eff} = 0.5 \times \left[\sqrt{\left(V_{gp} - V_{fb2}\right)^2 + \delta^2} - (V_{gp} - V_{fb2}) - \delta\right]$$
$$\delta = 0.01 \tag{3.32}$$

3.4.3 Gate-to-Chanel Current (I_{gc0}) and Gate-to-S/D (I_{gs} and I_{gd})

I_{gc0} is determined by ECB for NMOS and HVB (Hole tunneling from Valence Band) for PMOS at $V_{ds} = 0$, is formulated as

$$I_{gc0} = W_{eff} L_{eff} \cdot A \cdot T_{oxRatio} \cdot V_{gse} \cdot V_{aux}$$
$$\exp[(-B \cdot TOXE \cdot AIGC - BIGC \cdot V_{oxdepinv}) \cdot (1 + CIGC \cdot V_{oxdepinv})] \tag{3.33}$$

where A = 4.97232 A/V^2 for NMOS and 3.42537 A/V^2 for PMOS, B = 7.45669e11 $(g/F - s^2)^0.5$ for NMOS and 1.16645e12 $(g/F - s^2)^{0.5}$ for PMOS.

$$V_{aux} = n_q \cdot nVt \cdot (q_s + q_{deff}) \tag{3.34}$$

I_{gs} and I_{gd} – I_{gs} represents the gate tunneling current between the gate and the source diffusion region, while Igd represents the gate tunneling current between the gate and the drain diffusion region. I_{gs} and I_{gd} are determined by ECB for NMOS and HVB for PMOS, respectively.

$$I_{gs} = W_{eff} DLCIG \cdot A \cdot T_{oxRatioEdge} \cdot V_{gs} \cdot {V_{gs}}' \cdot i_{gtemp}$$
$$\cdot \exp\left[-B \cdot TOXE \cdot POXEDGE \cdot \left(AIGS - BIGS \cdot {V_{gs}}'\right) \cdot \left(1 + CIGS \cdot {V_{gs}}'\right)\right] \tag{3.35}$$

and

$$I_{gd} = W_{eff} DLCIGD \cdot A \cdot T_{oxRatioEdge} \cdot V_{gd} \cdot {V_{gd}}'$$
$$\cdot \exp\left[-B \cdot TOXE \cdot POXEDGE \cdot \left(AIGD - BIGD \cdot {V_{gd}}'\right) \cdot \left(1 + CIGD \cdot {V_{gd}}'\right)\right] \tag{3.36}$$

where A = 4.97232 A/V^2 for NMOS and 3.42537 A/V^2 for PMOS, B = 7.45669e11 $(g/F-s^2)^{0.5}$ for NMOS and 1.16645e12 $(g/F-s^2)^{0.5}$ for PMOS, and

$$T_{oxRatioEdge} = \left(\frac{TOXREF}{TOXE \cdot POXEDGE}\right)^{NTOX} \cdot \frac{1}{(TOXE \cdot POXEDGE)^2} \tag{3.37}$$
$$V'_{gs} = \sqrt{(V_{gs} - V_{fbsd})^2 + 10^{-4}} \tag{3.38}$$
$$V'_{gd} = \sqrt{(V_{gd} - V_{fbsd})^2 + 10^{-4}} \tag{3.39}$$

Vfbsd is the flat-band voltage between gate and S/D diffusions calculated as

If $NGATE > 0.0$

$$V_{fbsd} = -devsign \cdot \frac{k_B T}{q} log\left(\frac{NGATE}{NSD}\right) + VFBSDOFF \tag{3.40}$$

Else $V_{fbsd} = 0.0$.

Partition of I_{gc}: To consider the drain bias effect, I_{gc} is split into two components, I_{gcs} and I_{gcd}, that is $I_{gc} = I_{gcs} + I_{gcd}$, and

$$I_{gcs} = I_{gc0} \cdot \frac{PIGCD \cdot V_{dseffx} + exp(-PIGCD \cdot V_{dseffx}) - 1 + 10^{-4}}{PIGCD \cdot {V_{dseffx}}^2 + 2 \cdot 10^{-4}} \tag{3.41}$$

and

$$I_{gcd} = I_{gc0} \cdot \frac{1 - (PIGCD \cdot V_{dseffx} + 1) \cdot exp(-PIGCD \cdot V_{dseffx}) + 10^{-4}}{PIGCD \cdot {V_{dseffx}}^2 + 2 \cdot 10^{-4}} \tag{3.42}$$

where

$$V_{dseffx} = \sqrt{V_{dseff} + 0.01} - 0.1 \tag{3.43}$$

At $V_{ds} = 0$, $I_{gcs} = I_{gcd} = \frac{1}{2} I_{gc0}$. Thus I_{gc0} is the gate to channel current I_{gc} at $V_{ds} = 0$.

3.5 Body Contact Current

In BSIM-SOI 100.x.x, a body resistor is connected between the body (bi node) and the body contact (b node) if the transistor has a body-tie. The body resistance ($R_{bi,b}$) is modeled by

Linear Body Resistor (BODYMOD=1)

$$R_{bi,b} = \left(R_{body} \frac{W'_{eff}}{L_{eff}} \right) || \left(R_{halo} \frac{W'_{eff}}{2} \right), R_{bodyext} = R_{bsh} N_{rb} \tag{3.44}$$

Here $R_{bi,b}$ and $R_{bodyext}$ represent the intrinsic and extrinsic body resistance respectively. R_{body} is the intrinsic body sheet resistance, R_{halo} accounts for the effect of halo implant, N_{rb} is the number of square from the body contact to the device edge and R_{bsh} is the sheet resistance of the body contact diffusion.

Non-linear Body Resistor (BODYMOD=2) A bias-dependent body resistance sub-model is provided in BSIM-SOI 100.x.x to capture the variation of $R_{bi,b}$ with terminal voltages:

$$R_{bi,b} = \frac{W_{EFF}^2}{U_B \cdot Q_{body}} \tag{3.45}$$

where U_B is the majority carrier's mobility in body, and

$$Q_{body} = q \cdot N_{EFF} \cdot T_{SI} \cdot W_{EFF} \cdot L_{EFF} - Q_B; \tag{3.46}$$

is the total mobile majority charge in the neutral body region. Here N_{EFF} is the effective channel doping including the effect of halo implants, and T_{SI} is the thin film thickness. The total bulk charge Q_B includes the (front) gate induced bulk charge , the junction depletion charges and the back-gate induced bulk charge.

Body Contact Current $I_{bi,b}$ The body contact current $I_{bi,b}$ is defined as the current flowing through the body resistor:

$$I_{bi,b} = \frac{V_{bi,b}}{R_{bi,b} + R_{bodyext}} \tag{3.47}$$

where $V_{bi,b}$ is the voltage across the bi node and b node. Notice that $I_{bi,b} = 0$ if the transistor has a floating body.

The choice of BODYMOD = 0 makes the device a floating-body.

3.6 Body Contact Parasitics

The effective channel width may change due to the body contact. Hence the following equations are used:

$$W_{eff} = W_{drawn} - N_{bc}dW_{bc} - (2 - N_{bc})dW \tag{3.48}$$

$$W_{eff}{}' = W_{drawn} - N_{bc}dW_{bc} - (2 - N_{bc})dW' \tag{3.49}$$

$$W_{diod} = W_{eff}{}' + PDBCP \tag{3.50}$$

$$W_{dios} = W_{eff}{}' + PSBCP \tag{3.51}$$

Here dW_{bc} is the width offset for the body contact isolation edge. N_{bc} is the number of body contact isolation edge. For example: N_{bc}=0 for floating body devices, N_{bc}=1 for T-gate structures and N_{bc}=2 for H-gate structures. PDBCP/PSBCP represents the parasitic perimeter length for body contact at drain/source side. The body contact parasitics may affect the I-V significantly for narrow width devices.

After introducing all the mechanisms that contribute the body current, we can express the nodal equation (KCL) for the body node as

$$(I_{bs} + I_{bd}) + I_{bp} - I_{ii} - (I_{dgidl} + I_{sgisl}) - I_{gb} = 0 \tag{3.52}$$

Eq. 3.52 is important since it determines the body potential through the balance of various body current components. The I-V characteristics can then be correctly predicted after this critical body potential can be well anchored.

4 MOS C-V Model

BSIM-SOI 100.x.x approaches capacitance modeling for both the SOIMOD by adding SOI-specific capacitive effect to the core C-V model, the body charges belonged to the floating body node will be our emphasis. The model incorporates features listed below with the SOI-specific features.

- Separate effective channel length and width for IV and CV models.
- The CV model is not piece-wise (i.e. divided into inversion, depletion, and accumulation). Instead, a single equation is used for each nodal charge covering all regions of operation. This ensures continuity of all derivatives and enhances convergence properties.
- Bias independent fringing capacitances are added between the gate and source as well as the gate and drain. A sidewall source/drain to substrate (under the buried oxide) fringing capacitance is added.
- A source/drain-buried oxide-Si substrate parasitic MOS capacitor is added.
- Body-to-back-gate coupling is added.
- Parasitic gate capacitance model is improved by the new body contact model.

A good intrinsic charge model is important in bulk MOSFETs because intrinsic capacitance comprises a sizable portion of the overall capacitance, and because a well-behaved charge model is required for robust large circuit simulation convergence. In analog applications there are devices biased near the threshold voltage. Thus, a good charge/surface potential model must be well-behaved in transition regions as well.

A good physical charge model of SOI MOSFETs is even more important than in bulk. This is because transient behavior of the floating body depends on capacitive currents. Also, due to the floating body node, convergence issues in PD SOI are more volatile than in bulk, so that charge/surface potential smoothness and robustness are important. An example is that a large negative guess of body potential by SPICE during iterations can force the transistor into depletion, and a smooth transition between depletion and inversion is required. Therefore the gate/source/drain/backgate to body capacitive coupling is important in SOI FET.

4.1 Intrinsic Charges

SOIMOD=1, use charges derived from core surface potential based model while SOIMOD=0 uses similar expressions to BSIM-BULK 107.1.0 for inversion charge and bulk charge. Details with charge expression is described in Sec. 11.

4.2 Source/Drain Junction Charges

Beside the junction depletion capacitance (the bottom junction capacitance, sidewall junction capacitance along the isolation edge, and sidewall junction capacitance along the gate edge) considered in BSIM-BULK 107.1.0 [3], the diffusion capacitance, which is important in the forward body-bias regime, is also included in BSIM-SOI 100.x.x. The source/drain junction charges Q_{jswg}/Q_{jdwg} can therefore be expressed as

$$\begin{aligned} Q_{jswg} &= Q_{bsdep} + Q_{bsdif} \\ Q_{jdwg} &= Q_{bddep} + Q_{bddif} \end{aligned} \tag{4.1}$$

The depletion charges/capacitances have similar expressions as in BSIM-BULK 107.1.0 [Refer to Sub Sec. 6.1].

The diffusion charges Q_{bsdif}/Q_{bddif} can be modeled by

$$\begin{aligned} Q_{bsdif} &= \tau \frac{W'_{eff}}{N_{seg}} T_{si} J_{sbjt} \left[1 + L_{dif0}\left(L_{bj0}\left(\frac{1}{L_{eff}} + \frac{1}{L_n}\right)\right)^{N_{dif}}\right] \left[exp\left(\frac{V_{bs}}{n_{dios} V_t}\right) - 1\right] \frac{1}{\sqrt{E_{hlis} + 1}} \\ Q_{bddif} &= \tau \frac{W'_{eff}}{N_{seg}} T_{si} J_{dbjt} \left[1 + L_{dif0}\left(L_{bj0}\left(\frac{1}{L_{eff}} + \frac{1}{L_n}\right)\right)^{N_{dif}}\right] \left[exp\left(\frac{V_{bd}}{n_{diod} V_t}\right) - 1\right] \frac{1}{\sqrt{E_{hlid} + 1}} \end{aligned} \tag{4.2}$$

The parameter τ represents the transit time of the injected minority carriers in the body. The parameters L_{dif0} and N_{dif} are provided to better fit the data.

4.3 Extrinsic Capacitances

Expressions for extrinsic (parasitic) capacitances that are common in bulk and SOI MOSFETs are taken directly from BSIM-BULK 107.1.0. They are source/drain-to-gate overlap capacitance and source/drain-to-gate fringing capacitance [Refer to Sec. 11]. Additional SOI-specific parasitics added are substrate-to-source sidewall capacitance (C_{essw}), and substrate-to-drain sidewall capacitance (C_{edsw}), substrate-to-source bottom capacitance (C_{esb}) and substrate-to-drain bottom capacitance (C_{edb}).

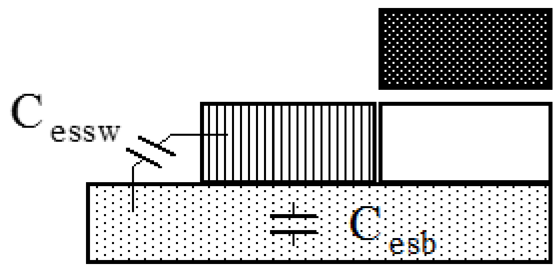

Figure 4: SOI MOSFET extrinsic charge components. C_{essw} is the substrate-to-source sidewall capacitance. C_{esb} is the substrate-to-source bottom capacitance.

Substrate-to-source bottom capacitance (C_{esb}): In SOI, there is a parasitic source/drain-buried oxide-Si substrate parasitic MOS structure with a bias dependent capacitance. If $V_{s,d}$=0, this MOS structure might be in accumulation. However, if $V_{s,d}$=V_{dd}, the MOS structure is in depletion with a much smaller capacitance, because the Si substrate is lightly doped. The bias dependence of this capacitance is similar to high frequency MOS depletion capacitance as shown in Fig. 4. It might be substantial in devices with large source/drain diffusion areas. BSIM-SOI 100.x.x models it not by piece-wise expressions. It ensures infinite-differentiability of charge and capacitance. The substrate-to-source bottom capacitance (per unit source/drain area) C_{esb} is:

$$C_{esb} = \frac{(C_{box} - C_{min})}{2}.tanh(ACESB.V_{es} + BCESB) + \frac{(C_{box} + C_{min})}{2} \tag{4.3}$$

ACESB and BCESB are fitting parameters for C_{esb}.

Substrate-to-source sidewall capacitance (C_{essw}):
The sidewall source/drain to substrate capacitance (per unit source/drain perimeter length) can be expressed by

$$C_{s/d,esw} = C_{sdesw} log\left(CFRCOEFF \cdot \left(1 + \frac{T_{si}}{T_{box}}\right)\right) \tag{4.4}$$

which depends on the silicon film thickness T_{si} and the buried oxide thickness T_{box}. The parameter C_{sdesw} represents the fringing capacitance per unit length.

4.4 Body Contact Parasitics

The parasitic capacitive coupling due to the body contact is considered in both the SOIMODs. The instance parameter AGBCP represents the parasitic gate-to-body overlap area due to the body contact, and AEBCP represents the parasitic substrate-to-body overlap area. The effect may be significant for small area devices.

There are four instance parameters used to calculate parasitic capacitances associated with body contacts. They are: PSBCP, PDBCP, AGBCP and AEBCP. It is worth pointing out that PSBCP and PDBCP represent additional gate perimeter to the source and drain and must be specified on a per finger basis, while AGBCP and AEBCP represent addition gate area and addition area of body over the box and must be specified on a total transistor basis.

BSIM-SOI 100.x.x also considers the P^+ implantation for body contact (shown in Fig. 5), which will induce parasitic P^+-poly gate/NMOS.

In BSIM-SOI 100.x.x, the instance parameter AGBCP represents the parasitic gate-to-body overlap area due to the body contact. This parameter only applies for the same-type gate. For the opposite-type gate, the charge will be overestimated by AGBCP. Charge model has to be modified to include the effect of P^+/P region in this case.

The higher V_{FB} in the P^+/P region lowers the gate charge and the net gate charge is the sum of N^+/P and P^+/P regions as shown below. One new instance parameter AGBCP2 is introduced to account for the opposite-type parasitic capacitance. The final charge could be expressed as following:

$$\text{Total Charge } = WL \times N^+/NMOS + AGBCP \times N^+/NMOS + AGBCP2 \times P^+/NMOS \tag{4.5}$$

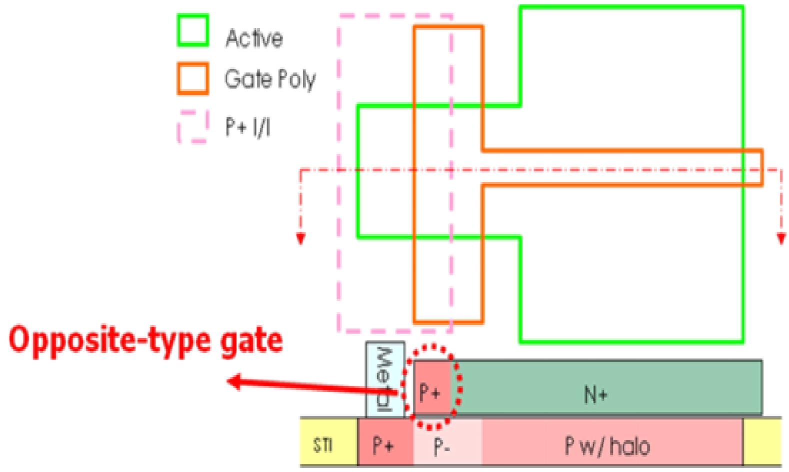

Figure 5: Parasitic capacitance in opposite-type gate.

In this case, there is a new instant parameters agbcp2, which is similar to agbcp and specified on a total transistor basis.

The body (bi_2) of the opposite-type gate parasitic FET is connected to the external body contact via a very small resistance RBODYAGBCP2, which has been defined as a model parameter.

The calculations for this body-contact-induced opposite-type gate parasitic FET are done when the device is body-contacted and the MODAGBCP2 flag is set to 1. It is advised to set MODAGBCP2 = 1 for body-tied devices where the contribution of this parasitic FET can be tuned with the AGBCP2 parameter.

5 Model Equations

5.1 Physical constants

Physical quantities are in M.K.S units unless specified otherwise.

$$q = 1.6 \times 10^{-19} C \tag{5.1}$$

$$\epsilon_0 = 8.8542 \times 10^{-12} \quad \frac{F}{m} \tag{5.2}$$

$$\epsilon_{sub} = EPSRSUB \cdot \epsilon_0 \quad \frac{F}{m} \tag{5.3}$$

$$\epsilon_{ox} = EPSROX \cdot \epsilon_0 \quad \frac{F}{m} \tag{5.4}$$

$$C_{ox} = \frac{3.9 \cdot \epsilon_0}{TOXE} \quad \frac{F}{m^2} \tag{5.5}$$

$$\epsilon_{ratio} = \frac{EPSRSUB}{3.9} \tag{5.6}$$

$$invSqrt2 = 7.0710678118654746e - 01 \tag{5.7}$$

$$oneSixth = 1.6666666666666667e - 01 \tag{5.8}$$

$$oneThird = 3.3333333333333333e - 01 \tag{5.9}$$

$$twoThirds = 6.6666666666666667e - 01 \tag{5.10}$$

$$pe = 4.6051701859880916e + 02 \tag{5.11}$$

$$pe1 = 2.3025850929940458e + 02 \tag{5.12}$$

$$Ede = 1.0e - 200 \tag{5.13}$$

$$Ede1 = 1.0e - 100 \tag{5.14}$$

$$\tag{5.15}$$

5.2 Effective Channel Length & Width

$$\Delta L = LINT + \frac{LL}{(L_{new})^{LLN}} + \frac{LW}{(W_{new})^{LWN}} + \frac{LWL}{(L_{new})^{LLN} \cdot (W_{new})^{LWN}} \tag{5.16}$$

$$\Delta W = WINT + \frac{WL}{(L_{new})^{WLN}} + \frac{WW}{(W_{new})^{WWN}} + \frac{WWL}{L_{new}^{WLN} \cdot (W_{new})^{WWN}} \tag{5.17}$$

$$\Delta L_1 = LINT + \frac{LL}{(L_{new} + DLBIN)^{LLN}} + \frac{LW}{(W_{new} + DWBIN)^{LWN}} + \tag{5.18}$$

$$\frac{LWL}{(L_{new} + DLBIN)^{LLN} \cdot (W_{new} + DWBIN)^{LWN}} \tag{5.19}$$

$$\Delta W_1 = WINT + \frac{WL}{(L_{new} + DLBIN)^{WLN}} + \frac{WW}{(W_{new} + DWBIN)^{WWN}} + \tag{5.20}$$

$$\frac{WWL}{(L_{new} + DLBIN)^{WLN} \cdot (W_{new} + DWBIN)^{WWN}} \tag{5.21}$$

$$L_{new} = L * LMLT + XL; \tag{5.22}$$

$$W_{new} = \frac{W}{NF} * WMLT + XW; \tag{5.23}$$

$$\Delta L_{CV} = DLC \tag{5.24}$$

$$\Delta W_{CV} = DWC \tag{5.25}$$

$$L_{eff} = L * LMLT + XL - 2\Delta L \tag{5.26}$$

$$W_{eff} = W * WMLT + XW - 2\Delta W \tag{5.27}$$

$$L_{eff,CV} = L * LMLT + XL - 2\Delta L_{CV} \tag{5.28}$$

$$W_{eff,CV} = W * WMLT + XW - 2\Delta W_{CV} \tag{5.29}$$

$$L_{eff,Bin} = L * LMLT + XL - 2\Delta L_1 \tag{5.30}$$

$$W_{eff,Bin} = W * WMLT + XW - 2\Delta W_1 \tag{5.31}$$

5.3 Binning Calculations

For a given L and W, each model parameter $PARAM_i$ is calculated as a function of PARAM, and length dependent term, LPARAM, width dependent term, WPARAM, area dependent term, PPARAM:

$$PARAM_i = PARAM + LPARAM \cdot BINL + WPARAM \cdot BINW + PPARAM \cdot BINWL \tag{5.32}$$

BINUNIT is the binning unit selector. When BINUNIT=1,

$$BINL = \frac{1e^{-6}}{L_{eff} + DLBIN} \tag{5.33}$$

$$BINW = \frac{1e^{-6}}{W_{eff} + DWBIN} \tag{5.34}$$

when BINUNIT=0,

$$BINL = \frac{1.0}{L_{eff} + DLBIN} \tag{5.35}$$

$$BINW = \frac{1.0}{W_{eff} + DWBIN} \tag{5.36}$$

and

$$BINWL = BINL \cdot BINW \tag{5.37}$$

For the list of binable parameters, please refer to the complete parameter list at the end of this technical note.

5.4 Global geomertical scaling

Following scaling formulation is used in global scaling -

$$\begin{aligned} PARAM[L] = PARAM \cdot \Bigg[1 + PARAML \cdot \left(\frac{1}{L_{eff}^{PARAMLEXP}} - \frac{1}{LLONG^{PARAMLEXP}}\right) \\ + PARAMW \cdot \left(\frac{1}{W_{eff}^{PARAMWEXP}} - \frac{1}{WWIDE^{PARAMWEXP}}\right) \\ + PARAMWL \cdot \left(\frac{1}{(L_{eff} \cdot W_{eff})^{PARAMWLEXP}}\right)\Bigg] \end{aligned} \tag{5.38}$$

LLONG is the length of extracted long channel device and WWIDE is the width for extracted wide device. They are used to ensure that scaling parameters do not affect long-wide fitting. We will not mention LLONG and WWIDE part again but all of the following scaling equation use above kind of formulation.

$$NDEP[L] = NDEP \cdot \Bigg[1 + NDEPL1 \cdot \frac{1}{L_{eff}^{NDEPLEXP1}} + NDEPL2 \cdot \frac{1}{L_{eff}^{NDEPLEXP2}} + NDEPW \cdot \frac{1}{W_{eff}^{NDEPWEXP}} + NDEPWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{NDEPWLEXP}}\Bigg] \tag{5.39}$$

$$VFB[L] = VFB \cdot \Bigg[1 + VFBL \cdot \frac{1}{L_{eff}^{VFBLEXP}} + VFBW \cdot \frac{1}{W_{eff}^{VFBWEXP}} + VFBWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{VFBWLEXP}}\Bigg] \tag{5.40}$$

$$NFACTOR[L] = NFACTOR \cdot \Bigg[1 + NFACTORL \cdot \frac{1}{L_{eff}^{NFACTORLEXP}} + NFACTORW \cdot \frac{1}{W_{eff}^{NFACTORWEXP}} + NFACTORWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{NFACTORWLEXP}}\Bigg] \tag{5.41}$$

$$CDSCD[L] = CDSCD \cdot \Bigg[1 + CDSCDL \cdot \frac{1}{L_{eff}^{CDSCDLEXP}}\Bigg] \tag{5.42}$$

$$CDSCB[L] = CDSCB \cdot \Bigg[1 + CDSCBL \cdot \frac{1}{L_{eff}^{CDSCBLEXP}}\Bigg] \tag{5.43}$$

$$\tag{5.44}$$

if (MOBSCALE=0) (5.45)

$$U0[L] = \begin{cases} U0 \cdot \left[1 - U0L \cdot \frac{1}{L_{eff}^{U0LEXP}}\right] & U0LEXP > 0 \\ U0 \cdot [1 - U0L] & \text{Otherwise} \end{cases} \tag{5.46}$$

else (5.47)

$$U_0[L] = U_0 \cdot [1 - UP1 \cdot \exp^{\frac{-L_{eff}}{LP1}} - UP2 \cdot \exp^{\frac{-L_{eff}}{LP2}}] \tag{5.48}$$

$$UA[L] = UA \cdot \left[1 + UAL \cdot \frac{1}{L_{eff}^{UALEXP}} + UAW \cdot \frac{1}{W_{eff}^{UAWEXP}} + UAWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{UAWLEXP}}\right] \tag{5.49}$$

$$EU[L] = EU \cdot \left[1 + EUL \cdot \frac{1}{L_{eff}^{EULEXP}} + EUW \cdot \frac{1}{W_{eff}^{EUWEXP}} + EUWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{EUWLEXP}}\right] \tag{5.50}$$

$$UD[L] = UD \cdot \left[1 + UDL \cdot \frac{1}{L_{eff}^{UDLEXP}}\right] \tag{5.51}$$

$$UC[L] = UC \cdot \left[1 + UCL \cdot \frac{1}{L_{eff}^{UCLEXP}} + UCW \cdot \frac{1}{W_{eff}^{UCWEXP}} + UCWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{UCWLEXP}}\right] \tag{5.52}$$

$$ETA0[L] = ETA0 \cdot \left[\frac{1}{L_{eff}^{DSUB}}\right] \tag{5.53}$$

$$ETAB[L] = ETAB \cdot \left[\frac{1}{L_{eff}^{ETABEXP}}\right] \tag{5.54}$$

$$PDIBLC[L] = PDIBLC \cdot \left[1 + PDIBLCL \cdot \frac{1}{L_{eff}^{PDIBLCLEXP}}\right] \tag{5.55}$$

$$DELTA[L] = DELTA \cdot \left[1 + DELTAL \cdot \frac{1}{L_{eff}^{DELTALEXP}}\right] \tag{5.56}$$

$$FPROUT[L] = FPROUT \cdot \left[1 + FPROUTL \cdot \frac{1}{L_{eff}^{FPROUTLEXP}}\right] \tag{5.57}$$

$$PCLM[L] = PCLM \cdot \left[1 + PCLML \cdot \frac{1}{L_{eff}^{PCLMLEXP}}\right] \tag{5.58}$$

$$\begin{aligned} VSAT[L] = VSAT \cdot \Bigg[&1 + VSATL \cdot \frac{1}{L_{eff}^{VSATLEXP}} + VSATW \cdot \frac{1}{W_{eff}^{VSATWEXP}} \\ &+ VSATWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{VSATWLEXP}}\Bigg] \end{aligned} \tag{5.59}$$

$$PSAT[L] = PSAT \cdot \left[1 + PSATL \cdot \frac{1}{L_{eff}^{PSATLEXP}}\right] \tag{5.60}$$

$$PTWG[L] = PTWG \cdot \left[1 + PTWGL \cdot \frac{1}{L_{eff}^{PTWGLEXP}}\right] \tag{5.61}$$

$$ALPHA0[L] = ALPHA0 \cdot \left[1 + ALPHA0L \cdot \frac{1}{L_{eff}^{ALPHA0LEXP}}\right] \tag{5.62}$$

$$AGIDL[L] = AGIDL \cdot \left[1 + AGIDLL \cdot \frac{1}{L_{eff}} + AGIDLW \cdot \frac{1}{W_{eff}}\right] \tag{5.63}$$

$$AGISL[L] = AGISL \cdot \left[1 + AGISLL \cdot \frac{1}{L_{eff}} + AGISLW \cdot \frac{1}{W_{eff}}\right] \tag{5.64}$$

$$AIGC[L] = AIGC \cdot \left[1 + AIGCL \cdot \frac{1}{L_{eff}} + AIGCW \cdot \frac{1}{W_{eff}}\right] \tag{5.65}$$

$$AIGS[L] = AIGS \cdot \left[1 + AIGSL \cdot \frac{1}{L_{eff}} + AIGSW \cdot \frac{1}{W_{eff}}\right] \tag{5.66}$$

$$AIGD[L] = AIGD \cdot \left[1 + AIGDL \cdot \frac{1}{L_{eff}} + AIGDW \cdot \frac{1}{W_{eff}}\right] \tag{5.67}$$

$$PIGCD[L] = PIGCD \cdot \left[1 + PIGCDL \cdot \frac{1}{L_{eff}}\right] \tag{5.68}$$

$$\begin{aligned} NDEPCV[L] = NDEPCV \cdot \Bigg[&1 + NDEPCVL1 \cdot \frac{1}{L_{eff}^{NDEPCVLEXP1}} \\ &+ NDEPCVL2 \cdot \frac{1}{L_{eff}^{NDEPCVLEXP2}} + NDEPCVW \cdot \frac{1}{W_{eff}^{NDEPCVWEXP}} \\ &+ NDEPCVWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{NDEPCVWLEXP}}\Bigg] \end{aligned} \tag{5.69}$$

$$VFBCV[L] = VFBCV \cdot \left[1 + VFBCVL \cdot \frac{1}{L_{eff}^{VFBCVLEXP}} \right.$$
$$\left. + VFBCVW \cdot \frac{1}{W_{eff}^{VFBCVWEXP}} + VFBCVWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{VFBCVWLEXP}} \right] \tag{5.70}$$

$$VSATCV[L] = VSATCV \cdot \left[1 + VSATCVL \cdot \frac{1}{L_{eff}^{VSATCVLEXP}} \right.$$
$$\left. + VSATCVW \cdot \frac{1}{W_{eff}^{VSATCVWEXP}} + VSATCVWL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{VSATCVWLEXP}} \right] \tag{5.71}$$

$$PCLMCV[L] = PCLMCV \cdot \left[1 + PCLMCVL \cdot \frac{1}{L_{eff}^{PCLMCVLEXP}}\right] \tag{5.72}$$

$$K2[L] = K2 \cdot \left[1 + K2L \cdot \frac{1}{L_{eff}^{K2LEXP}} + K2W \cdot \frac{1}{W_{eff}^{K2WEXP}} + K2WL \cdot \frac{1}{(L_{eff} \cdot W_{eff})^{K2WLEXP}}\right] \tag{5.73}$$

$$PRWB[L] = PRWB \cdot \left[1 + PRWBL \cdot \frac{1}{L_{eff}^{PRWBLEXP}}\right] \tag{5.74}$$

$$RSW[L] = RSW \cdot \left[1 + RSWL \cdot \frac{1}{L_{eff}^{RSWLEXP}}\right] \tag{5.75}$$

$$RDW[L] = RDW \cdot \left[1 + RDWL \cdot \frac{1}{L_{eff}^{RDWLEXP}}\right] \tag{5.76}$$

$$RDSW[L] = RDSW \cdot \left[1 + RDSWL \cdot \frac{1}{L_{eff}^{RDSWLEXP}}\right] \tag{5.77}$$

5.5 Terminal Voltages

BSIM-SOI 100.x.x is a body referenced model.

$$V_t = \frac{K.T}{q} \tag{5.78}$$

$$V_g = V_g - V_b \tag{5.79}$$

$$V_d = V_d - V_b \tag{5.80}$$

$$V_s = V_s - V_b \tag{5.81}$$

$$V_e = V_e - V_b \tag{5.82}$$

$$V_{gs} = V_g - V_s \tag{5.83}$$

$$V_{gd} = V_g - V_d \tag{5.84}$$

$$V_{gb} = V_g - V_b \tag{5.85}$$

$$V_{ds} = V_d - V_s \tag{5.86}$$

$$V_{es} = V_e - V_s \tag{5.87}$$

$$V_{ed} = V_e - V_d \tag{5.88}$$

$$V_{sb\_noswap} = V_s \tag{5.89}$$

$$V_{dsx} = \left[\frac{2}{AVDSX}.ln(1 + exp^{AVDSX.V_{DS}}) - V_{DS} - \frac{2}{AVDSX}.ln(2)\right] \tag{5.90}$$

$$V_{bsx} = -\left[V_s + \frac{1}{2}(V_{ds} - V_{dsx})\right] \tag{5.91}$$

5.6 Physical Quantities Calculations

$$\phi_b = \ln\left(\frac{n_{body}}{n_i}\right) \tag{5.92}$$

$$\gamma_0 = \frac{\sqrt{2 \cdot q \cdot \epsilon_{si} \cdot NDEP}}{C_{ox}\sqrt{nV_t}} \tag{5.93}$$

$$\gamma_g = \frac{\sqrt{2 \cdot q \cdot \epsilon_{si} \cdot NGATE}}{C_{ox}\sqrt{nV_t}} \tag{5.94}$$

$$\gamma^{'} = \gamma_0 \cdot \sqrt{nV_t} \tag{5.95}$$

$$\gamma_g^{'} = \gamma_g \cdot \sqrt{nV_t} \tag{5.96}$$

$$\delta_{PD} = \frac{NDEP}{NGATE} \tag{5.97}$$

$$\left(\frac{\gamma_0}{\gamma_g}\right)^2 = \left(\frac{\frac{\sqrt{2 \cdot q \cdot \epsilon_{si} \cdot NDEP}}{C_{ox}\sqrt{nV_t}}}{\frac{\sqrt{2 \cdot q \cdot \epsilon_{si} \cdot NGATE}}{C_{ox}\sqrt{nV_t}}}\right)^2 = \frac{NDEP}{NGATE} = \delta_{PD} \tag{5.98}$$

$$\gamma = \frac{\gamma_0}{1 + \delta_{PD}} \tag{5.99}$$

$$CB = \frac{epssi}{TSI} \tag{5.100}$$

$$CBOX = \frac{epsox}{TBOX} \tag{5.101}$$

$$RC = \frac{(CBOX + CDSBS)}{CB} \tag{5.102}$$

$$RT = \frac{TOXE}{TBOX} \tag{5.103}$$

$$\gamma_{sb} = \frac{\gamma_0}{RT} \tag{5.104}$$

$$\mathrm{phisf} = 2 \cdot \frac{\mathrm{phib}}{\mathrm{n}} + \mathrm{vs} \tag{5.105}$$

$$\mathrm{phidf} = 2 \cdot \frac{\mathrm{phib}}{\mathrm{n}} + \mathrm{vdeff} \tag{5.106}$$

5.7 Calculations for SOIMOD=1 DDSOI mode - Dynamic Depletion SOI Mode

For dynamic depletion device, the surface potential for the partial depletion case is solved independently in BSIM-SOI 100.x.x model. The decoupled back gate surface potential is given by [9],

$$f = \frac{\mathrm{RT}^2\left(\mathrm{phisb0} - \mathrm{vgfbb}\right)^2}{{\gamma_0}^2} - \mathrm{e}^{-\mathrm{phisb0}} - \mathrm{phisb0} + 1 \tag{5.107}$$

for compact modeling purpose the initial guess derivation suitable for back gate surface potential is as below,

$$\mathrm{x1_{csb}} = \frac{5.0}{4.0} \tag{5.108}$$

$$\mathrm{limit_{sb}} = 10^{-7} \cdot \left(1 + \gamma_{\mathrm{sb}}\sqrt{2})\right) \tag{5.109}$$

if $\mid \mathrm{vgfbb} \mid < \mathrm{limit_{sb}}$:

$$\mathrm{x1_{sb}} = 1 + \gamma_{\mathrm{sb}} \cdot \sqrt{2} \tag{5.110}$$

$$\mathrm{pd_{sb}} = -\frac{\mathrm{vgfbb}}{\gamma_{\mathrm{sb}}} \cdot \left(1 + \gamma_{\mathrm{sb}} \cdot \left(\frac{-\mathrm{vgfbb}}{6.0 \cdot \sqrt{2} \cdot \mathrm{x1_{sb}} \cdot \mathrm{x1_{sb}}}\right)\right) \tag{5.111}$$

elseif $\mathrm{vgfbb} < -\mathrm{limit_{sb}}$

$$\mathrm{PD_{yg}} = -\mathrm{vgfbb} \tag{5.112}$$

$$\mathrm{PD_z} == \frac{\mathrm{x1_{csb}} \cdot \mathrm{PD_{yg}}}{\gamma_{\mathrm{sb}}} \tag{5.113}$$

$$\mathrm{PD}_\eta = 0.5 \cdot \left(\mathrm{PD_z} + 10.0 - \sqrt{(\mathrm{PD_z} - 6.0) \cdot (\mathrm{PD_z} - 6.0) + 64.0}\right) \tag{5.114}$$

$$\mathrm{PD_a} = (\mathrm{PD_{yg}} - \mathrm{PD}_\eta) \cdot (\mathrm{PD_{yg}} - \mathrm{PD}_\eta) + {\gamma_{\mathrm{sb}}}^2 \cdot (\mathrm{PD}_\eta + 1.0) \tag{5.115}$$

$$\mathrm{PD_c} = 2.0 \cdot (\mathrm{PD_{yg}} - \mathrm{PD}_\eta) - {\gamma_{\mathrm{sb}}}^2 \tag{5.116}$$

$$\mathrm{PD}_\tau = \log\left(\frac{\mathrm{PD_a}}{{\gamma_{\mathrm{sb}}}^2}\right) - \mathrm{PD}_\eta \tag{5.117}$$

$$\mathrm{PD_v} = \mathrm{PD_a} + \mathrm{PD_c} \tag{5.118}$$

$$\mathrm{PD}_{\mu} = \frac{\mathrm{PD_v}^2}{\mathrm{PD}_{\tau}} \tag{5.119}$$

$$\mathrm{PD_{y0}} = \mathrm{PD}_{\eta} + \frac{\mathrm{PD_a} \cdot \mathrm{PD_v}}{\mathrm{PD}_{\mu} + \mathrm{PD_c} \cdot \left(\frac{\mathrm{PD_c}^2}{3} - \mathrm{PD_a}\right) \cdot \frac{\mathrm{PD_v}}{\mathrm{PD}_{\mu}}} \tag{5.120}$$

$$\mathrm{PD_{D0}} = \exp\left(\mathrm{PD_{y0}}\right) \tag{5.121}$$

$$\mathrm{PD_{temp}} = \mathrm{PD_{yg}} - \mathrm{PD_{y0}} \tag{5.122}$$

$$\mathrm{PD_p} = 2.0 \cdot \mathrm{PD_{temp}} + {\gamma_{\mathrm{sb}}}^2 \cdot \left(\mathrm{PD_{D0}} - 1.0\right) \tag{5.123}$$

$$\mathrm{PD_q} = \mathrm{PD_{temp}} \cdot \mathrm{PD_{temp}} + {\gamma_{\mathrm{sb}}}^2 \cdot \left(\mathrm{PD_{y0}} + 1.0 - \mathrm{PD_{D0}}\right) \tag{5.124}$$

$$\mathrm{PD_{xi}} = 1.0 - {\gamma_{\mathrm{sb}}}^2 \cdot \frac{\mathrm{PD_{D0}}}{2} \tag{5.125}$$

$$\mathrm{PD_{temp1}} = \mathrm{PD_p} \cdot \mathrm{PD_p} - 4.0 \cdot \left(\mathrm{PD_{xi}} \cdot \mathrm{PD_q}\right) \tag{5.126}$$

$$\mathrm{PD_w} = 2.0 \cdot \left(\frac{\mathrm{PD_q}}{\left(\mathrm{PD_p} + \sqrt{\mathrm{PD_{temp}}}\right.}\right) \tag{5.127}$$

$$\mathrm{pd_{sb}} = -\left(\mathrm{PD_{y0}} + \mathrm{PD_w}\right) \tag{5.128}$$

else

$$\mathrm{PD_{Afac}} = \left(\mathrm{x1_{sb}} \cdot \mathrm{x1_{csb}} \cdot \mathrm{inv_{xg1sb}} - 1.0\right) \cdot \mathrm{inv_{xg1sb}} \tag{5.129}$$

$$\mathrm{PD_{xbar}} = \frac{\mathrm{vgfbb}}{\mathrm{xi_{sb}}} \cdot \left(1.0 + \mathrm{PD_{Afac}} \cdot \mathrm{vgfbb}\right) \tag{5.130}$$

$$\mathrm{PD_{temp2}} = \exp\left(-\mathrm{PD_{xbar}}\right) \tag{5.131}$$

$$\mathrm{PD_{w1}} = 1.0 - \mathrm{PD_{temp2}} \tag{5.132}$$

$$\mathrm{PD_{x0}} = \mathrm{vgfbb} + \frac{{\gamma_{\mathrm{sb}}}^2}{2} - \gamma_{\mathrm{sb}} \cdot \sqrt{\left(\mathrm{vgfbb} + {\gamma_{\mathrm{sb}}}^2 \cdot 0.25 - \mathrm{PD_{w1}}\right)} \tag{5.133}$$

$$\mathrm{PD_D01} = \exp\left(-\mathrm{PD_{x0}}\right) \tag{5.134}$$

$$\mathrm{PD_{p1}} = 2.0 \cdot \left(\mathrm{vgfbb} - \mathrm{PD_{x0}}\right) + {\gamma_{\mathrm{sb}}}^2 \cdot \left(1.0 - \mathrm{PD_{D01}}\right) \tag{5.135}$$

$$\mathrm{PD_{q1}} = \left(\mathrm{vgfbb} - \mathrm{PD_x0}\right)^2 - {\gamma_{\mathrm{sb}}}^2 \cdot \left(\mathrm{PD_{x0}} - 1.0 + \mathrm{PD_{D01}}\right) \tag{5.136}$$

$$\mathrm{PD_{xi1}} = 1.0 - \frac{{\gamma_{\mathrm{sb}}}^2}{2} \cdot \mathrm{PD_{D01}} \tag{5.137}$$

$$\mathrm{PD_{temp3}} = \mathrm{PD_{p1}} \cdot \mathrm{PD_{p1}} - 4.0 \cdot \left(\mathrm{PD_{xi1}} \cdot \mathrm{PD_{q1}}\right) \tag{5.138}$$

$$\mathrm{PD_u} = 2.0 \cdot \frac{\mathrm{PD_{q1}}}{\mathrm{PD_{p1}} + \sqrt{\left(\mathrm{PD_{temp3}}\right)}} \tag{5.139}$$

$$\mathrm{pd_{sb}} = \mathrm{PD_{x0}} + \mathrm{PD_u} \tag{5.140}$$

The above initial guess provides smooth derivatives for back gate surface potential, the accuracy of the initial guess is further improved by the following procedure,

$$\text{if } \mid \mathrm{vgfbb} \mid < \mathrm{limit_{sb}} :$$

$$phisb0 = \frac{1}{\mathrm{xi_{sb}}}\,\mathrm{v_{gfbb}}\left(\frac{\sqrt{2}\,\cdot\gamma_{\mathrm{sb}}\,\mathrm{v_{gfbb}}}{12\,\cdot\mathrm{x1_{sb}}^2} - 1\right) \tag{5.141}$$

$$\text{else} \tag{5.142}$$

$$\mathrm{f} = \frac{\mathrm{RT}^2\,(\mathrm{pd_{sb}} - \mathrm{vgfbb})^2}{\gamma_{\mathrm{sb}}{}^2} - \mathrm{e}^{-\mathrm{pd_{sb}}} - \mathrm{pd_{sb}} + 1 \tag{5.143}$$

$$\mathrm{f}' = \mathrm{e}^{-\mathrm{pd_{sb}}} + \frac{\mathrm{RT}^2\,\left(2\,\mathrm{pd_{sb}} - 2\,\mathrm{vgfbb}\right)}{\gamma_{\mathrm{sb}}{}^2} - 1 \tag{5.144}$$

$$\mathrm{phisb0} = \mathrm{pd_{sb}} - \frac{\mathrm{f}}{\mathrm{f}'} \tag{5.145}$$

Coupled surface potential for the Dynamically depleted FET is given by [9], (BSIM-SOI 4.6.0),

$$x1 = 1.0 + \gamma_0 \cdot \frac{1}{\sqrt{2}} \tag{5.146}$$

$$\psi_{\mathrm{sf}} = 2 \cdot \frac{\mathrm{phib}}{\mathrm{n}} + \mathrm{vs} \tag{5.147}$$

$$\mathrm{T0} = -\mathrm{DVBD1} \cdot \frac{\mathrm{Leff}}{\mathrm{litl}} \tag{5.148}$$

$$\mathrm{T1} = \mathrm{DVBD0} \cdot \left(\mathrm{e}^{\frac{\mathrm{T0}}{2}} + 2.0 \cdot \mathrm{e}^{\mathrm{T0}}\right) \cdot (\mathrm{V_{ds}} + \mathrm{VSCE}) \tag{5.149}$$

$$\psi^* = (1.0 + \frac{\mathrm{LPE1}}{\mathrm{Leff}}) \cdot \frac{\mathrm{q} \cdot \mathrm{NDEP} \cdot \mathrm{TSI}^2}{2 \cdot \epsilon_{\mathrm{si}} \cdot \mathrm{nVt}} + \frac{\mathrm{VBSA}}{\mathrm{nVt}} - \mathrm{RC} \cdot \mathrm{vgfbb} \cdot \mathrm{CDSBS1_i} \tag{5.150}$$

$$\mathrm{phics} = \psi^* + (1 + \mathrm{RC}) \cdot \mathrm{phisb0} \tag{5.151}$$

$$\mathrm{vg_{FD}} = \psi^* + \gamma_0 \cdot \sqrt{(\mathrm{e}^{-\psi^*} + \psi^* - 1.0)} \tag{5.152}$$

$$f = (\mathrm{vgfb} - \psi)^2 - \mathrm{RT}^2 \left(\mathrm{vgfbb} - \frac{\mathrm{phisb0} + \frac{\mathrm{nVt}\log\left(\frac{e^{-\frac{\mathrm{phics}-\mathrm{nVt}\,\psi}{\mathrm{nVt}}}+1}{e^{-\frac{\mathrm{phics}-\mathrm{nVt}\,\mathrm{spvfb}}{\mathrm{nVt}}}+1}\right)}{\mathrm{RC}+1}}{\mathrm{nVt}} \right)^2$$

$$+\gamma_0{}^2 \left(e^{-\frac{\mathrm{phisb0} + \frac{\mathrm{nVt}\log\left(\frac{e^{-\frac{\mathrm{phics}-\mathrm{nVt}\,\psi}{\mathrm{nVt}}}+1}{e^{-\frac{\mathrm{phics}-\mathrm{nVt}\cdot\mathrm{spvfb}}{\mathrm{nVt}}}+1}\right)}{\mathrm{RC}+1}}{\mathrm{nVt}}} - e^{-\psi} - \psi \right)$$

$$+\gamma_0{}^2 \left(\frac{\mathrm{phisb0} + \frac{\mathrm{nVt}\log\left(\frac{e^{-\frac{\mathrm{phics}-\mathrm{nVt}\,\psi}{\mathrm{nVt}}}+1}{e^{-\frac{\mathrm{phics}-\mathrm{nVt}\,\mathrm{spvfb}}{\mathrm{nVt}}}+1}\right)}{\mathrm{RC}+1}}{\mathrm{nVt}} + e^{-\psi_{\mathrm{sf}}} \left(\psi - e^{\psi} + \frac{\psi^2}{\psi^2+2} + 1 \right) \right) \tag{5.153}$$

for compact modeling purpose the initial guess derivation suitable for coupled surface potential of dynamic depletion region is as below,

$$limit = 10^{-3} \cdot \mathrm{x1} \tag{5.154}$$

$if\ \ \mathrm{vgfbb}\ < \mathrm{vg}_{\mathrm{FD}};$ Initial guess for partial depletion operation

$$\text{if} \mid \mathrm{vgfbb} \mid < \text{limit} \tag{5.155}$$

$$\mathrm{T0} = \frac{1}{\mathrm{x1}^2} \cdot \frac{1}{6 \cdot \sqrt{2}} \tag{5.156}$$

$$\mathrm{sp}_{\mathrm{dd}} = \mathrm{vgfb} \cdot \frac{1}{\mathrm{x1}} \cdot (1.0 + \mathrm{vgfb} \cdot (1.0 - e^{-\psi_{\mathrm{sf}}}) \cdot \gamma_0 \cdot \mathrm{T0}) \tag{5.157}$$

else if vgfbb $< -$limit

$$\mathrm{SPS}_{\mathrm{yg}} = -\mathrm{vgfb} \tag{5.158}$$

$$\mathrm{SPS}_{\mathrm{ysub}} = 1.25 \cdot \left(\mathrm{SPS}_{\mathrm{yg}} \cdot \frac{1}{\mathrm{x1}} \right) \tag{5.159}$$

$$\mathrm{SPS}_{\mathrm{eta}} = 0.5 \cdot (\mathrm{SPS}_{\mathrm{ysub}} + 10.0) - 0.5 \cdot \left(\sqrt{((\mathrm{SPS}_{\mathrm{ysub}} - 6.0) \cdot (\mathrm{SPS}_{\mathrm{ysub}} - 6.0) + 64.0)} \right) \tag{5.160}$$

$$\mathrm{SPS_{temp1}} = \mathrm{SPS_{yg}} - \mathrm{SPS_{eta}} \tag{5.161}$$

$$\mathrm{SPS_{a}} = \mathrm{SPS_{temp1}} \cdot \mathrm{SPS_{temp1}} + {\gamma_0}^2 \cdot (\mathrm{SPS_{eta}} + 1.0) \tag{5.162}$$

$$\mathrm{SPS_{c}} = 2.0 \cdot \mathrm{SPS_{temp1}} - {\gamma_0}^2 \tag{5.163}$$

$$\mathrm{SPS_{tau}} = -\mathrm{SPS_{eta}} + \log\left(\mathrm{SPS_{a}} \cdot \frac{1}{{\gamma_0}^2}\right) \tag{5.164}$$

$$\mathrm{sigma0}\left(\mathrm{SPS_{a}}, \mathrm{SPS_{c}}, \mathrm{SPS_{tau}}, \mathrm{SPS_{eta}}, \mathrm{SPS_{y0}}\right) \tag{5.165}$$

$$\mathrm{SPS_{delta0}} = \mathrm{e}^{\mathrm{SPS_{y0}}} \tag{5.166}$$

$$\mathrm{SPS_{delta1}} = \frac{1}{\mathrm{SPS_{delta0}}} \tag{5.167}$$

$$\mathrm{SPS_{temp2}} = \frac{1}{(2.0 + \mathrm{SPS_{y0}} \cdot \mathrm{SPS_{y0}})} \tag{5.168}$$

$$\mathrm{SPS_{xi0}} = \mathrm{SPS_{y0}} \cdot \mathrm{SPS_{y0}} \cdot \mathrm{SPS_{temp2}} \tag{5.169}$$

$$\mathrm{SPS_{xi1}} = 4.0 \cdot (\mathrm{SP_{Sy}0} \cdot \mathrm{SPS_{temp2}} \cdot \mathrm{SPS_{temp2}}) \tag{5.170}$$

$$\mathrm{SPS_{xi2}} = (8.0 \cdot \mathrm{SPS_{temp2}} - 12.0 \cdot \mathrm{SPS_{xi0}}) \cdot \mathrm{SPS_{temp2}} \cdot \mathrm{SPS_{temp2}} \tag{5.171}$$

$$\mathrm{SPS_{temp3}} = \mathrm{SPS_{yg}} - \mathrm{SPS_{y0}} \tag{5.172}$$

$$\mathrm{SPS_{temp4}} = (\mathrm{e}^{-\psi_{\mathrm{sf}}}) \cdot \mathrm{SPS_{delta1}} \tag{5.173}$$

$$\begin{aligned}\mathrm{SPS_{pC}} &= 2.0 \cdot \mathrm{SPS_{temp3}} \\ &+{\gamma_0}^2 \cdot (\mathrm{SPS_{delta0}} - 1.0 - \mathrm{SPS_{temp4}} + (\mathrm{e}^{-\psi_{\mathrm{sf}}}) \cdot (1.0 - \mathrm{SPS_{xi1}}))\end{aligned} \tag{5.174}$$

$$\begin{aligned}\mathrm{SPS_{qC}} &= \mathrm{SP_{Stemp3}} \cdot \mathrm{SPS_{temp3}} \\ &\quad -{\gamma_0}^2 \cdot (\mathrm{SPS_{delta0}} - \mathrm{SPS_{y0}} - 1.0 + \mathrm{SPS_{temp4}} \\ &\quad +(\mathrm{e}^{-\psi_{\mathrm{sf}}}) \cdot (\mathrm{SPS_{y0}} - 1.0 - \mathrm{SPS_{xi0}}))\end{aligned} \tag{5.175}$$

$$\mathrm{SPS_{temp5}} = 2.0 - {\gamma_0}^2 \cdot (\mathrm{SPS_{delta0}} + \mathrm{SPS_{temp4}} - (\mathrm{e}^{-\psi_{\mathrm{sf}}}) \cdot \mathrm{SPS_{xi2}}) \tag{5.176}$$

$$\mathrm{SPS_{temp6}} = \mathrm{SPS_{pC}} \cdot \mathrm{SPS_{pC}} - 2.0 \cdot (\mathrm{SPS_{qC}} \cdot \mathrm{SPS_{temp5}}) \tag{5.177}$$

$$\mathrm{sp_{dd}} = -\mathrm{SPS_{y0}} - 2.0 \cdot \frac{\mathrm{SPS_{qC}}}{(\mathrm{SPS_{pC}} + \sqrt{\mathrm{SPS_{temp6}}})} \tag{5.178}$$

else

$$\mathrm{SP_{xg1}} = \frac{1}{(1.25 + \gamma_0 \cdot 7.324648775608221\mathrm{e}-001)} \tag{5.179}$$

$$\mathrm{SPSA_{fac}} = (\mathrm{x1} \cdot 1.25 \cdot \mathrm{SP_{xg1}} - 1.0) \cdot \mathrm{SP_{xg1}} \tag{5.180}$$

$$\mathrm{SPS_{xbar}} = \mathrm{vgfb1} \cdot \frac{1}{\mathrm{x1}} \cdot (1.0 + \mathrm{SPSA_{fac}} \cdot \mathrm{vgfb1}) \tag{5.181}$$

$$\mathrm{SPS_{temp}} = \mathrm{e}^{-\mathrm{SPS_{xbar}}} \tag{5.182}$$

$$\mathrm{SPS_{w}} = 1.0 - \mathrm{SPS_{temp}} \tag{5.183}$$

$$\mathrm{SPS_{x1}} = \mathrm{vgfb1} + {\gamma_0}^2 \cdot 0.5 - \mathrm{gam} \cdot \sqrt{(\mathrm{vgfb1} + {\gamma_0}^2 \cdot 0.25 - \mathrm{SPS_w})} \tag{5.184}$$

$$\mathrm{SPS_{bx}} = \mathrm{e}^{\psi_{\mathrm{sf}}} + 3.0 \tag{5.185}$$

$$\begin{aligned}\mathrm{SPS_{eta}} &= \mathrm{MINF}(\mathrm{SPS_{x1}}, \mathrm{SPS_{bx}}, 5.0)\\ &\quad -0.5 \cdot (\mathrm{SPS_{bx}} - \sqrt{(\mathrm{SPS_{bx}} \cdot \mathrm{SPS_{bx}} + 5.0)})\end{aligned} \tag{5.186}$$

$$\mathrm{SPS_{temp}} = \mathrm{vgfb1} - \mathrm{SPS_{eta}} \tag{5.187}$$

$$\mathrm{SPS_{temp1}} = \mathrm{e}^{-\mathrm{SP_{Se}ta}} \tag{5.188}$$

$$\mathrm{SPS_{temp2}} = \mathrm{frac1}(2.0 + \mathrm{SPS_{eta}} \cdot \mathrm{SPS_{eta}}) \tag{5.189}$$

$$\mathrm{SPS_{xi0}} = \mathrm{SPS_{eta}} \cdot \mathrm{SPS_{eta}} \cdot \mathrm{SPS_{temp2}} \tag{5.190}$$

$$\mathrm{SPS_{xi1}} = 4.0 \cdot (\mathrm{SPS_{eta}} \cdot \mathrm{SPS_{temp2}} \cdot \mathrm{SPS_{temp2}}) \tag{5.191}$$

$$\mathrm{SPS_{xi2}} = (8.0 \cdot \mathrm{SPS_{temp2}} - 12.0 \cdot \mathrm{SPS_{xi0}}) \cdot \mathrm{SPS_{temp2}} \cdot \mathrm{SPS_{temp2}} \tag{5.192}$$

$$\begin{aligned}&\mathrm{SPS_{a}} = \mathrm{MAX}(1.0\mathrm{e}-40, \mathrm{SPS_{temp}} \cdot \mathrm{SPS_{temp}}\\ &-{\gamma_0}^2 \cdot \left(\mathrm{SPS_{temp1}} + \mathrm{SPS_{eta}} - 1.0 - \mathrm{e}^{-\psi_{\mathrm{sf}}} \cdot (\mathrm{SPS_{eta}} + 1.0 + \mathrm{SPS_{xi0}})\right)\end{aligned} \tag{5.193}$$

$$\mathrm{SPS_{b}} = 1.0 - 0.5 \cdot ({\gamma_0}^2 \cdot \left(\mathrm{SPS_{temp1}} - \mathrm{e}^{-\psi_{\mathrm{sf}}} \cdot \mathrm{SPS_{xi2}}\right) \tag{5.194}$$

$$\mathrm{SPS_{c}} = 2.0 \cdot \mathrm{SPS_{temp}} + {\gamma_0}^2 \cdot (1.0 - \mathrm{SPS_{temp1}} - \mathrm{exp_{ns}} \cdot (1.0 + \mathrm{SPS_{xi1}})) \tag{5.195}$$

$$\mathrm{SPS_{tau}} = \left(2 \cdot \frac{\mathrm{phib}}{\mathrm{n}} + \mathrm{vs}\right) - \mathrm{SPS_{eta}} + \log\left(\frac{\mathrm{SP_{Sa}}}{{\gamma_0}^2}\right)$$

$$\mathrm{SPS_{tau}} = (\mathrm{phisf}) - \mathrm{SP_{Se}ta} + \log\left(\frac{\mathrm{SPS_a}}{{\gamma_0}^2}\right)$$

$$\mathrm{sigma1}\,(\mathrm{SP_{Sa}}, \mathrm{SP_{Sb}}, \mathrm{SP_{Sc}}, \mathrm{SP_{St}au}, \mathrm{SP_{Se}ta}, \mathrm{SP_{Sx}0}) \tag{5.196}$$

$$\mathrm{SPS_{delta0}} = \mathrm{e}^{\psi_{\mathrm{sf}}} \cdot \mathrm{SPS_{delta0}} \tag{5.197}$$

$$\tag{5.198}$$

$\mathrm{if}(\mathrm{SPS_{x0}} < \mathrm{pe1})$

$$\mathrm{SPS_{delta0}} = \mathrm{e}^{\mathrm{SPS_{x0}}} \tag{5.199}$$

$$\mathrm{SPS_{delta1}} = \frac{1}{\mathrm{SPS_{delta0}}} \tag{5.200}$$

else

$\mathrm{if}(\mathrm{SPS_{x0}} > (\mathrm{phisf}) - \mathrm{pe1})$

$$\mathrm{SPS_{delta0}} = \mathrm{e}^{(\mathrm{SP_{S}x0} - (\mathrm{phisf}))} \tag{5.201}$$

$$\mathrm{SPS_{delta1}} = \frac{\mathrm{exp_{n}s}}{\mathrm{SPS_{delta0}}} \tag{5.202}$$

else

$$\mathrm{SPS_{delta0}} = \frac{\mathrm{Ede1}}{\mathrm{PE3}((\mathrm{phisf}) - \mathrm{SPS_{x0}} - \mathrm{pe1})} \tag{5.203}$$

$$\mathrm{SPS_{delta1}} = \frac{\mathrm{Ede1}}{\mathrm{PE3}(\mathrm{SPS_{x0}} - \mathrm{pe1})} \tag{5.204}$$

end

end

$$\mathrm{SPS_{temp}} = \frac{1.0}{(2.0 + \mathrm{SPS_{x0}} \cdot \mathrm{SPS_{x0}})} \tag{5.205}$$

$$\mathrm{SPS_{xi0}} = \mathrm{SPS_{x0}} \cdot \mathrm{SPS_{x0}} \cdot \mathrm{SPS_{temp}} \tag{5.206}$$

$$\mathrm{SPS_{xi1}} = 4.0 \cdot (\mathrm{SPS_{x0}} \cdot \mathrm{SPS_{temp}} \cdot \mathrm{SPS_{temp}}) \tag{5.207}$$

$$\mathrm{SPS_{xi2}} = (8.0 * \mathrm{SPS_{temp}} - 12.0 \cdot \mathrm{SPS_{xi0}}) \cdot \mathrm{SPS_{temp}} \cdot \mathrm{SPS_{temp}} \tag{5.208}$$

$$\mathrm{SPS_{temp}} = \mathrm{vgfb1} - \mathrm{SPS_{x0}} \tag{5.209}$$

$$\mathrm{SPS_{pC}} = 2.0 \cdot \mathrm{SPS_{temp}}$$

$$+{\gamma_0}^2 \cdot (1.0 - \mathrm{SPS_{delta1}} + \mathrm{SPS_{delta0}} - (\mathrm{exp_{ns}}) \cdot (1.0 + \mathrm{SPS_{xi1}}))$$

$$\mathrm{SPS_{qC}} = \mathrm{SPS_{temp}} \cdot \mathrm{SPS_{temp}}$$

$$-{\gamma_0}^2 \cdot (\mathrm{SPS_{delta1}} + \mathrm{SPS_{x0}} - 1.0 + \mathrm{SPS_{delta0}} - (\mathrm{exp_{ns}}) \cdot (\mathrm{SPS_{x0}} + 1.0 + \mathrm{SPS_{xi0}})) \tag{5.210}$$

$$\mathrm{SPS_{temp}} = 2.0 - {\gamma_0}^2 (\mathrm{SPS_{delta1}} + \mathrm{SPS_{delta0}} - (\mathrm{exp_{ns}}) \cdot \mathrm{SPS_{xi2}}) \tag{5.211}$$

$$\mathrm{SPS_{temp}} = \mathrm{SPS_{pC}} \cdot \mathrm{SPS_{pC}} - 2.0 \cdot (\mathrm{SPS_{qC}} \cdot \mathrm{SPS_{temp}}) \tag{5.212}$$

$$\mathrm{sp_{dd}} = \mathrm{SPS_{x0}} + 2.0 \cdot \left(\frac{\mathrm{SPS_{qC}}}{(\mathrm{SP_{Sp}C} + \mathrm{sqrt}(\mathrm{SPS_{temp}}))} \right) \tag{5.213}$$

else $\quad$ −Initial guess for full depletion operation

$$\mathrm{zeta} = \mathrm{RT} \tag{5.214}$$

$$\mathrm{zeta2} = \mathrm{zeta} \cdot \mathrm{zeta} \tag{5.215}$$

$$\mathrm{u_{crit}} = \mathrm{phic_{star}} - \mathrm{phisb0} \cdot \frac{1}{\mathrm{nVt}} \tag{5.216}$$

$$\mathrm{usf_{FD}} = \mathrm{vgfb1} - \gamma_0 \cdot \sqrt{(\mathrm{e}^{-\mathrm{u_{crit}}} + \mathrm{u_{crit}} - 1)} \tag{5.217}$$

$$\mathrm{SPS_{bx}} = (\mathrm{phisf}) + 3.0 \tag{5.218}$$

$$\mathrm{SPS_{Z0}} = \mathrm{MINF}(\mathrm{usf_{FD}}, \mathrm{SPS_{bx}}, 40) \tag{5.219}$$

$$\mathrm{a_{fd}} = (\mathrm{vgfb1} - \mathrm{SPS_{Z0}}) \cdot (\mathrm{vgfb1} - \mathrm{SPS_{Z0}}) - {\gamma_0}^2 \cdot (\mathrm{phic_{star}})$$

$$-\mathrm{zeta2} \cdot (\mathrm{vgfbb} - \mathrm{SPS_{Z0}} + \mathrm{phic_{star}}) \cdot (\mathrm{vgfbb} - \mathrm{SPS_{Z0}} + \mathrm{phic_{star}}) \tag{5.220}$$

$$\mathrm{c_{fd}} = 2 \cdot (\mathrm{vgfb1} - \mathrm{SPS_{Z0}}) - 2 \cdot \mathrm{zeta2} \cdot (\mathrm{vgfbb} - \mathrm{SPS_{Z0}} + \mathrm{phic_{star}}) \tag{5.221}$$

$$\mathrm{c_{fd2}} = \mathrm{c_{fd}} \cdot \mathrm{c_{fd}} \tag{5.222}$$

$$\mathrm{d_{fd}} = 1 - \mathrm{zeta2} \tag{5.223}$$

$\mathrm{if}(\mathrm{a_{fd}} < 0)\mathrm{begin}$

$$\mathrm{a_{fd}} = 0 \tag{5.224}$$

end

$$\mathrm{tau_{fd}} = \mathrm{phisf} - \mathrm{SPS_{Z0}} + \log\left(\mathrm{a_{fd}} \cdot \frac{1}{{\gamma_0}^2}\right) \tag{5.225}$$

$$\mathrm{v_{fd}} = \mathrm{a_{fd}} + \mathrm{c_{fd}} \tag{5.226}$$

$$\mathrm{v_{fd2}} = \mathrm{v_{fd}} \cdot \mathrm{v_{fd}} \tag{5.227}$$

$$\mathrm{mu_{fd}} = \frac{\mathrm{v_{fd2}}}{\mathrm{tau_{fd}}} + 0.5 \cdot \mathrm{c_{fd2}} - \mathrm{a_{fd}} \cdot \mathrm{d_{fd}} \tag{5.228}$$

$$\mathrm{temp1_{fd}} = \frac{\mathrm{c_{fd}} \cdot \mathrm{v_{fd}}}{\mathrm{mu_{fd}}} \tag{5.229}$$

$$\mathrm{temp2_{fd}} = \left(\frac{1}{3} \cdot \mathrm{c_{fd2}} - \mathrm{a_{fd}} \cdot \mathrm{d_{fd}}\right) \tag{5.230}$$

$$\mathrm{u_{first_r}} = \frac{(\mathrm{v_{fd}} \cdot \mathrm{a_{fd}})}{(\mathrm{mu_f d} + \mathrm{temp1_{fd}} \cdot \mathrm{temp2_{fd}})} \tag{5.231}$$

$$\mathrm{u0_{first_F D}} = \mathrm{SPS_{Z0}} + \mathrm{u_{first_r}} \tag{5.232}$$

$$\mathrm{delta0_{fd}} = \mathrm{e}^{\mathrm{u0_{first_F D}} - \mathrm{phisf}} \tag{5.233}$$

$$\tag{5.234}$$

$$
\begin{aligned}
&\mathrm{p_{fd}} = 2 \cdot (\mathrm{vgfb1} - \mathrm{u0_{first_FD}}) - 2 \cdot \mathrm{zeta2} \cdot (\mathrm{vgfbb} - \mathrm{u0_{first_FD}} + \mathrm{phic_{star}}) \\
&+{\gamma_0}^2 \cdot \mathrm{delta0_{fd}} \qquad (5.235) \\
\mathrm{q_{fd}} &= (\mathrm{vgfb1} - \mathrm{u0_{first_FD}}) \cdot (\mathrm{vgfb1} - \mathrm{u0_{first_FD}}) - \mathrm{zeta2} \cdot (\mathrm{vgfbb} - \mathrm{u0_{first_FD}} + \mathrm{phic_{star}}) \\
&\cdot(\mathrm{vgfbb} - \mathrm{u0_{first_FD}} + \mathrm{phic_{star}}) - {\gamma_0}^2 \cdot (\mathrm{phic_{star}} + \mathrm{delta0_{fd}}) \\
&\mathrm{temp3_{fd}} = 2 \cdot \mathrm{q_{fd}} \cdot (2 - 2 \cdot \mathrm{zeta2} - {\gamma_0}^2 \cdot \mathrm{delta0_{fd}}) \qquad (5.236) \\
&\mathrm{w_{fd}} = \frac{(2 \cdot \mathrm{q_{fd}})}{\left(\mathrm{p_{fd}} + \sqrt{(\mathrm{p_{fd}} \cdot \mathrm{p_{fd}} - \mathrm{temp3_{fd}})}\right)} \qquad (5.237) \\
&\mathrm{sp_{dd}} = \mathrm{u0_{first_FD}} + \mathrm{w_{fd}} \qquad (5.238) \\
&\mathrm{end} \qquad (5.239) \\
& \qquad (5.240)
\end{aligned}
$$

The above inital guess provides smooth derivatives for coupled surface potnetial, the accuracy of the coupled initial guess is further improved by the following procedure,

$$
\begin{aligned}
&\mathrm{if} \mid \mathrm{vgfb} \mid < \mathrm{limit}: \\
&\mathrm{T0} = \frac{1}{\mathrm{x1}^2} \cdot \frac{1}{6} \frac{1}{\sqrt{2}} \qquad (5.241) \\
&\psi = \mathrm{vgfb1} \cdot \frac{1}{\mathrm{X1}} \cdot (1.0 + \mathrm{vgfb1} * (1.0 - (\mathrm{exp_{ns}})) \cdot \gamma_0 \cdot \mathrm{T0}) \qquad (5.242) \\
& \qquad (5.243)
\end{aligned}
$$

else

$$
\psi = \mathrm{sp_{dd}} - \frac{\mathrm{f}}{\mathrm{f}'} \cdot \left(1 + \frac{\mathrm{f}' \cdot \mathrm{f}''}{2 \cdot \mathrm{f}' \cdot \mathrm{f}'}\right) \qquad (5.244)
$$

where, f is coupled Poisson's equation (5.153), and f‘, f“ are its first and second derivatives. With three iteration to final coupled Poisson's equation solutions accuracy of nV is achieved compared to the numerical solution. Similar procedure is followed to calculate the drain side surface potential.

5.7.1 Drain current and Charges Calculations for SOIMOD=1

source side variable calculations:

if $\psi_{\mathrm{s}} < 0$

$$\mathrm{qis} = 0.0 \tag{5.245}$$

$$\mathrm{xgm} = \mathrm{vgfb} - \psi_{\mathrm{s}} \tag{5.246}$$

$$\mathrm{Voxm} = \mathrm{xgm} \cdot \mathrm{nVt} \tag{5.247}$$

$$\mathrm{Vdsat} = \mathrm{Vdsat}_{\mathrm{lim}} \tag{5.248}$$

$$\mathrm{vdeff} = 0.0 \tag{5.249}$$

else

$$\mathrm{delta}_{1\mathrm{s}} = 0.0 \tag{5.250}$$

$$\mathrm{tempc} = \frac{1.0}{2.0 + \psi_{\mathrm{s}} \cdot \psi_{\mathrm{s}}} \tag{5.251}$$

$$\mathrm{xi0s} = \psi_{\mathrm{s}} \cdot \psi_{\mathrm{s}} \cdot \mathrm{tempc} \tag{5.252}$$

$$\mathrm{xi1s} = 4.0 \cdot (\psi_{\mathrm{s}} \cdot \mathrm{tempc} \cdot \mathrm{tempc}) \tag{5.253}$$

$$\mathrm{xi2s} = (8.0 \cdot \mathrm{tempc} - 12.0 \cdot \mathrm{xi0s}) \cdot \mathrm{tempc} \cdot \mathrm{tempc} \tag{5.254}$$

$$\mathrm{delta}_{1\mathrm{s}} = \mathrm{e}^{\psi_{\mathrm{s}}} \tag{5.255}$$

$$\mathrm{Es1} = \frac{1.0}{\mathrm{delta}_1 \mathrm{s}} \tag{5.256}$$

$$\mathrm{delta}_{1\mathrm{s}} = \mathrm{e}^{-\mathrm{phisf}} \cdot \mathrm{delta}_{1\mathrm{s}} \tag{5.257}$$

$$\mathrm{Ds} = \mathrm{delta}_{1\mathrm{s}} - \mathrm{e}^{-\mathrm{phisf}} \cdot (\psi_{\mathrm{s}} + 1.0 + \mathrm{xi0s}) \tag{5.258}$$

$$\mathrm{Ps} = (\mathrm{vgfb} - \psi_{\mathrm{s}}) * (\mathrm{vgfb} - \psi_{\mathrm{s}}) \cdot \frac{1}{{\gamma_0}^2} - \mathrm{Ds} \tag{5.259}$$

$$\mathrm{sqms} = \sqrt{\mathrm{Ps}} \tag{5.260}$$

$$\mathrm{alphas} = 1.0 + 0.5 \cdot \frac{\mathrm{gam} \cdot (1.0 - \mathrm{Es1})}{\mathrm{sqms}} \tag{5.261}$$

$$\mathrm{xgs} = \mathrm{gamma}_0 \cdot \sqrt{\mathrm{Ps} + \mathrm{Ds}} \tag{5.262}$$

$$\mathrm{qis} = \frac{{\gamma_0}^2 \cdot \mathrm{Ds} \cdot \mathrm{nVt}}{\mathrm{xgs} + \gamma \cdot \mathrm{sqms}} \tag{5.263}$$

$$\text{qbs} = \text{sqms} \cdot \gamma \cdot \text{nVt} \tag{5.264}$$

$$\text{Em} = \text{Es1} \tag{5.265}$$

$$\text{Ed} = \text{Em} \tag{5.266}$$

$$\text{Dm} = \text{Ds} \tag{5.267}$$

$$\text{Dd} = \text{Dm} \tag{5.268}$$

end

Drain side variable calculations:

$$\text{x}_{\text{ds}} = \psi_{\text{d}} - \psi_{\text{s}} \tag{5.269}$$

$$\text{dps} = \text{x}_{\text{ds}} \cdot \text{nVt} \tag{5.270}$$

$$\text{xi0d} = \frac{{\psi_{\text{d}}}^2}{2.0 + \psi_{\text{d}} \cdot \psi_{\text{d}}} \tag{5.271}$$

$$\text{Ed} = \text{e}^{-\psi_{\text{d}}} \tag{5.272}$$

$$\text{Dd} = \text{e}^{-\text{phidf}} \cdot \frac{1.0}{\text{Ed} - \psi_{\text{d}} - 1.0 - \text{xi0d}} \tag{5.273}$$

$$\text{xi0d} = \frac{{\psi_{\text{d}}}^2}{2.0 + {\psi_{\text{d}}}^2} \tag{5.274}$$

$$\text{Ed} = \text{e}^{-\psi_{\text{d}}} \tag{5.275}$$

$$\text{Dd} = \text{e}^{-\text{phidf}} \cdot \frac{1}{1.0/\text{Ed} - \psi_{\text{d}} - 1.0 - \text{xi0d}} \tag{5.276}$$

$$\text{Pd} = (\text{vgfb1} - \psi_{\text{d}})^2 \cdot \frac{1.0}{{\gamma_0}^2} - \text{Dd} \tag{5.277}$$

$$\text{sqd} = \sqrt{\text{Pd}} \tag{5.278}$$

$$\text{xgd} = \gamma \cdot \sqrt{\text{Pd} + \text{Dd}} \tag{5.279}$$

$$\text{qid} = \frac{\gamma^2 \cdot \text{Dd} \cdot \text{nVt}}{\text{xgd} + \gamma \cdot \text{sqd}} \tag{5.280}$$

$$\text{qbd} = \text{sqd} \cdot \text{gam} \cdot \text{nVt} \tag{5.281}$$

Average surface potential calculations:

$$\mathrm{x_m} = 0.5 \cdot (\psi_\mathrm{s} + \psi_\mathrm{d}) \tag{5.282}$$

$$\mathrm{tempc} = \mathrm{abs}(\mathrm{Ed} \cdot \mathrm{Es1}) \tag{5.283}$$

$$\mathrm{Em} = \sqrt{\mathrm{tempc}} \tag{5.284}$$

$$\mathrm{D_{bar}} = 0.5 \cdot (\mathrm{Ds} + \mathrm{Dd}) \tag{5.285}$$

$$\mathrm{Dm} = \mathrm{D_{bar}} + 0.125 \cdot \left(\mathrm{x_{ds}} \cdot \mathrm{x_{ds}} \cdot (\mathrm{Em} - 2.0 \cdot \frac{1}{\gamma^2})\right) \tag{5.286}$$

$$\mathrm{Pm} = (\mathrm{vgfb} - \mathrm{x_m})^2 \cdot \frac{1}{\gamma^2} - \mathrm{Dm} \tag{5.287}$$

$$\mathrm{xgm} = \gamma \cdot \sqrt{\mathrm{Dm} + \mathrm{Pm}} \tag{5.288}$$

$$\mathrm{sqm} = \sqrt{\mathrm{Pm}} \tag{5.289}$$

Poly depletion effect: In presence of poly depletion effect (PDEMOD=1) the updated average surface potential is as given below,

$$\mathrm{pdef} = \frac{2 \cdot \mathrm{Cox} \cdot \mathrm{Cox} \cdot \mathrm{nVt}}{(\mathrm{q} \cdot \epsilon_\mathrm{si} \cdot \mathrm{NGATE})} \tag{5.290}$$

$$\mathrm{d0} = 1.0 - \mathrm{Em} + 2.0 \cdot (\mathrm{xgm} \cdot \frac{1}{\gamma^2}) \tag{5.291}$$

$$\mathrm{eta_p} = \frac{1.0}{\sqrt{(1.0 + \mathrm{pdef} \cdot \mathrm{xgm})}} \tag{5.292}$$

$$\mathrm{tempc} = \frac{\mathrm{eta_p}}{\mathrm{eta_p} + 1.0} \tag{5.293}$$

$$\mathrm{x_{pm}} = \mathrm{pdef} \cdot (\mathrm{tempc} \cdot \mathrm{tempc} \cdot \mathrm{xgm} \cdot \mathrm{xgm}) \cdot \frac{\mathrm{Dm}}{\mathrm{Dm} + \mathrm{Pm}} \tag{5.294}$$

$$\mathrm{p_{pd}} = 2.0 \cdot (\mathrm{xgm} - \mathrm{x_{pm}}) + \gamma^2 \cdot (1.0 - \mathrm{Em} + \mathrm{Dm}) \tag{5.295}$$

$$\mathrm{q_{pd}} = \mathrm{x_{pm}} \cdot (\mathrm{x_{pm}} - 2.0 \cdot \mathrm{xgm}) \tag{5.296}$$

$$\mathrm{xi_{pd}} = 1.0 - 0.5 \cdot \left(\gamma^2 \cdot (\mathrm{Em} + \mathrm{Dm})\right) \tag{5.297}$$

$$\mathrm{u_{pd}} = \frac{\mathrm{q_{pd}} \cdot \mathrm{p_{pd}}}{(\mathrm{p_{pd}}^2 - \mathrm{xi_{pd}} \cdot \mathrm{q_{pd}})} \tag{5.298}$$

$$\mathrm{x_m} = \mathrm{x_m} + \mathrm{u_{pd}} \tag{5.299}$$

$$\mathrm{km} = \mathrm{e}^{\mathrm{u_{pd}}} \tag{5.300}$$

$$\mathrm{Em} = \frac{\mathrm{Em}}{\mathrm{km}} \tag{5.301}$$

$$\mathrm{Dm} = \mathrm{Dm} \cdot \mathrm{km} \tag{5.302}$$

$$\mathrm{Pm} = (\mathrm{vgfb} - \mathrm{x_m} + \mathrm{u_{pd}}) \cdot (1/\gamma^2) - \frac{\mathrm{Dm}}{\mathrm{km}} \tag{5.303}$$

$$\mathrm{xgm} = \mathrm{gam} \cdot \sqrt{\mathrm{Dm} + \mathrm{Pm}} \tag{5.304}$$

$$\mathrm{km0} = 1.0 - \mathrm{Em} + 2.0 \cdot \left(\mathrm{xgm} \cdot \mathrm{eta_p} \cdot (1/\gamma^2)\right) \tag{5.305}$$

$$\mathrm{x_{ds}} = \frac{\mathrm{x_{ds}} \cdot \mathrm{km} \cdot (\mathrm{d0} + \mathrm{D_{bar}})}{(\mathrm{km0} + \mathrm{km} \cdot \mathrm{D_{bar}})} \tag{5.306}$$

$$\mathrm{dps} = \mathrm{x_{ds}} \cdot \mathrm{nVt} \tag{5.307}$$

$$\mathrm{sqm} = \sqrt{\mathrm{Pm}} \tag{5.308}$$

Continuing the average potential calculation,

$$\alpha_{\mathrm{DD}} = \frac{(\mathrm{qis} - \mathrm{qid})}{\mathrm{dps}} \tag{5.309}$$

$$\mathrm{qim} = \mathrm{nVt} \cdot \left(\frac{\gamma^2 \cdot \mathrm{Dm}}{(\mathrm{xgm} + \gamma \cdot \mathrm{sqm})}\right) \tag{5.310}$$

$$\mathrm{qbm} = \mathrm{sqm} \cdot \gamma \cdot \mathrm{nVt} \tag{5.311}$$

$$\mathrm{Voxm} = \mathrm{xgm} \cdot \mathrm{nVt} \tag{5.312}$$

These calculations will be used in final drain current expression as given in eq. (5.478)

5.7.2 Calculations for SOIMOD=0
PDSOI mode - Partial Depletion Mode

Pinch-off Potential with Poly Depletion:

In accumulation and inversion under depletion approximation, the bulk charge is given as [10]

$$Q_b = -sign(\psi_s) \cdot \gamma' \cdot C_{ox} \cdot \sqrt{V_t.(e^{-\frac{\psi_s}{V_t}} - 1) + \psi_s} \tag{5.313}$$

From potential balance equation including poly depletion,

$$V_G = V_{FB} + \psi_S - \frac{Q_i + Q_b}{C_{ox}} + \left(\frac{Q_i + Q_b}{\gamma_g^{'}.C_{ox}}\right)^2 \tag{5.314}$$

At pinch off,$\psi_S = \psi_P$ and $Q_i = 0$. Substituting in (5.313) and (5.314),

$$V_G - V_{FB} = \psi_P + \gamma^{'} \cdot \sqrt{V_t.(e^{-\frac{\psi_P}{V_t}} - 1) + \psi_P} + (\frac{\gamma^{'}}{\gamma_g^{'}})^2 \left(V_t.(e^{-\frac{\psi_P}{V_t}} - 1) + \psi_P\right) \tag{5.315}$$

$$= \psi_P + \gamma^{'} \cdot \sqrt{V_t.(e^{-\frac{\psi_P}{V_t}} - 1) + \psi_P} + \delta_{PD}\left(V_t.(e^{-\frac{\psi_P}{V_t}} - 1) + \psi_P\right) \tag{5.316}$$

Normalizing it,

$$v_g - v_{fb} = \psi_p + \gamma_0 \cdot \sqrt{e^{-\psi_p} + \psi_p - 1} + \delta_{PD}\left(e^{-\psi_p} - 1 + \psi_p\right) \tag{5.317}$$

Explicit expression for ψ_p can be derived from above relation in the asymptotic form by inspecting the behavior in three different regions. First consider the depletion and inversion region of operation where $\psi_p >> 0$ so that $e^{-\psi_p}$ is very small. Let $\zeta_1 = e^{-\psi_p}$

$$v_g - v_{fb} = \psi_p + \gamma_0 \cdot \sqrt{\psi_p + \zeta_1 - 1} + \delta_{PD}\left(\zeta_1 - 1 + \psi_p\right) \tag{5.318}$$

Let

$$\sqrt{\psi_p + \zeta_1 - 1} = x \tag{5.319}$$

or

$$\psi_p = x^2 + 1 - \zeta_1 \tag{5.320}$$

Thus

$$v_g - v_{fb} = x^2 + 1 - \zeta_1 + \gamma_0.x + \delta_{PD}.x^2 \tag{5.321}$$

or

$$x^2 + \frac{\gamma_0}{1 + \delta_{PD}} \cdot x + \frac{1 - \zeta_1}{1 + \delta_{PD}} - \frac{v_g - v_{fb}}{1 + \delta_{PD}} = 0 \tag{5.322}$$

This gives

$$x = \left[\sqrt{\frac{v_g - v_{fb} - 1 + \zeta_1}{1+\delta_{PD}} + \left(\frac{\gamma_0}{2\cdot(1+\delta_{PD})}\right)^2} - \frac{\gamma_0}{2\cdot(1+\delta_{PD})}\right] \tag{5.323}$$

$$\psi_p = x^2 + 1 - \zeta_1 = \left[\sqrt{\frac{v_g - v_{fb} - 1 + \zeta_1}{1+\delta_{PD}} + \left(\frac{\gamma_0}{2\cdot(1+\delta_{PD})}\right)^2} - \frac{\gamma_0}{2\cdot(1+\delta_{PD})}\right]^2 + 1 - \zeta_1 \tag{5.324}$$

$$= \left[\sqrt{\frac{v_g - v_{fb} - 1 + \zeta_1}{1+\delta_{PD}} + \left(\frac{\gamma}{2}\right)^2} - \frac{\gamma}{2}\right]^2 + 1 - \zeta_1 \tag{5.325}$$

where $\gamma = \frac{\gamma_0}{1+\delta_{PD}}$
Similarly,
when ψ_p is close to 0

$$\psi_{p0} = \left[\frac{v_g - v_{fb}}{2} - 3(1+\frac{\gamma}{\sqrt{2}})\right] + \sqrt{\left[\frac{v_g - v_{fb}}{2} - 3(1+\frac{\gamma}{\sqrt{2}})\right]^2 + 6(v_g - v_{fb})} \tag{5.326}$$

and in accumulation where $\psi_p << 0$ ($\zeta_2 = \psi_p$),

$$\psi_p = -\ln\left[1 - \zeta_2 + \left(\frac{v_g - v_{fb} - \zeta_2}{\gamma}\right)^2\right] \tag{5.327}$$

Thus the pinch off potential is expressed as

$$\psi_p = \begin{cases} -\ln\left[1 - \psi_{p0} + \left(\frac{v_g - v_{fb} - \psi_{p0}}{\gamma}\right)^2\right] & \text{if } v_g - v_{fb} < 0 \\ 1 - e^{-\psi_{p0}} + \left[\sqrt{v_g - v_{fb} - 1 + e^{-\psi_{p0}} + \left(\frac{\gamma}{2}\right)^2} - \frac{\gamma}{2}\right]^2 & \text{otherwise} \end{cases} \tag{5.328}$$

Note : Derivatives of ψ_p are continuous in all regions.

5.7.3 Normalized Charge Density

Inversion Charge [11], [12] : Normalized inversion charge density at source/drain is derived similar to BSIM-SOI 107.1.0 [3] and can be obtained as follows.

Charge sheet model approximates inversion charge density as

$$Q_i = -\gamma'.C_{ox}.\sqrt{V_t}\left[\sqrt{\frac{\psi_S}{V_t} + e^{\frac{\psi_S - 2.\phi_F - V_{ch}}{V_t}}} - \sqrt{\frac{\psi_S}{V_t}}\right] \tag{5.329}$$

Using inversion charge linearization [12],

$$Q_i = n_q.C_{ox} \cdot (\psi_S - \psi_P) \tag{5.330}$$

or

$$\psi_S = \psi_P + \frac{Q_i}{n_q.C_{ox}} \tag{5.331}$$

Substituting ψ_S from (5.331) in (5.329),

$$-\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}} = \left[\sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t} + e^{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}} - 2.\phi_F - V_{ch}}{V_t}}} - \sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t}}\right] \tag{5.332}$$

rearranging,

$$\left[-\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}} + \sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t}}\right]^2 = \left[\sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t} + e^{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}} - 2.\phi_F - V_{ch}}{V_t}}}\right]^2 \tag{5.333}$$

$$e^{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}} - 2.\phi_F - V_{ch}}{V_t}} = \left(-\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}}\right)^2 - 2.\left(\frac{Q_i}{\gamma.C_{ox}.\sqrt{V_t}}\right) \cdot \sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t}} \tag{5.334}$$

This reduces to

$$\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}} - 2.\phi_F - V_{ch}}{V_t} = \ln\left[\left(-\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}}\right)^2 - 2.\left(\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}}\right) \cdot \sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t}}\right] \tag{5.335}$$

$$= \ln\left[-\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}}\left(-\frac{Q_i}{\gamma'.C_{ox}.\sqrt{V_t}} + 2 \cdot \sqrt{\frac{\psi_P + \frac{Q_i}{n_q.C_{ox}}}{V_t}}\right)\right] \tag{5.336}$$

Normalizing inversion charge to $-2V_t.n_q.C_{ox}$, all voltages to V_t ,

$$\psi_p - 2.q_i - 2.\phi_f - v_{ch} = \ln\left[\frac{2n_q.q_i}{\gamma_0}\left(\frac{2.n_q.q_i}{\gamma_0} + 2 \cdot \sqrt{\psi_p - 2q_i}\right)\right] \tag{5.337}$$

which gives

$$\ln{(q_i)} + \ln\left[\frac{2n_q}{\gamma_0}(q_i\frac{2n_q}{\gamma_0} + 2\sqrt{\psi_p - 2q_i})\right] + 2q_i = \psi_p - 2\phi_f - v_{ch} \tag{5.338}$$

This is a general equation which can be solved to give normalized inversion charge density. The procedure of obtaining initial guess for the solution of above equation for weak inversion is described below [13]. Note that to generalized the process, subscript "i" is dropped from the term q_i

Let $v = \psi_p - 2\phi_f - v_{ch} - \ln(\frac{4n_q\sqrt{\psi_p}}{\gamma}) = \ln q + 2q$

$$v = \ln q + 2q \tag{5.339}$$

$$= \ln q + 2e^{\ln q} \tag{5.340}$$

$$= \ln q + \frac{1}{F(\ln q)} \tag{5.341}$$

Here in second term q has been used as $\ln(e^q)$. The function F is defined as

$$F = \frac{1}{2e^{\ln q}} \tag{5.342}$$

$$= \frac{1}{2e^(\ln q + \ln q_t - \ln q_t)} \tag{5.343}$$

$$= \frac{1}{2q_t e^{\ln\frac{q}{q_t}}} \tag{5.344}$$

$$= \frac{1}{2q_t}e^{-\Delta} \tag{5.345}$$

Where $\Delta = \ln\frac{q}{q_t}$. Expanding (5.345) around $\Delta = 0$ using Taylor series expansion (as $|2q| << |\ln q|$),

$$F = \frac{1}{2q_t}.[1 - e^{-0}.\Delta]] \tag{5.346}$$

$$= \frac{1}{2q_t}(1 - \ln\frac{q}{q_t}) \tag{5.347}$$

substituting in (5.341),

$$v = \ln q + \frac{2q_t}{1 - \ln q + \ln q_t} \tag{5.348}$$

This equation is solved for q. Let,

$$\ln q = x \tag{5.349}$$

$$v = x + \frac{2q_t}{1 + \ln q_t - x} \tag{5.350}$$

$$v(1 + \ln q_t) - vx - x(1 + \ln q_t) + x^2 - 2q_t = 0 \tag{5.351}$$

$$x = \frac{v + (1 + \ln q_t) - \sqrt{(v + (1 + \ln q_t)^2 - 4v(1 + \ln q_t) + 8q_t}}{2} \tag{5.352}$$

For subthreshold region, normalized inversion charge density will be $|q| << 1$ and $|\ln q| >> |2q|$. The initial value is taken at a point where $|\ln q| = 2.|2q|$ which gives

$$q_t = 0.301 \tag{5.353}$$

$$1 + \ln q_t = -0.201491 \tag{5.354}$$

substituting in (5.352),

$$x = \frac{v - 0.201491 - \sqrt{(v - 0.201491)^2 - 4v(-0.201491) + 8(0.301)}}{2} \tag{5.355}$$

$$x = \ln q = \frac{v - 0.201491 - \sqrt{(v + 0.402982)v + 2.446562}}{2} \tag{5.356}$$

Once the initial guess is known, the final value is obtained by using analytical method as shown below

$$n_{q0} = 1 + \frac{\gamma}{2\sqrt{\psi_p}} \tag{5.357}$$

$$v = \psi_p - 2\phi - v_{ch} - ln\left(4.0 \cdot \frac{n_{q0}}{\gamma} \cdot \sqrt{\psi_p}\right) \tag{5.358}$$

$$ln_{q0} = \frac{1}{2}\left[v - 0.201491 - \sqrt{v \cdot (v + 0.402982) + 2.446562}\right] \tag{5.359}$$

$$q_0 = e^{ln_{q0}} \tag{5.360}$$

if $ln_{q0} <= -80.0$

$$q_{s/d} = f = q_0 \cdot \left[1 + \psi_p - 2\phi - v_{ch} - ln_{q0} - \ln\left(2 \cdot \frac{n_{q0}}{\gamma}\left(2 \cdot q_0 \cdot \frac{n_{q0}}{\gamma} + 2 \cdot \sqrt{\psi_p}\right)\right)\right] \tag{5.361}$$

In this equation, if $\ln q_0$ becomes very large and negative then $q_0 = e^{\ln q_0}$ may be out of range of precision limit of the simulator. Therefore it is approximated as follows
if $\ln q_0 < -110$, $q_0 = e^{-100}$
if $\ln q_0 > -90$, $q_0 = e^{\ln q_0}$
else $q_0 = exp(-100 + 20(\frac{5}{64} + \frac{z}{2} + z^2(\frac{15}{16} - z^2(1.25 - z^2))))$
where $z = \frac{\ln q_0 + 100}{20}$.
The above polynomial provides smooth derivatives for q. For the derivation of polynomial coefficients, refer to Appendix A.
For $\ln q_0 > -80$

$$f = 2q_0 + \ln(2q_0 \frac{n_q}{\gamma}(2q_0 \frac{n_q}{\gamma} + 2\sqrt{\psi_p}) - (v_p - 2\phi_f - v_{ch}) \tag{5.362}$$

$$f' = 2 + \frac{1}{q_0} + \frac{\frac{n_{q0}}{\gamma} - \frac{1}{\sqrt{\psi_p}}}{\frac{n_{q0}}{\gamma} \cdot q_0 + \sqrt{\psi_p}} \tag{5.363}$$

$$q_1 = q_0 - \frac{f}{f'} \tag{5.364}$$

The accuracy of this initial guess is further improved by following procedure

$$f = 2q_1 + \ln(2q_1 \frac{n_q}{\gamma}(2q_1 \frac{n_q}{\gamma} + 2\sqrt{\psi_p}) - (v_p - 2\phi_f - v_{ch}) \tag{5.365}$$

$$f' = 2 + \frac{1}{q_1} + \frac{\frac{n_{q1}}{\gamma} - \frac{1}{\sqrt{\psi_p}}}{\frac{n_{q1}}{\gamma} \cdot q_1 + \sqrt{\psi_p}} \tag{5.366}$$

Applying Halley's method,

$$f'' = -\frac{1}{q_1^2} - \frac{1}{\left[(\psi_p)^{\frac{3}{2}}\right] \cdot \left[\frac{n_{q0}}{\gamma} \cdot q_1 + \sqrt{\psi_p}\right]} - \left[\frac{\frac{n_{q0}}{\gamma} - \frac{1}{\sqrt{\psi_p}}}{\frac{n_{q0}}{\gamma} \cdot q_1 + \sqrt{\psi_p}}\right]^2 \tag{5.367}$$

$$q_{s/d} = q_1 - \frac{f}{f'} \cdot \left(1 + \frac{f \cdot f''}{2 \cdot f'^2}\right) \tag{5.368}$$

5.8 Short Channel Effects

Asymmetry Mode

There are some devices with asymmetric drain and source structures. This is modeled as follows:

Weighting Function for forward and reverse modes

$$T_0 = tanh(ASYMP * q * Vds_{noswap}/K \cdot T); \tag{5.369}$$
$$wf = 0.5 + 0.5 * T_0; \tag{5.370}$$
$$wr = 1.0 - wf; \tag{5.371}$$
$$if(ASYMMOD! = 0) \tag{5.372}$$
$$PARAM_A = PARAM_I * wr + PARAM_I * wf \tag{5.373}$$
$$else \tag{5.374}$$
$$PARAM_A = PARAM_I \tag{5.375}$$
$$\tag{5.376}$$

ASYMP is a model parameter and its default value is 0.6. This parameter improves flexibility in asymmetry mode.

Asymmetry mode affect following parameters: CDSCD, ETA0, PDIBLC, PCLM, PSAT, VSAT, PTWG, U0, UA, UC, UD, UCS

Vt Roll-off, DIBL, and Subthreshold Slope Degradation (Ref.: BSIM4 Model, BSIM-IMG Technical manual)

Scale Length λ:

$$\lambda_f = \sqrt{TSI \cdot \epsilon_{ratio} \cdot TOXE} \tag{5.377}$$
$$\lambda_s = \sqrt{TSI \cdot \left(\epsilon_{ratio} \cdot TOXE + \frac{3}{8} \cdot TSI\right)} \tag{5.378}$$
$$t_{eff} = TSI + \epsilon_{ratio} * (TOXE + TBOX) \tag{5.379}$$
$$T_0 = \frac{V_{gfb} \cdot TBOX \cdot \epsilon_{ratio} + V_{efbb} \cdot (TOXE \cdot \epsilon_{ratio} + TSI)}{t_{eff}} \tag{5.380}$$
$$x_\lambda = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}\left[ASCL + BSCL \cdot T_0\right]$$
$$\lambda = \lambda_s + x_\lambda(\lambda_f - \lambda_s) \tag{5.381}$$

$$\psi_{st} = 0.4 + PHIN + \frac{kT}{q} \cdot \ln \frac{NDEP}{n_i} \tag{5.382}$$

$$PhistVbs = \psi_{st} - V_{bsx} \tag{5.383}$$

$$X_{dep} = \sqrt{\frac{2 \cdot \epsilon_{sub} \cdot PhistVbs}{q \cdot NDEP}} \tag{5.384}$$

$$\tag{5.385}$$

if (SOIMOD=0)

$$n = 1 + \frac{CIT + NFACTOR + CDSCD \cdot V_{dsx} - CDSCB \cdot V_{bsx}}{C_{ox}} \tag{5.386}$$

$$\tag{5.387}$$

else

$$V_{esxpos} = 0.5 \cdot \left(V_{esx} + \sqrt{(V_{esx}^2 + 4 \cdot \delta_2^2)} \right) \tag{5.388}$$

$$\delta_1 = (CDSCD + CSECSED \cdot V_{esxpos} - CBCBD \cdot V_{bsx}) \cdot V_{dsx} \tag{5.389}$$

$$\theta_{SCE} = \frac{0.5}{cosh\left(DVT1 \cdot \frac{L_{eff}}{\lambda} \right) - 1} \tag{5.390}$$

$$\begin{aligned}
C_{dsc} =& CSECSE0 \cdot V_{esx} + CSECSE0P \cdot V_{esx}^2 + CBCB0 \cdot V_{bsx} + CBCB0P \cdot V_{bsx}^2 \\
&+ \theta_{SCE} \cdot (CDSC + CSECSE \cdot V_{esx} + CSECSEP \cdot V_{esx}^2 \\
&+ CBCB \cdot V_{bsx} + CBCBP \cdot V_{bsx}^2 + \delta_1) \\
&n = 1 + \frac{CIT + NFACTOR + C_{dsc}}{C_{ox} + C_B \parallel C_{BOX}}
\end{aligned} \tag{5.391}$$

end

$$V_t = \frac{k_b \cdot T}{q} \tag{5.392}$$

$$nV_t = n \cdot V_t \tag{5.393}$$

$$\Delta V_{th,VDNUD} = K1 \cdot (\sqrt{\phi_{st} - V_{bs}} - \sqrt{\phi_{st}}) - K2 \cdot V_{bsx} \tag{5.394}$$

$$\Delta V_{th,DIBL} = -(ETA0 + ETAB \cdot V_{bsx}) \cdot V_{dsx} \tag{5.395}$$

$$\Delta V_{th,DITS} = -n\frac{KT}{q} \cdot \ln\left(\frac{L_{eff}}{L_{eff} + DVTP0 \cdot (1 + \exp(-DVTP1 \cdot V_{ds})}\right) \tag{5.396}$$

$$- \left(DVTP5 + \frac{DVTP2}{L_{eff}^{DVTP3}}\right) \cdot tanh\left(DVTP4 \cdot V_{dsx}\right) \tag{5.397}$$

$$\Delta V_{th,all} = \Delta V_{th,VNUD} + \Delta V_{th,DIBL} + \Delta V_{th,DITS} \tag{5.398}$$

$$V_{gfb} = V_g - V_{fb} - \Delta V_{th,all} \tag{5.399}$$

Note: Short channel effect and Reverse short channel effect are modeled using NDEPL1, NDELEXP1, NDEPL2 and NDEPLEXP2 parameters. Width scaling of V_{th} is modeled using NDEPW and NDEPWEXP parameters.

5.9 Drain Saturation Voltage

The drain saturation voltage model is calculated after the source-side charge (q_s $/q_{is}$) has been calculated. V_{dseff} is subsequently used to compute the drain-side charge (q_d/q_{id}).

Electric Field Calculations

Electric Field is in MV/cm

$$\eta = \begin{cases} \frac{1}{2} \cdot ETAMOB & \text{for NMOS} \\ \frac{1}{3} \cdot ETAMOB & \text{for PMOS} \end{cases} \tag{5.400}$$

$$E_{effs} = 10^{-8} \cdot \left(\frac{q_{bs} + \eta \cdot q_{is}}{\epsilon_{ratio} \cdot TOXE}\right) \tag{5.401}$$

Drain Saturation Voltage (V_{dsat}) Calculations: SOIMOD=0 (Ref. BSIM4 & BSIM-BULK 107.1.0 Model)

$$D_{mobs} = 1 + (UA + UC \cdot V_{bsx}) \cdot (E_{effs})^{EU} + \frac{UD}{\left[\frac{1}{2} \cdot \left(1 + \frac{q_{is}}{q_{bs}}\right)\right]^{UCS}} \tag{5.402}$$

$$T_{11} = PSATB \cdot V_{bsx}$$
$$T_{12} = \sqrt{0.1 + T_{11} \cdot T_{11}}$$
$$T_0 = 0.5 * (1 - T_{11} + \sqrt{(1 - T_{11}) \cdot (1 - T_{11}) + T_{12}}) \tag{5.403}$$
$$\lambda_C = \frac{2 \cdot U0 \cdot nV_t}{(D_{mobs})^{PSAT} \cdot VSAT \cdot L_{eff}} \cdot [1 + PTWG \cdot \frac{10 \cdot PSATX \cdot qs \cdot T_0}{10 \cdot PSATX + qs \cdot T_0}] \tag{5.404}$$
$$q_{dsat} = \frac{\lambda_C}{2} \cdot \frac{q_s^2 + q_s}{1 + \frac{\lambda_C}{2} \cdot (1 + q_s)} \tag{5.405}$$
$$v_{dsat} = \psi_p - \frac{2\phi_b}{n} - 2q_{dsat} - \ln\left[\frac{2q_{dsat} \cdot n_q}{gam} \cdot \left(\frac{2q_{dsat} \cdot n_q}{gam} + \frac{gam}{n_q - 1}\right)\right] \tag{5.406}$$

Drain Saturation Voltage (V_{dsat}) Calculations: SOIMOD=1 (ref. BSIM4 & BSIM-CMG)

$$D_{mobs} = 1 + (UA + UC \cdot V_{bsx}) \cdot (E_{effm})^{EU} + \frac{UD}{\left[\frac{1}{2} \cdot \left(1 + \frac{q_{is}}{q_{bs}}\right)\right]^{UCS}} \tag{5.407}$$

$$T_0 = 1 + PRWG \cdot q_{is}$$
$$T_1 = PRWB \cdot (\sqrt{\phi_{st} - V_{bs}} - \sqrt{\phi_{st}}) \tag{5.408}$$
$$T_2 = \frac{1}{T_0} + T_1 \tag{5.409}$$
$$T_3 = \frac{1}{2}\left[T_2 + \sqrt{T_2^2 + 0.01}\right] \tag{5.410}$$
$$rdstemp = TRatio^{PRT_i}; \tag{5.411}$$

$$\text{If } RDSMOD = 0 \text{ then} \tag{5.412}$$
$$R_{ds,s} = \frac{1}{(W_{eff0}(\mu m))^{WR_i}} \cdot (RDSWMIN(T) + RDSW(T) \cdot T3) \cdot NF \cdot rdstemp$$
$$\text{else if } RDSMOD = 1 \text{ then}$$
$$R_{ds,s} = 0$$
$$\text{else if } RDSMOD = 2 \text{ then}$$
$$R_{ds,s} = \left(RS_{geo} + RD_{geo} + \frac{NF}{(W_{eff0}(\mu m))^{WR_i}} \cdot (RDSWMIN(T) + RDSW(T) \cdot T3)\right) \cdot rdstemp \tag{5.413}$$

$$E_{sat} = \frac{2 \cdot VSAT(T)}{\mu_0(T)/D_{mobs}} \tag{5.414}$$

$$E_{satL} = E_{sat} \cdot L_{eff} \tag{5.415}$$

Here, RS_{geo} and RD_{geo} are geometry dependent (bias independent) part of source and drain resistances. In RDSMOD=2 they are included in $R_{ds,s}$ calculation

If $R_{ds,s} = 0$ then

$$V_{dsat} = \frac{E_{satL} \cdot KSATIV_i \cdot (q_{is} + 2\frac{kT}{q})}{E_{satL} + KSATIV_i \cdot (q_{is} + 2\frac{kT}{q})} \tag{5.416}$$

else

$$WVC_{ox} = W_{eff0} \cdot VSAT(T) \cdot C_{ox} \tag{5.417}$$

$$T_a = 2 \cdot WVC_{ox} \cdot R_{ds,s} \tag{5.418}$$

$$T_b = KSATIV_i \cdot (q_{is} + 2\frac{kT}{q}) \cdot (1 + 3 \cdot WVC_{ox} \cdot R_{ds,s}) + E_{satL} \tag{5.419}$$

$$T_c = KSATIV_i \cdot (q_{is} + 2\frac{kT}{q}) \tag{5.420}$$

$$\times \left(E_{satL} + T_a \cdot KSATIV_i \cdot (q_{is} + 2\frac{kT}{q})\right) \tag{5.421}$$

$$V_{dsat} = \frac{\left(T_b - \sqrt{T_b^2 - 2T_aT_c}\right)}{T_a} \tag{5.422}$$

5.10 Mobility degradation with vertical field: Same for both the SOIMOD

(Ref. BSIM4 Model)

$$E_{effm} = 10^{-8} \cdot \left(\frac{q_{ba} + \eta \cdot q_{ia}}{\epsilon_{ratio} \cdot TOXE}\right) \tag{5.423}$$

Where q_{ia}/q_{im} and q_{ba}/q_{bm} are the average inversion charge and bulk charge densities respectively.

$$D_{mob} = 1 + (UA + UC \cdot V_{bsx}) \cdot (E_{effm})^{EU} + \frac{UD}{\left[\frac{1}{2} \cdot \left(1 + \frac{q_{ia}}{q_{ba}}\right)\right]^{UCS}} \tag{5.424}$$

The D_{mob} goes into denominator of mobility expression.

5.11 Output Conductance

The Output conductance model is taken from BSIM4 [14]

Channel Length Modulation (CLM)

$$E_{sat} = \frac{2 \cdot VSAT}{\frac{U0}{D_{mob}}} \tag{5.425}$$

$$F = \begin{cases} 1 & \text{for } FPROUT \leq 0 \\ \frac{1}{1 + \frac{FPROUT \cdot \sqrt{L_{eff}}}{qia + 2 \cdot nV_t}} & \text{for } FPROUT > 0 \end{cases} \tag{5.426}$$

$$C_{clm} = \begin{cases} PCLM \cdot \left(1 + PCLMG \cdot \frac{q_{ia}}{E_{sat} \cdot L_{eff}}\right) \frac{1}{F} & \text{for } PCLMG > 0 \\ \frac{PCLM}{\cdot \left(1 - PCLMG \cdot \frac{q_{ia}}{E_{sat} \cdot L_{eff}}\right)} \frac{1}{F} & \text{for } PCLMG < 0 \end{cases} \tag{5.427}$$

$$V_{asat} = V_{dssat} + E_{satL} \tag{5.428}$$

$$M_{CLM} = 1 + C_{clm} \ln\left[1 + \frac{V_{ds} - V_{dseff}}{V_{asat}} \cdot \frac{1}{C_{clm}}\right] \tag{5.429}$$

Note: V_{dsat} is used in place of V_{dssat} in SOIMOD =1.

Drain Induced Barrier Lowering (DIBL)

$$PVAGfactor = \begin{cases} 1 + PVAG \cdot \frac{q_{im}}{E_{sat} L_{eff}} & \text{for } PVAG > 0 \\ \frac{1}{1 - PVAG \cdot \frac{q_{im}}{E_{sat} L_{eff}}} & \text{for } PVAG < 0 \end{cases} \tag{5.430}$$

$$\theta_{rout} = PDIBLC \tag{5.431}$$

$$V_{ADIBL} = \frac{q_{ia} + 2kT/q}{\theta_{rout}} \cdot \left(1 - \frac{V_{dssat}}{V_{dssat} + q_{ia} + 2kT/q}\right) \cdot PVAGfactor \cdot \frac{1}{1 + PDIBLCB \cdot V_{bsx}} \tag{5.432}$$

$$M_{DIBL} = \left(1 + \frac{V_{ds} - V_{dseff}}{V_{ADIBL}}\right) \tag{5.433}$$

Note: 1. Length scaling parameters for PDIBLC are PDIBLCL and PDIBLCLEXP.

2. V_{dsat} is used in place of V_{dssat} in SOIMOD =1.

Drain Induced Threshold Shift (DITS)

$$V_{ADITS} = \frac{1}{PDITS} \cdot F \cdot [1 + (1 + PDITSL \cdot L_{eff}) \exp(PDITSD \cdot V_{ds})] \tag{5.434}$$

$$M_{DITS} = \left(1 + \frac{V_{ds} - V_{dseff}}{V_{ADITS}}\right) \tag{5.435}$$

Substrate Current induced Body Effect (SCBE)

$$litl = \sqrt{(\epsilon_{sub}/\epsilon_{ox}) \cdot TOXE \cdot XJ} \tag{5.436}$$

$$V_{ASCBE} = \frac{L_{eff}}{PSCBE2} \cdot \exp\left(\frac{PSCBE1 \cdot litl}{V_{ds} - V_{dseff}}\right) \tag{5.437}$$

$$M_{SCBE} = \left(1 + \frac{V_{ds} - V_{dseff}}{V_{ASCBE}}\right) \tag{5.438}$$

$$M_{oc} = M_{DIBL} \cdot M_{CLM} \cdot M_{DITS} \cdot M_{SCBE} \tag{5.439}$$

M_{oc} is multiplied to I_{ds} in the final drain current expression.

5.12 Velocity Saturation

Current Degradation Due to Velocity Saturation: SOIMOD=1 [15]

$$\zeta_{sb} = \begin{cases} 1.0 + PSATB_i \cdot Vbsx; & \text{for } PSATB_i > 0 \\ \frac{1}{1.0 - PSATB_i \cdot Vbsx} & \text{for } PSATB_i < 0 \end{cases} \tag{5.440}$$

$$T0 = qim \cdot \zeta_{sb} \tag{5.441}$$

$$wsat = \frac{100 \cdot T0}{100 + T0} \tag{5.442}$$

$$\Delta L = ALP \cdot ln\left(\frac{1 + \frac{V_{ds} - \Delta\Psi}{VP}}{1 + \frac{V_{dseff} - \Delta\Psi}{VP}}\right) \tag{5.443}$$

$$\frac{\Delta L}{L} = \frac{1}{1 + \frac{\Delta L}{L} + \left(\frac{\Delta L}{L}\right)^2} \tag{5.444}$$

$$DmobL = Dmob \cdot \frac{\Delta L}{L} \tag{5.445}$$

$$T_1 = \begin{cases} 1.0 + PSAT_i \cdot wsat; & \text{for } PSAT_i > 0 \\ \frac{1}{1.0 - PSAT_i \cdot wsat} & \text{for } PSAT_i < 0 \end{cases} \tag{5.446}$$

$$thesat1 = THESAT_i \cdot \frac{T1}{DmobL} \tag{5.447}$$

$$zsat = \begin{cases} (thesat1 \cdot \Delta\Psi)^2; & NMOS \\ \frac{zsat}{(1 + thesat1 \cdot \Delta\Psi)}; & PMOS \\ \end{cases} \tag{5.448}$$

$$Dvsat = \frac{DmobL}{2} \cdot (1.0 + \sqrt{(1.0 + 2.0 \cdot zsat)}; \tag{5.449}$$

$$D_{tot} = D_{mob} \cdot D_{vsat} \cdot D_r \tag{5.450}$$

where D_r is the effect of internal resistance (R_{dsi}) on current, defined as

$$D_r = \begin{cases} 1 & \text{if } RDSMOD = 1 \\ 1 + U0 \cdot C_{ox} \cdot \frac{W_{eff}}{L_{eff}} \cdot q_{ia} \cdot R_{dsi} & \text{if } RDSMOD = 0 \end{cases} \tag{5.451}$$

Current Degradation Due to Velocity Saturation: SOIMOD=0

$$T_1 = 2 \cdot \lambda_C \cdot (q_s - q_{deff}) \tag{5.452}$$

$$\lambda_C = \frac{2 \cdot U0 \cdot nV_t}{(D_{mobs})^{\frac{1}{PSAT}} \cdot VSAT \cdot L_{eff}} \cdot [1 + PTWG \cdot \frac{10 \cdot PSATX \cdot qs \cdot T_0}{10 \cdot PSATX + qs \cdot T_0}] \tag{5.453}$$

$$D_{vsat} = \frac{1}{2}\left[\sqrt{1+T_1^2} + \frac{1}{T_1} \cdot \ln(T_1 + \sqrt{1+T_1^2})\right] \tag{5.454}$$

$$D_{ptwg} = D_{vsat} \tag{5.455}$$

$$D_{tot} = D_{mob} \cdot D_{vsat} \cdot D_r \tag{5.456}$$

$$\tag{5.457}$$

where D_r is the effect of internal resistance (R_{dsi}) on current, defined as

$$D_r = \begin{cases} 1 & \text{if } RDSMOD = 1 \\ 1 + U0 \cdot C_{ox} \cdot \frac{W_{eff}}{L_{eff}} \cdot q_{ia} \cdot R_{dsi} & \text{if } RDSMOD = 0 \end{cases} \tag{5.458}$$

Non-Saturation Effect Some devices do not exhibit prominent or abrupt velocity saturation. The parameters A1 and A2 are used to tune this non-saturation effect to better the $I_{d,sat}$ or $g_{m,sat}$ fitting.

$$T_0 = A_1 + A_2/(q_{ia} + 2.0 * n * V_{tm}) \tag{5.459}$$

$$dqi = q_s - q_{deff} \tag{5.460}$$

$$T_1 = T_0 * dqi * dqi \tag{5.461}$$

$$T_2 = T_1 + 1.0 - 0.001 \tag{5.462}$$

$$T_3 = -1.0 + 0.5 * (T_2 + \sqrt{T_2 * T_2 + 0.004}) \tag{5.463}$$

$$N_{sat} = 0.5 * (1.0 + \sqrt{1.0 + T_3}) \tag{5.464}$$

$$N_{sat} = 0.5 * [(N_{sat} + 1.0) - \sqrt{(N_{sat} - 1.0)^2 + 0.25 * 0.01^2}] + 0.25 * 0.01 \tag{5.465}$$

$$I_{ds} = I_{ds}/N_{sat} \tag{5.466}$$

5.13 Effective Mobility

$$\mu_{eff} = \frac{U0}{D_{tot}} \tag{5.467}$$

5.14 Drain Current Model

5.14.1 Without Velocity Saturation: SOIMOD=1

The drain current expression is derived as follows,

$$I_{ds} = I_{drift} + I_{diff} \tag{5.468}$$

$$I_{ds} = -W_{eff}.Q_i \cdot \mu_{eff} \cdot C_{ox}\frac{d\psi_s}{dx} + W \cdot \mu_{eff} \cdot V_t\frac{dQ_i}{dx} \tag{5.469}$$

x is the position along the channel length. After integration,

$$I_{ds} = -\frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot \mu_{eff}\left(\int_{\psi_s}^{\psi_d} Q_i \cdot d\psi + \cdot V_t \cdot Q_s - Q_d\right) \tag{5.470}$$

After symmetric charge lineariation [9],

$$Q_i = Q_{im} - \alpha_{DD} \cdot u_f \tag{5.471}$$

where,

$$u_f = \psi_s - \psi_m \tag{5.472}$$

$$\psi_m = \frac{\psi_s + \psi_d}{2} \tag{5.473}$$

$$\alpha_{DD} = \frac{Q_s - Q_d}{\psi_d - \psi_s} \tag{5.474}$$

Substituting eq. (5.472), (5.473) and (5.474) in eq. (5.470), and performing integration, the drain current expression is given by,

$$I_{ds} = -\frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot \mu_{eff}\left(Q_{im} + V_t \cdot \alpha_{DD}\right) \cdot (\psi_d - \psi_s) \tag{5.475}$$

5.14.2 Including Velocity Saturation: SOIMOD=1

As the device is getting smaller and smaller, the lateral electric field strength and therefore kinetic energy of the carriers increases. On reaching optical phonon energy levels, they releases optical phonon by virtue of reduction in kinetic energy and therefore loses velocity [7]. The effect of velocity saturation on mobility is derived in similar manner as given in [15],

$$I_{ds} = \quad -\left(W_{eff}.Q_i \cdot \mu_{eff} \cdot C_{ox}\frac{d\psi_s}{dx} + W \cdot \mu_{eff} \cdot V_t\frac{dQ_i}{dx}\right) \cdot \frac{1}{\sqrt{1 + (\frac{\mu_{eff}}{V_{sat}} \cdot \frac{d\psi_s}{dx})^2}} \tag{5.476}$$

Following [15] and symmetric charge linearization it is possible to arrive at the following velocity saturation factor,

$$D_{vsat} = \frac{Dmob \cdot D_{\Delta L}}{2} \cdot \left(1 + \sqrt{(1 + 2 \cdot Zsat)}\right) \tag{5.477}$$

Zsat is given in eq. (5.448) Drain current including velocity saturation is given by,

$$I_{ds} = -\frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot \mu_{eff}\left(Q_{im} + V_t \cdot \alpha_{DD}\right) \cdot \frac{(\psi_d - \psi_s)}{D_{vsat}} \tag{5.478}$$

5.14.3 Without Velocity Saturation: SOIMOD=0

The drain current expression is derived as follows,

$$I_{ds} = I_{drift} + I_{diff} \tag{5.479}$$

$$I_{ds} = -W_{eff}.Q_i \cdot \mu_{eff}\frac{d\psi_s}{dx} + W \cdot \mu_{eff} \cdot V_t\frac{dQ_i}{dx} \tag{5.480}$$

from charge linearization, $\psi_s = \psi_p + \frac{Q_i}{n_q.C_{ox}}$. Thus

$$I_{ds} = \mu_{eff}.W_{eff} \cdot \left[-\frac{Q_i}{n_q.C_{ox}} + V_t\right]\frac{dQ_i}{dx} \tag{5.481}$$

normalizing inversion charge to $-2n_qC_{ox}V_t$ and using $\xi = \frac{x}{L}$,

$$I_{ds} = \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot \left[-\frac{(-2.n_q.C_{ox}.V_t.q)}{n_q.C_{ox}} + V_t\right]\frac{d(-2.n_q.C_{ox}.V_t.q)}{d\xi} \tag{5.482}$$

$$= -2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot (2q+1)\frac{dq}{d\xi} \tag{5.483}$$

Total drain current,

$$I_{DS} = \int_0^1 I_{ds}\ d\xi = -2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot \int_{q_s}^{q_d} (2q+1)dq \tag{5.484}$$

which gives

$$I_{DS} = 2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot [(q_s - q_{deff})(q_s + q_{deff} + 1)] \tag{5.485}$$

n_q is the slope factor in charge based model and nV_t is $n.\frac{KT}{q}$ with n given by (5.386).

$$I_{DS} = 2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot [(q_s - q_{deff})(q_s + q_{deff} + 1)] \tag{5.486}$$

5.14.4 Including Velocity Saturation: SOIMOD=0

As the device is getting smaller and smaller, the lateral electric field strength and therefore kinetic energy of the carriers increases. On reaching optical phonon energy levels, they releases optical phonon by virtue of reduction in kinetic energy and therefore loses velocity [16]. The effect of velocity saturation on mobility is captured as follows

$$\mu = \frac{\mu_{eff}}{\sqrt{1 + (\frac{E}{E_c})^2}} \tag{5.487}$$

$$= \frac{\mu_{eff}}{\sqrt{1 + (\frac{1}{E_c} \cdot \frac{d\psi_s}{dx})^2}} \tag{5.488}$$

from (5.483) and (5.488),

$$I_{ds} = -2 \cdot n_q \cdot \frac{\mu_{eff}}{\sqrt{1 + (\frac{1}{E_c} \cdot \frac{d\psi_s}{dx})^2}} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot (2q+1)\frac{dq}{d\xi} \tag{5.489}$$

$$= z \cdot \frac{(2q+1)\frac{dq}{d\xi}}{\sqrt{1 + (\frac{1}{E_c} \cdot \frac{d\psi_s}{dx})^2}} \tag{5.490}$$

with $z = -2\mu_{eff} \cdot n_q \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2$

Total current,

$$I_{DS} = \int_0^1 I_{ds}\ d\xi = z \cdot \int_{q_s}^{q_d} \frac{(2q+1)}{\sqrt{1 + (\frac{1}{E_c} \cdot \frac{d\psi_s}{dx})^2}} dq \tag{5.491}$$

$$I_{DS}\int_0^1 \sqrt{1+\left(\frac{1}{E_c}\cdot\frac{d\psi_s}{dx}\right)^2}d\xi = z\cdot\int_{q_s}^{q_d}(2q+1)dq \tag{5.492}$$

from (5.518),

$$\int_{q_s}^{q_d}(2q+1)dq = -(q_s - q_{deff})(q_s + q_{deff}+1) \tag{5.493}$$

Now consider the LHS of (5.492). Using charge linearization, $\psi_s = \psi_p + \frac{Q_i}{n_q.C_{ox}}$,

$$\frac{1}{E_c}\frac{d\psi_s}{dx} = \frac{1}{E_c.n_q.C_{ox}}\frac{Q_i}{dx} = -\frac{2V_t}{E_c.L}\frac{dq}{d\xi} = -\lambda_c\cdot\frac{dq}{d\xi} \tag{5.494}$$

Let

$$D_{vsat} = \int \sqrt{1+\left(\frac{1}{E_c}\cdot\frac{d\psi_s}{dx}\right)^2}d\xi$$

It is evaluated by assuming that lateral electric field ($-\frac{d\psi_s}{d\xi}$) increases linearly from 0 at source to $2\cdot\left(\frac{\psi_{s,D}-\psi_{s,S}}{L}\right)$ at drain [17] i.e.

$$-\frac{d\psi_s}{dx} = 2\cdot\frac{\psi_{s,D}-\psi_{s,S}}{L}\cdot\frac{x}{L} = 2\cdot\frac{\psi_{s,D}-\psi_{s,S}}{L}\cdot\xi \tag{5.495}$$

From charge linearization (5.331),

$$\psi_{s,S} = \psi_P + \frac{Q_S}{n_q.C_{ox}} = \psi_P - 2V_t.q_s \tag{5.496}$$

$$\psi_{s,D} = \psi_P + \frac{Q_D}{n_q.C_{ox}} = \psi_P - 2V_t.q_d \tag{5.497}$$

$$\psi_{s,D} - \psi_{s,S} = 2.V_t(q_s - q_d) \tag{5.498}$$

substituting in (5.495),

$$-\frac{d\psi_s}{dx} = 2\cdot\frac{2.V_t(q_s-q_d)}{L^2}\cdot x = 2\cdot\frac{2.V_t(q_s-q_d)}{L}\cdot\xi \tag{5.499}$$

$$-\frac{1}{E_c}\frac{d\psi_s}{dx} = 2\cdot\frac{2.V_t}{E_cL}\cdot(q_s-q_d)\xi = 2\lambda_c(q_s-q_d)\xi \tag{5.500}$$

where $\lambda_c = \frac{2V_t}{E_c.L}$. Thus D_{vsat} can be given as

$$D_{vsat} = \int \sqrt{1 + \left(\frac{1}{E_c} \cdot \frac{d\psi_s}{dx}\right)^2} d\xi \tag{5.501}$$

$$= \int \sqrt{1 + (2\lambda_c(q_s - q_d)\xi)^2} d\xi = \int \sqrt{1 + (2\lambda_c.\Delta q \cdot \xi)^2} d\xi \tag{5.502}$$

$$= \frac{1}{2}\left[\sqrt{1 + (2.\lambda_c.\Delta q)^2} + \frac{1}{2.\lambda_c.\Delta q}.\ln\left(2.\lambda_c.\Delta q + \sqrt{1 + (2.\lambda_c.\Delta q)^2}\right)\right] \tag{5.503}$$

with $\Delta q = q_s - q_d$. From (5.492), (5.493) and (5.503),

$$I_{DS} = 2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot [(q_s - q_{deff})(q_s + q_{deff} + 1)].M_{oc} \tag{5.504}$$

where $\mu_{eff} = \frac{U0}{D_{tot}}$ and $D_{tot} = D_{mod}.D_{vsat}.D_r$

5.15 Threshold Voltage Model

5.15.1 Long Channel Threshold Voltage

The drain current under the standard drift-diffusion formalism can be represented as

$$I_{ds} = I_{drift} + I_{diff} \tag{5.505}$$

$$I_{ds} = - W.Q_i \cdot \mu\frac{d\psi_s}{dx} + W \cdot \mu \cdot V_t\frac{dQ_i}{dx} \tag{5.506}$$

Defining threshold voltage as the gate voltage at which $I_{drift} = I_{diff}$, which leads to $q_i = \frac{1}{2}$. Here, we define threshold voltage as the gate voltage at which normalized inversion charge density at the source is $q_s = \frac{1}{2}$. Pinch-off potential corresponding to $q_s = \frac{1}{2}$, $\psi_{p,th}$, is calculated using $q_s = \frac{1}{2}$ in (5.338) leading to

$$\psi_{p,th} = \left[\ln(0.5) + 1\right] + \ln\left[\frac{2n_q}{\gamma_0}(\frac{n_q}{\gamma_0} + 2\sqrt{\psi_{p,th} - 1})\right] + 2\phi_f + v_s \tag{5.507}$$

To obtain $\psi_{p,th}$ in explicit form, we make an assumption that at threshold voltage,

$$\psi_{p,th} - 1 = 2\phi_f + v_s \tag{5.508}$$

The other bias dependent term n_q is also approximated by

$$n_q = 1 + \frac{\gamma}{2\sqrt{2\phi_f}} \tag{5.509}$$

This gives,

$$\psi_{p,th} = \Bigg[\ln(0.5) + 1\Bigg] + \ln\Bigg[\frac{2n_q}{\gamma_0}(\frac{n_q}{\gamma_0} + 2\sqrt{2\phi_f - v_b})\Bigg] + 2\phi_f + v_s \tag{5.510}$$

After $\psi_{p,th}$ is obtained, next step is to calculate threshold voltage.The potential balance equation in conjunction with Poisson's equation and Gauss law for the MOSFET is given as ()

$$V_G = V_{FB} + \psi_S - \frac{Q_{in} + Q_{dep}}{C_{ox}} \tag{5.511}$$

At pinch-off, $\psi_S = \psi_P$, and $Q_{in} = 0$

$$V_G = V_{FB} + \psi_P + \gamma\sqrt{\psi_P} \tag{5.512}$$

Thus [18]

$$V_{TH,long} = V_{FB} + \psi_{p,th}.V_t - \gamma\sqrt{\psi_{p,th}.V_t} \tag{5.513}$$

Short Channel Threshold Voltage

$$V_{TH} = V_{TH,long} - \Delta V_{th,all} \tag{5.514}$$

5.16 MNUD model to increase $I_{DS} - V_{DS}$ Fitting Flexibility

$$MNUD = 1 + K_0\left[\frac{(q_s - q_{deff})}{(M_0 + q_s + q_{deff})}\right]^2 \tag{5.515}$$

$$I_{DS} = 2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot [(q_s - q_{deff})(q_s + q_{deff} + 1)].M_{oc}/MNUD \tag{5.516}$$

5.17 MNUD1 model to increase $g_m - I_D$ Fitting Flexibility

$$MNUD1 = exp\left[-\frac{C0}{[(C0SI + C0SISAT \cdot (q_s - q_{deff})^2)(q_s + q_{deff}) + 2 \cdot n \cdot V_t]}\right] \tag{5.517}$$

$$I_{DS} = 2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot [(q_s - q_{deff})(q_s + q_{deff} + 1)].M_{oc}/MNUD \cdot MNUD1 \tag{5.518}$$

5.18 AbulkIV model to increase $I_{DS} - V_{DS}$ Fitting Flexibility

$$T1 = \frac{Leff}{Leff + sqrt(XJ_i * Xdep)} \tag{5.519}$$

$$AbulkIV = 1 + \frac{A0 * T1 - AGS * q_s^{AGS1} * V_T * T1}{1 + KETA * V_{bsx}} \tag{5.520}$$

$$Vdssat = Vdssat/AbulkIV \tag{5.521}$$

$$T7 = pow(Vds/Vdssat, 1/Delta\_t) \tag{5.522}$$

$$T8 = pow(1 + T7, -Delta\_t) \tag{5.523}$$

$$Vdseff = Vds * T8 \tag{5.524}$$

$$Vdeff = (Vdseff + Vs) * inv\_Vt \tag{5.525}$$

Calculate qdeff using new Vdeff

5.19 Subthreshold Hump Module

The Edge FET Model current is added to the main current I_{ds}, when the parameter EDGEFET is 1, for SOIMOD=0 and 1 both [19].

$$
\begin{aligned}
I_{ds} = -NF \cdot \frac{WEDGE}{L_{eff}} \cdot C_{ox} \cdot \mu_{eff} \left(Q_{im} + V_t \cdot \alpha_{DD} \right) \cdot \\
(\psi_d - \psi_s) \cdot \frac{DdL \cdot D_{vsat}}{D_r} \cdot M_{oc} \qquad \text{SOIMOD=1}
\end{aligned}
\tag{5.526}
$$

$$
\begin{aligned}
Ids, EDGE = 2 \cdot NF \cdot nq \cdot \mu_{eff} \cdot WEDGE/L_{eff} \cdot C_{ox} \cdot \\
nV_t \cdot ((q_s - q_{deff}) \cdot (1 + q_s + q_{deff})) \cdot M_{oc} \qquad \text{SOIMOD=0}
\end{aligned}
\tag{5.527}
$$

Now the total current will be $I_{Total} = Ids_{EDGE} + I_{ds}$. The parameters associated with subthreshold hump model are: WEDGE, DGAMMA, DGAMMAL, DGAMMALEXP, DVTEDGE, NFACTOREDGE, CITEDGE, CDSCDEDGE, CDSCBEDGE, ETA0EDGE, ETABEDGE, KT1EDGE, KT1LEDGE, KT2EDGE, KT1EXPEDGE, TNFACTOREDGE, TETA0EDGE, DVT0EDGE, DVT1EDGE, DVT2EDGE.

5.20 Modeling of Sub-Surface Leakage Drain Current

The flag for the sub-surface leakage model is SSLMOD [20].

$$
T1 = \left(\frac{NDEP}{1e23} \right)^{SSLEXP1} \qquad T2 = \left(\frac{300}{T} \right)^{SSLEXP2} \tag{5.528}
$$

$$
T3 = \frac{devsign.SSL5.V_{bs}}{Vt} \tag{5.529}
$$

$$
SSL0\_NT = SSL0.exp(-T1T2) \qquad SSL0\_NT = SSL1.T2.T1 \tag{5.530}
$$

$$
T5 = SSL3.tanh\left(exp(devsign.SSL4.(V_{gb} - VTH - V_{sb})) \right) \tag{5.531}
$$

$$
\begin{aligned}
I_{ssl} = sigvds.NF.W_{eff}.SSL0\_NT.exp(T3). \\
exp(-SSL1\_NT.L + \frac{T5}{Vt}) \left[exp(SSL2.\frac{V_{dsx}}{Vt}) - 1 \right]
\end{aligned}
\tag{5.532}
$$

5.21 Gate resistance

5.21.1 Gate Electrode Electrode and Intrinsic-Input Resistance (IIR) Model

General Description: BSIM-SOI 100.x.x provides four options for modeling gate electrode resistance (bias-independent) and intrinsic-input resistance (IIR, bias-dependent). The IIR model considers the relaxation-time effect due to the distributive RC nature of the channel region, and therefore describes the first-order non-quasi-static effect. Thus, the IIR model should not be used together with the charge-deficit NQS model at the same time. The model selector RGATEMOD is used to choose different options.

Model Option and Schematic: There are four model selectors for gate resistance network.

RGATEMOD = 0 (zero-resistance): In this case, no gate resistance is generated (see Figure 6).

RGATEMOD = 1 (constant-resistance): In this case, only the electrode gate resistance (bias-independent) is generated by adding an internal gate node. *Rgeltd* is given by

$$Rgeltd = \frac{RSHG \cdot (XGW + \frac{W_{effci}}{3 \cdot NGCON})}{NGCON \cdot (L_{drawn} - XGL) \cdot NF} \tag{5.533}$$

RGATEMOD = 2 (IIR model with variable resistance): In this case, the gate resistance is the sum of the electrode gate resistance *Rgeltd* (5.533) and the intrinsic-input resistance R_{ii} as given by (5.534).

$$\frac{1}{R_{ii}} = XRCRG1.NF \cdot \left(\frac{I_{ds}}{V_{dseff}} + XRCRG2 \cdot \frac{W_{eff}\mu_{eff}C_{oxeff}V_t}{L_{eff}} \right)$$

or

$$\frac{1}{R_{ii}} \approx XRCRG1.NF \cdot \left(\mu_{eff} \left(\frac{W_{eff}}{L_{eff}} \right) C_{ox} \cdot q_{ia} + XRCRG2 \cdot \frac{W_{eff}\mu_{eff}C_{oxeff}V_t}{L_{eff}} \right) \tag{5.534}$$

RGATEMOD = 3 (IIR model with two nodes): In this case, the gate electrode resistance *Rgeltd* is in series with the intrinsic-input resistance R_{ii} through two internal gate nodes, so that the overlap capacitance current will not pass through the intrinsic-input resistance.

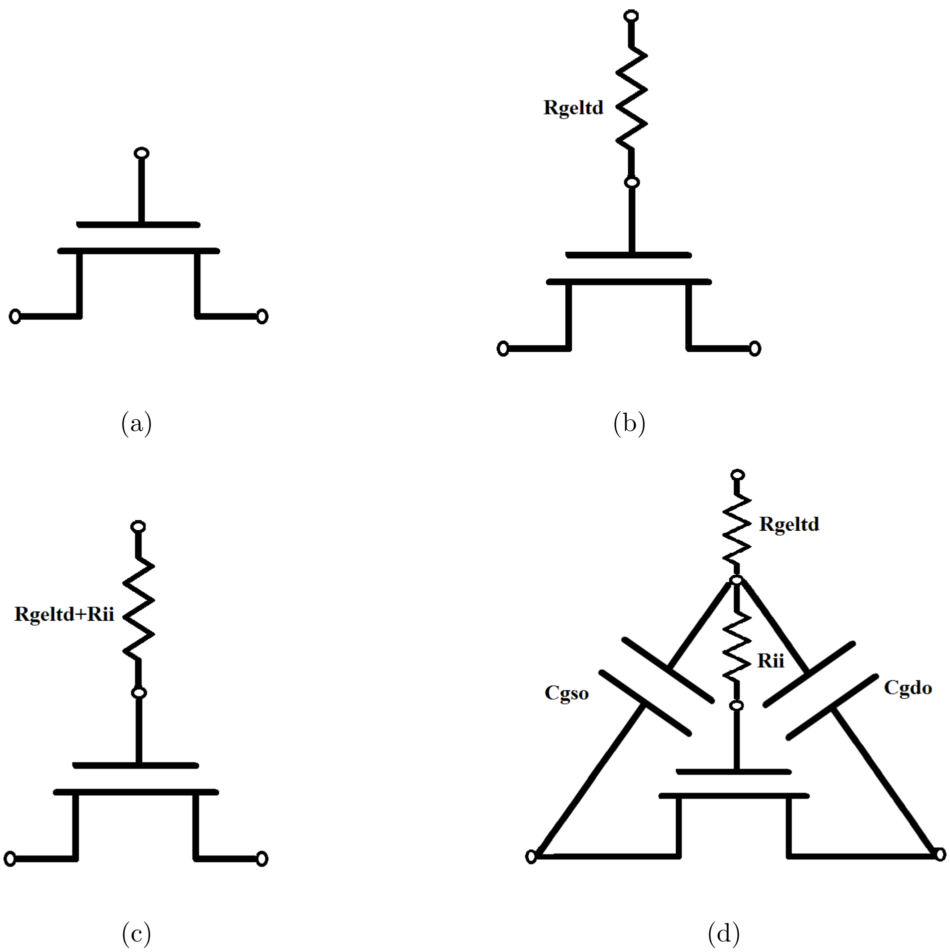

Figure 6: Gate resistance network for (a) $RGATEMOD = 0$ (b) $RGATEMOD = 1$ (b) $RGATEMOD = 2$ (d) $RGATEMOD = 3$.

5.22 Noise Modeling

The following noise sources in MOSFETs are modeled in BSIM-SOI 100.x.x for SPICE noise analysis: flicker noise (also known as 1/f noise), channel thermal noise and induced gate noise and their correlation, thermal noise due to physical resistances such as the source/drain, gate electrode, and substrate resistances, and shot noise due to the gate dielectric tunneling current.

Noise models in BSIM-SOI 100.x.x	**Origin**
Flicker noise model	BSIM4 Unified Model (FNOIMOD=0)
Thermal noise(TNOIMOD=0)	BSIM4 (TNOIMOD=0)
Thermal noise (TNOIMOD=1)	BSIM4 (TNOIMOD=2)
Gate current shot noise	BSIM4 gate current noise
Noise associated with parasitic resistances	BSIM4 parasitic resistance noise

5.22.1 Flicker Noise Models

FNOIMOD = 1 (Flicker Noise Model): The flicker noise model in BSIM-SOI 100.x.x for FNOIMOD = 0 is the same as FNOIMOD = 1 in BSIM4. The unified physical flicker noise model is smooth over all bias regions.

The physical mechanism for the flicker noise is trapping/detrapping-related charge fluctuation in oxide traps, which results in fluctuations of both mobile carrier numbers and mobilities in the channel. The unified flicker noise model captures this physical process. In the inversion region, the noise density is expressed as [21]

$$\begin{aligned} S_{id,inv}(f) &= \frac{kTq^2\mu_{eff}I_{ds}}{C_{oxe}L_{effNOI}^2 f^{EF}\cdot 10^{10}}\left(NOIA\cdot log\left(\frac{N_0+N^*}{N_l+N^*}\right)\right. \\ &\left. NOIB\cdot(N_0-N_l)+\frac{NOIC}{2}(N_0^2-N_l^2)\right) \\ &\frac{kTI_{ds}^2\Delta L_{clm}}{W_{eff}L_{effNOI}^2 f^{EF}\cdot 10^{10}}\left(\frac{NOIA+NOIB\cdot N_l+NOIC\cdot N_l^2}{(N_l+N^*)^2}\right) \end{aligned} \tag{5.535}$$

where $L_{effNOI}=L_{eff}-2\cdot LINTNOI$, μ_{eff} is the effective mobility at the given bias condition, and L_{eff} and W_{eff} are the effective channel length and width, respectively.

The parameter N_0 is the charge density at the source side given by

$$N_0 = \frac{2n_q C_{ox} V_t q_s}{q} \tag{5.536}$$

The parameter N_l is the charge density at the drain end given by

$$N_l = \frac{2n_q C_{ox} V_t q_{deff}}{q} \tag{5.537}$$

and N* is given by

$$N^* = \frac{V_t \big(C_{ox} + C_d + CIT\big)}{q} \tag{5.538}$$

where CIT is a model parameter from DC IV and C_d is the depletion capacitance.

ΔL_{clm} is the channel length reduction due to channel length modulation and given by

$$\begin{aligned} \Delta L_{clm} &= litl \cdot log\left(\frac{\frac{V_{ds}-V_{dseff}}{litl} + EM}{E_{sat}}\right) \\ E_{sat} &= \frac{2VSAT}{\mu_{eff}} \end{aligned} \tag{5.539}$$

In the subthreshold region, the noise density is written as

$$S_{id,subVt}(f) = \frac{NOIA \cdot k \cdot T \cdot I_{ds}^2}{W_{eff} L_{eff} f^{EF} N^{*2} \cdot 10^{10}} \tag{5.540}$$

The total flicker noise density is

$$S_{id}(f) = \frac{S_{id,inv} \cdot S_{id,subVt}}{S_{id,inv} + S_{id,subVt}} \tag{5.541}$$

FNOIMOD = 1 (Flicker Noise Model for Halo Transistors): For the noise modeling, it is now assumed that the transistor is composed of two transistors, channel transistor of length $L - LH$ and halo transistor of length LH connected in series and carries same current (IDS) as in single transistor configuration. The individual contribution of the halo and channel transistors to overall noise is obtained using small signal analysis and principle of superposition [22] [23].

Total drain current noise PSD becomes,

$$S_{ID} = S_{ID,1} + S_{ID,2} \tag{5.542}$$

$$= S_{ID,h}\left[\frac{g_{m,ch} + g_{d,ch}}{g_{m,ch} + g_{d,ch} + g_{d,h}}\right]^2 +$$

$$S_{ID,ch}\left[\frac{g_{d,h}}{g_{m,ch} + g_{d,ch} + g_{d,h}}\right]^2 \tag{5.543}$$

$$= S_{ID,h}.CF_h + S_{ID,ch}.CF_{ch} \tag{5.544}$$

We refer to the multiplying factors to $S_{ID,h}$ and $S_{ID,ch}$ in (5.543) as contribution factors (CF). Where,

$$g_{d,ch} = 2n_q \mu C_{ox}\frac{W}{L - L_h}V_t q_{d,ch} \tag{5.545}$$

$$g_{d,h} = 2n_q \mu C_{ox}\frac{W}{L_h}V_t q_{d,h} \tag{5.546}$$

$$g_{m,ch} = 2\mu C_{ox}\frac{W}{L - L_h}V_t(q_{s,ch} - q_{d,ch}) \tag{5.547}$$

To calculate $q_{d,h}$ and $q_{s,ch}$, the fact that the same current flows in one transistor and two transistor noise equivalent configuration is used,

$$i_h = \frac{I_{DS}}{-2n_q \mu C_{ox}\frac{W}{L_h}V_t^2} = (q_{s,h}^2 + q_{s,h}) - (q_{d,h}^2 + q_{d,h}) \tag{5.548}$$

$$i_{ch} = \frac{I_{DS}}{-2n_q \mu C_{ox}\frac{W}{L-L_h}V_t^2} = (q_{s,ch}^2 + q_{s,ch}) - (q_{d,ch}^2 + q_{d,ch}) \tag{5.549}$$

where i_h and i_{ch} are the normalized drain current of halo and channel transistor respectively. Since I_{DS} is known from DC modeling, the above equations are solved for,

$$q_{d,h} = -\frac{1}{2} + \frac{1}{2}\sqrt{1 + 4(q_{s,h}^2 + q_{s,h} - i_h)} \tag{5.550}$$

$$q_{s,ch} = -\frac{1}{2} + \frac{1}{2}\sqrt{1 + 4(q_{d,ch}^2 + q_{d,ch} + i_{ch})} \tag{5.551}$$

Here we have used the unified model presented in for halo and channel transistors separately, where S_{ID} is expressed as

$$S_{ID,h} = \frac{kTI_{DS}^2}{\gamma f W L_h^2} \int_0^{L_h} \frac{N_{t,h}^*(E_{F_n})}{N_h^2} dx \tag{5.552}$$

$$S_{ID,ch} = \frac{kTI_{DS}^2}{\gamma f W (L - L_h)^2} \int_{L_h}^{L} \frac{N_{t,ch}^*(E_{F_n})}{N_{ch}^2} dx \tag{5.553}$$

where apparent trap density $N_{t,ch(h)}^*(E_{F_n}) = A_{ch(h)} + B_{ch(h)} N_{ch(h)} + C_{ch(h)} N_{ch(h)}^2$, A, B, C are the noise parameters, γ is the tunneling parameter, k is the Boltzmann constant, and T is the temperature. FNOIMOD=1 is available only for SOIMOD=0.

Flicker Noise Tuning Flexibility:

$$S_{ID,new} = \frac{S_{ID,old}}{1 + NOIA1 \cdot (qs - qdeff)^(NOIAX)} \tag{5.554}$$

Flicker Noise due to drain and source resistance: Flicker noise due to source resistance (S_{id,R_s}) is given by

$$S_{id,R_s} = \frac{KFNS * W * (\frac{I_d}{W})^{AFNS}}{f^{BFNS}} \tag{5.555}$$

Flicker noise due to drain resistance (S_{id,R_d}) is given by

$$S_{id,R_d} = \frac{KFND * W * (\frac{I_d}{W})^{AFND}}{f^{BFND}} \tag{5.556}$$

AFNS, BFNS, KFNS, AFND, BFND and KFND are the model parameters.

Flicker Noise due to EDGEFET (EDGEFET=1): The Edge FET Model noise is added to the main flicker noise S_{ID}, when the parameter EDGEFET is 1

$$noia_{edge} = NOIA * NEDGE; \tag{5.557}$$

$$noib_{edge} = NOIB * NEDGE; \tag{5.558}$$

$$noic_{edge} = NOIC * NEDGE; \tag{5.559}$$

$$N_0 = \frac{2n_q C_{ox} V_t q_{s,edge}}{q} \tag{5.560}$$

The parameter N_l is the charge density at the drain end given by

$$N_l = \frac{2n_q C_{ox} V_t q_{deff,edge}}{q} \tag{5.561}$$

$$\begin{aligned} S_{id,inv,edge}(f) \quad = \quad & \frac{kTq^2 \mu_{eff} Ids_{edge}}{C_{oxe} L_{effNOI}^2 f^{EF} \cdot 10^{10}} \left(noia_{edge} \cdot log \left(\frac{N0 + N^*}{N1 + N^*} \right) \right. \\ & \left. noib_{edge} \cdot (N_0 - N_l) + \frac{noic_{edge}}{2} (N_0^2 - N_l^2) \right) \\ & \frac{kTIds_{edge}^2 \Delta L_{clm}}{W_{edge} LeffNOI_{edge}^2 f^{EF} \cdot 10^{10}} \left(\frac{noia_{edge} + noib_{edge} \cdot N_l + noic_{edge} \cdot N_l^2}{(N_l + N^*)^2} \right) \end{aligned} \tag{5.562}$$

$$S_{id,subVt,edge}(f) = \frac{noia_{edge} \cdot k \cdot T \cdot Ids_{edge}^2}{W_{edge} LeffNOI_{edge} f^{EF} N^{*2} \cdot 10^{10}} \tag{5.563}$$

$$S_{id,edge}(f) = \frac{S_{id,inv,edge} \cdot S_{id,subVt,edge}}{S_{id,inv,edge} + S_{id,subVt,edge}} \tag{5.564}$$

5.22.2 Channel Thermal Noise

There are two channel thermal noise models in BSIM-SOI 100.x.x. One is a charge-based model (default model) similar to that used in BSIM3v3.2 and BSIM4.7.0 (TNOIMOD=0). The other is the holistic model similar to BSIM4.7.0 (TNOIMOD=2). These two models can be selected through the model selector TNOIMOD.

TNOIMOD = 0 (Charge based Model): The noise current is given by

$$Q_{inv} = |Q_{s,intrinsic} + Q_{d,intrinsic}| \times NFIN_{total} \tag{5.565}$$

$$\overline{i_d^2} = \begin{cases} NTNOI \cdot \frac{4kT\Delta f}{R_{ds}+\frac{L_{eff}^2}{\mu_{eff}Q_{inv}}} & \text{if RDSMOD} = 0 \\ NTNOI \cdot \frac{4kT\Delta f}{L_{eff}^2} \cdot \mu_{eff}Q_{inv} & \text{if RDSMOD} = 1 \end{cases} \tag{5.566}$$

where $R_{ds}(V)$ is the bias-dependent LDD source/drain resistance, and the parameter NTNOI is introduced for more accurate fitting of short-channel devices. Q_{inv} is the total inversion charge in the channel.

TNOIMOD = 1 (Correlated Noise Model): In this thermal noise model (similar to TNOIMOD = 2 in BSIM4.7.0), all the short-channel effects and velocity saturation effect incorporated in the IV model are automatically included, hence the name "holistic thermal noise model". In this thermal noise model both the gate and the drain noise are implemented as current noise sources. The drain current noise flows from drain to source; whereas the induced gate current noise flows from the gate to the source. The correlation between the two noise sources is independently controllable and can be tuned using the parameter RNOIC, although the use of default value 0.395 is recommended when measured data is not available. The relevant formulations of TNOIMOD=2 are given below. For more details, see Ph.D. thesis of Darsen Lu and BSIM4 manual.

$$\beta_{tnoi} = RNOIA \cdot \left[1.0 + TNOIA \cdot L_{eff} \cdot \left(\frac{q_{ia}}{E_{sat,noi}L_{eff}}\right)^2\right] \tag{5.567}$$

$$\theta_{tnoi} = RNOIB \cdot \left[1.0 + TNOIB \cdot L_{eff} \cdot \left(\frac{q_{ia}}{E_{sat,noi}L_{eff}}\right)^2\right] \tag{5.568}$$

$$c_{tnoi} = RNOIC \cdot \left[1.0 + TNOIC \cdot L_{eff} \cdot \left(\frac{q_{ia}}{E_{sat,noi}L_{eff}}\right)^2\right] \tag{5.569}$$

$$\tag{5.570}$$

$$S_{id} = 4KT \cdot \mu C_{ox} \frac{W_{eff}}{L_{vsat}} V_t D_{ptwg} M_{oc} \left[\frac{q_s + q_{deff}}{2} \cdot \left(1 + \frac{\beta^2_{lowId}}{TNOIK2 + q_{ia}} \cdot \frac{V_{dseff}}{V_{dsat}} \right) \right.$$
$$\left. + (3 \cdot \beta^2_{tnoi}) \cdot \frac{(q_s - q_{deff})^2}{12 \left(\frac{1+q_s+q_{deff}}{2} \right)} \right] \tag{5.571}$$

$$mig = \frac{1}{12 \cdot NF \cdot W_{eff} \mu_{eff} \cdot D_{ptwg} M_{oc} C_{ox} \cdot V_t} \frac{L^3_{vsat}}{L^2_{eff}} \cdot \left[\frac{\frac{q_s+q_{deff}}{2}}{\left(\frac{1+q_s+q_{deff}}{2} \right)^2} \right.$$
$$\left. - \frac{\left(6 \left(\frac{q_s+q_{deff}}{2} \right) + \frac{1}{2} \right) (q_s - q_{deff})^2}{60 \left(\frac{1+q_s+q_{deff}}{2} \right)^4} + \frac{(q_s - q_{deff})^4}{144 \left(\frac{1+q_s+q_{deff}}{2} \right)^5} \right] \cdot (\frac{15}{4} \cdot \theta^2_{tnoi}) \tag{5.572}$$

$$S_{ig} = \frac{4KT \cdot [\omega \cdot C_{ox} \cdot W \cdot NF \cdot L]^2 \cdot mig}{1 + (\omega \cdot C_{0x} \cdot L \cdot W \cdot NF \cdot mig)^2} \tag{5.573}$$

$$S_{ig,id} = -j\omega \cdot 4KT \cdot \mu C_{ox} D_{ptwg} M_{oc} V_t (\frac{L_{vsat}}{L_{eff}}) \cdot \left[\frac{(q_s - q_{deff})}{12 \left(\frac{1+q_s+q_{deff}}{2} \right)} - \frac{(q_s - q_{deff})^3}{144 \left(\frac{1+q_s+q_{deff}}{2} \right)^3} \right] \cdot \frac{c_{tnoi}}{0.395} \tag{5.574}$$

$$c = \frac{S_{ig,id}}{\sqrt{S_{ig}} \cdot \sqrt{S_{id}}} \tag{5.575}$$

5.22.3 Gate Current Shot Noise

$$\overline{i^2_{gs}} = 2q(I_{gcs} + I_{gs}) \tag{5.576}$$
$$\overline{i^2_{gd}} = 2q(I_{gcd} + I_{gd}) \tag{5.577}$$
$$\overline{i^2_{gb}} = 2qI_{gbinv} \tag{5.578}$$

5.22.4 Resistor Noise

The noise associated with each parasitic resistors in BSIM6 are calculated

If $RDSMOD = 1$ then

$$\frac{\overline{i^2_{RS}}}{\Delta f} = 4kT \cdot \frac{1}{R_{source}} \tag{5.579}$$

$$\frac{\overline{i^2_{RD}}}{\Delta f} = 4kT \cdot \frac{1}{R_{drain}} \tag{5.580}$$

If $RGATEMOD = 1$ then

$$\frac{\overline{i^2_{RG}}}{\Delta f} = 4kT \cdot \frac{1}{R_{geltd}} \tag{5.581}$$

5.23 Self Heating Model

Effect of self heating is modeled by employing a thermal network consisting of thermal resistance (R_{th}) and capacitance (C_{th}) as shown in Fig.7. The voltage at thermal node T gives the rise in temperature, which is added to the ambient temperature, and all the temperature-sensitive variables in the model are updated accordingly [2].

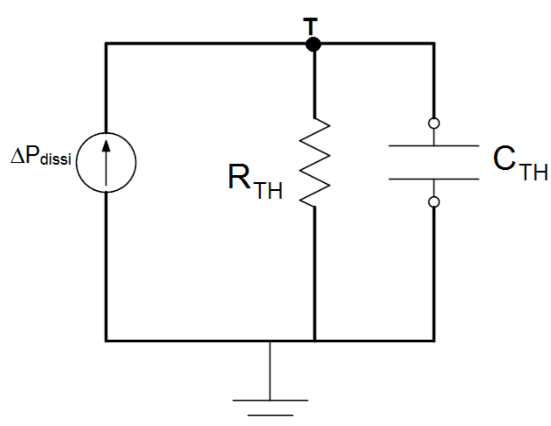

Figure 7: Thermal Network for Self Heating Model.

$$R_{th} = \frac{RTH0}{(WTH0 + Weff) \cdot NF}$$

$$C_{th} = CTH0 * (WTH0 + Weff) \cdot NF$$

6 Asymmetric MOS Junction Diode Models

6.1 Junction Diode CV Model

Source and drain junction capacitances consist of three components: the bottom junction capacitance, sidewall junction capacitance along the isolation edge, and sidewall junction capacitance along the gate edge. An analogous set of equations are used for both sides but each side has a separate set of model parameters.

Source/Body Junction Diode The source-side junction capacitance can be calculated by

$$C_{bs} = A_{seff} C_{jbs} + P_{seff} C_{jbssw} + W_{effcj} \cdot NF \cdot C_{jbsswg} \tag{6.1}$$

where C_{jbs} is the unit-area bottom S/B junciton capacitance, C_{jbssw} is the unit-length S/B junction sidewall capacitance along the isolation edge, and C_{jbsswg} is the unit-length S/B junction sidewall capacitance along the gate edge. The effective area and perimeters in (6.1) are given in Section Layout-Dependent Parasitics Models.

Cjbs is calculated by

$$C_{jbs} = \begin{cases} CJS(T) \cdot \left(1 - \frac{V_{bs}}{PBS(T)}\right)^{-MJS} & \text{if } \frac{V_{bs}}{PBS(T)} \le x_0 \\ CJS(T) \cdot \frac{1}{(1-x_0)^{MJS}} \cdot \left[1 + MJS\left(1 + \frac{\frac{V_{bs}}{PBS} - 1}{1-x_0}\right)\right] & \text{otherwise} \end{cases} \tag{6.2}$$

where the value of x_0 is taken as 0.9.

Cjbssw is calculated by

$$C_{jbssw} = \begin{cases} CJSWS(T) \cdot \left(1 - \frac{V_{bs}}{PBSWS(T)}\right)^{-MJSWS} & \text{if } \frac{V_{bs}}{PBSWS(T)} \leq x_0 \\ CJSWS(T) \cdot \frac{1}{(1-x_0)^{MJSWS}} \cdot \left[1 + MJSWS\left(1 + \frac{\frac{V_{bs}}{PBSWS(T)} - 1}{1-x_0}\right)\right] & \text{otherwise} \end{cases} \tag{6.3}$$

where the value of x_0 is taken as 0.9.

Cjbsswg is calculated by

$$C_{jbsswg} = \begin{cases} CJSWGS(T) \cdot \left(1 - \frac{V_{bs}}{PBSWGS(T)}\right)^{-MJSWGS} & \text{if } \frac{V_{bs}}{PBSWGS(T)} \leq x_0 \\ CJSWGS(T) \cdot \frac{1}{(1-x_0)^{MJSWGS}} \cdot \left[1 + MJSWGS\left(1 + \frac{\frac{V_{bs}}{PBSWGS(T)} - 1}{1-x_0}\right)\right] & \text{otherwise} \end{cases} \tag{6.4}$$

where the value of x_0 is taken as 0.9.

Drain/Body Junction Diode The drain-side junction capacitance can be calculated by

$$C_{bd} = A_{deff} C_{jbd} + P_{deff} C_{jbdsw} + W_{effcj} \cdot NF \cdot C_{jbdswg} \tag{6.5}$$

where Cjbd is the unit-area bottom D/B junciton capacitance, Cjbdsw is the unit-length D/B junction sidewall capacitance along the isolation edge, and Cjbdswg is the unit-length D/B junction sidewall capacitance along the gate edge. The effective area and perimeters in (6.5) are given in Section Layout-Dependent Parasitics Models.

Cjbd is calculated by

$$C_{jbd} = \begin{cases} CJD(T) \cdot \left(1 - \frac{V_{bs}}{PBD(T)}\right)^{-MJD} & \text{if } \frac{V_{bs}}{PBD(T)} \leq x_0 \\ CJD(T) \cdot \frac{1}{(1-x_0)^{MJD}} \cdot \left[1 + MJD\left(1 + \frac{\frac{V_{bs}}{PBD} - 1}{1-x_0}\right)\right] & \text{otherwise} \end{cases} \tag{6.6}$$

where the value of x_0 is taken as 0.9.

Cjbdsw is calculated by

$$C_{jbdsw} = \begin{cases} CJSWD(T) \cdot \left(1 - \frac{V_{bs}}{PBSWD(T)}\right)^{-MJSWS} & \text{if } \frac{V_{bs}}{PBSWD(T)} \leq x_0 \\ CJSWD(T) \cdot \frac{1}{(1-x_0)^{MJSWD}} \cdot \left[1 + MJSWD\left(1 + \frac{\frac{V_{bs}}{PBSWD(T)} - 1}{1-x_0}\right)\right] & \text{otherwise} \end{cases} \tag{6.7}$$

where the value of x_0 is taken as 0.9.

Cjbdswg is calculated by

$$C_{jbdswg} = \begin{cases} CJSWGD(T) \cdot \left(1 - \frac{V_{bs}}{PBSWGD(T)}\right)^{-MJSWGD} & \text{if } \frac{V_{bs}}{PBSWGD(T)} \leq x_0 \\ CJSWGD(T) \cdot \frac{1}{(1-x_0)^{MJSWGD}} \cdot \left[1 + MJSWGD\left(1 + \frac{\frac{V_{bs}}{PBSWGD(T)} - 1}{1-x_0}\right)\right] & \text{otherwise} \end{cases} \tag{6.8}$$

where the value of x_0 is taken as 0.9.

7 Layout dependent Parasitics Models

7.1 Layout-Dependent Parasitics Models

BSIM-SOI 100.x.x provides a comprehensive and versatile geometry/layout-dependent parasitics model taken from BSIM4. It supports modeling of series (such as isolated, shared, or merged source/ drain) and multi-finger device layout or a combination of these two configurations. Note that the narrow-width effect in the per-finger device with multi-finger configuration is accounted for by this model. A complete list of model parameters and selectors can be found at the end.

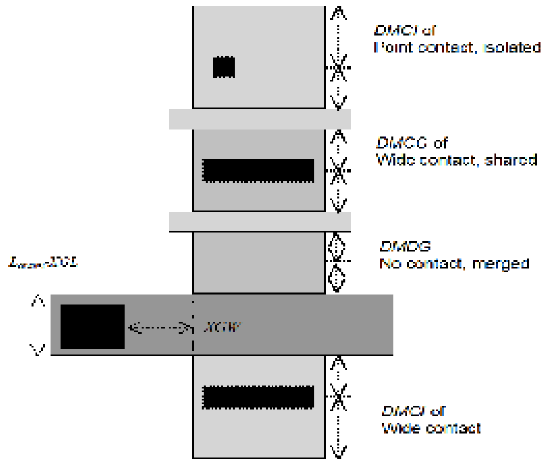

Figure 8: Definition for layout parameters.

7.1.1 Geometry Definition

Figure 8 schematically shows the geometry definition for various source/drain connections and source/drain/gate contacts. The layout parameters shown in this figure will be used to calculate resistances and source/drain perimeters and areas.

7.1.2 Model Formulation and Options

Effective Junction Perimeter and Area: In the following, only the source-side case is illustrated. The same approach is used for the drain side. The effective junction perimeter on the source side is calculated by

If (PS is given)

    if (**perMod**=0)

        P_{seff}=PS

    else

Else

    P_{seff} computed from NF, DWJ, **geoMod**, DMCG, DMCI, DMDG, DMCGT, RSH, and MIN.

The effective junction area on the source side is calculated by

If (AS is given)

    $A_{seff} = AS$

Else

A_{seff} computed from NF, DWJ, **geoMod**, DMCG, DMCI, DMDG, DMCGT, RSH, and MIN.

In the above, P_{seff} and A_{seff} will be used to calculate junction diode IV and CV. P_{seff} does not include the gate-edge perimeter.

Source/Drain Diffusion Resistance: The source diffusion resistance is calculated by

If(number of sources NRS is given)

ELSE if(**rgeoMod**=0)

Source diffusion resistance R_{sdiff} is not generated.

Else

R_{sdiff} computed from NF, DWJ, **geoMod**, DMCG, DMCI, DMDG, DMCGT, RSH, and MIN.

where the number of source squares NRS is an instance parameter. Similarly, the drain diffusion resistance is calculated by

If(number of sources NRD is given)

ELSE if(**rgeoMod**=0)

Drain diffusion resistance R_{ddiff} is not generated.

Else

R_{ddiff} computed from NF, DWJ, **geoMod**, DMCG, DMCI, DMDG, DMCGT, RSH, and MIN.

Gate Electrode Resistance: The gate electrode resistance with multi-finger configuration is modeled by

$$Rgeltd = \frac{RSHG \cdot \left(XGW + \frac{W_{effci}}{3NGCON}\right)}{NGCON \cdot \left(L_{drawn} - XGL\right) \cdot NF} \tag{7.1}$$

Option for Source/Drain Connections: Table 1 lists the options for source/drain connections through the model selector **geoMod**. For multi-finger devices, all inside S/D diffusions are assumed shared.

Option for Source/Drain Contacts: Table 2 lists the options for source/drain contacts through the model selector **rgeoMod**.

geomod	**End Source**	**End drain**	**Note**
0	isolated	isolated	NF=Odd
1	isolated	shared	NF=Odd, Even
2	shared	shared	NF=Odd, Even
3	shared	isolated	NF=Odd, Even
4	isolated	merged	NF=Odd
5	shared	merged	NF=Odd, Even
6	merged	isolated	NF=Odd
7	merged	shared	NF=Odd, Even
8	merged	merged	NF=Odd
9	sha/iso	shared	NF=Even
10	shared	sha/iso	NF=Even

Table 1: geoMod options.

rgeoMod	**End-source contact**	**End-drain contact**
0	No R_{sdiff}	No R_{ddiff}
1	wide	wide
2	wide	point
3	point	wide
4	point	point
5	wide	merged
6	point	merged
7	merged	wide
8	merged	point

Table 2: **rgeoMod** options.

8 Temperature dependence Models

8.1 Temperature Dependence Model

Accurate modeling of the temperature effects on MOSFET characteristics is important to predict circuit behavior over a range of operating temperatures (T). The operating temperature might be different from the nominal temperature (TNOM) at which the BSIM-SOI 100.x.x model parameters are extracted. This chapter presents the BSIM-SOI 100.x.x temperature dependence models for threshold voltage, mobility, saturation velocity, source/drain resistance, and junction diode IV and CV.

8.1.1 Length Scaling of Temperature parameters

$$UTE = UTE \cdot \left(1 + UTEL\frac{1}{L_{eff}}\right) \tag{8.1}$$

$$UA1 = UA1 \cdot \left(1 + UA1L\frac{1}{L_{eff}}\right) \tag{8.2}$$

$$UD1 = UD1 \cdot \left(1 + UD1L\frac{1}{L_{eff}}\right) \tag{8.3}$$

$$AT = AT \cdot \left(1 + ATL\frac{1}{L_{eff}}\right) \tag{8.4}$$

$$PTWGT = PTWGT \cdot \left(1 + PTWGTL\frac{1}{L_{eff}}\right) \tag{8.5}$$

$$\tag{8.6}$$

8.1.2 Temperature Dependence of Threshold Voltage

The temperature dependence of V_{th} is modeled by

$$
\begin{aligned}
V_{th}(T) &= V_{th}(TNOM) + \left(KT1_i + KT2_i \cdot V_{breff}\right) \cdot \Big(\Big(\frac{T}{TNOM}\Big)^{KT1EXP} - 1\Big) \\
V_{fb}(T) &= V_{fb}(TNOM) - KT1 \cdot \Big(\frac{T}{TNOM} - 1\Big) && (8.7) \\
VFBSDOFF(T) &= VFBSDOFF(TNOM) \cdot [1 + TVFBSDOFF \cdot (T - TNOM)] \\
NFACTOR(T) &= NFACTOR(TNOM) + TFACTOR \cdot \Big(\frac{T}{TNOM} - 1\Big) && (8.8) \\
ETA0(T) &= ETA0(TNOM) + TETA0\Big(\frac{T}{TNOM} - 1\Big) && (8.9)
\end{aligned}
$$

8.1.3 Temperature Dependence of Mobility

$$
\begin{aligned}
U0(T) &= U0(TNOM) \cdot (T/TNOM)^{UTE} && (8.10) \\
UA(T) &= UA(TNOM)[1 + UA1 \cdot (T - TNOM)] && (8.11) \\
UC(T) &= UC(TNOM)[1 + UC1 \cdot (T - TNOM)] && (8.12) \\
UD(T) &= UD(TNOM) \cdot (T/TNOM)^{UD1} && (8.13) \\
UCS(T) &= UCS(TNOM) \cdot (T/TNOM)^{UCSTE} && (8.14) \\
EU(T) &= EU(TNOM) \cdot (1 + EU1 \cdot (\frac{T}{TNOM} - 1)) && (8.15) \\
& && (8.16)
\end{aligned}
$$

8.1.4 Temperature Dependence of Saturation Velocity

$$VSAT(T) = VSAT(TNOM) \cdot (T/TNOM)^{-AT} \tag{8.17}$$

8.1.5 Temperature Dependence of LDD Resistance

$$rdstemp = (T/TNOM)^{PRT} \tag{8.18}$$

RDSMOD = 0 (internal source/drain LDD resistance)

$$RDSW(T) = RDSW(TNOM) \cdot rdstemp \tag{8.19}$$
$$RDSWMIN(T) = RDSWMIN(TNOM) \cdot rdstemp \tag{8.20}$$

RDSMOD = 1 (external source/drain LDD resistance)

$$RDW(T) = RDW(TNOM) \cdot rdstemp \tag{8.21}$$
$$RDWMIN(T) = RDWMIN(TNOM) \cdot rdstemp) \tag{8.22}$$
$$RSW(T) = RSW(TNOM) \cdot rdstemp \tag{8.23}$$
$$RSWMIN(T) = RSWMIN(TNOM) \cdot rdstemp \tag{8.24}$$

8.1.6 Temperature Dependence of Junction Diode IV

- **Source-side diode** The source-side diode is turned off if both A_{seff} and P_{seff} are zero. Otherwise, the source-side saturation current is given by

$$I_{sbs} = A_{seff} J_{ss}(T) + P_{seff} J_{ssws}(T) + W_{effcj} \cdot NF \cdot J_{sswgs}(T) \tag{8.25}$$

where

$$\begin{aligned} J_{ss}(T) &= JSS(TNOM) \cdot exp\left(\frac{\frac{E_g(TNOM)}{v_t(TNOM)} - \frac{E_g(T)}{v_t(T)} + XTIS \cdot ln\left(\frac{T}{TNOM}\right)}{NJS}\right) \\ J_{ssws}(T) &= JSSWS(TNOM) \cdot exp\left(\frac{\frac{E_g(TNOM)}{v_t(TNOM)} - \frac{E_g(T)}{v_t(T)} + XTIS \cdot ln\left(\frac{T}{TNOM}\right)}{NJS}\right) \\ J_{sswgs}(T) &= JSSWGS(TNOM) \cdot exp\left(\frac{\frac{E_g(TNOM)}{k_b \cdot TNOM} - \frac{E_g(T)}{k_b \cdot T} + XTIS \cdot ln\left(\frac{T}{TNOM}\right)}{NJS}\right) \end{aligned} \tag{8.26}$$

where Eg is given in Temperature Dependences of Eg and ni.

- **Drain-side diode** The drain-side diode is turned off if both A_{seff} and P_{seff} arezero. Otherwise, the drain-side saturation current is given by

$$I_{sbd} = A_{deff} J_{sd}(T) + P_{deff} J_{sswd}(T) + W_{effcj} \cdot NF \cdot J_{sswgd}(T) \tag{8.27}$$

where

$$\begin{aligned}
J_{sd}(T) &= JSD(TNOM) \cdot exp\left(\frac{\frac{E_g(TNOM)}{k_b \cdot TNOM} - \frac{E_g(T)}{k_b \cdot T} + XTID \cdot ln\left(\frac{T}{TNOM}\right)}{NJD} \right) \\
J_{sswd}(T) &= JSSWD(TNOM) \cdot exp\left(\frac{\frac{E_g(TNOM)}{k_b \cdot TNOM} - \frac{E_g(T)}{k_b \cdot T} + XTID \cdot ln\left(\frac{T}{TNOM}\right)}{NJD} \right) \\
J_{sswgd}(T) &= JSSWGD(TNOM) \cdot exp\left(\frac{\frac{E_g(TNOM)}{k_b \cdot TNOM} - \frac{E_g(T)}{k_b \cdot T} + XTID \cdot ln\left(\frac{T}{TNOM}\right)}{NJD} \right)
\end{aligned} \tag{8.28}$$

8.1.7 Temperature Dependence of Junction Diode CV

- Source-side diode: The temperature dependences of zero-bias unit-length/area junction capacitances on the source side are modeled by

$$\begin{aligned}
CJS(T) &= CJS(TNOM) + TCJ \cdot (T - TNOM) \\
CJSWS(T) &= CJSWS(TNOM) + TCJSW \cdot (T - TNOM) \\
CJSWGS(T) &= CJSWGS(TNOM) + TCJSWG \cdot (T - TNOM)
\end{aligned} \tag{8.29}$$

The temperature dependences of the built-in potentials on the source side are modeled by

$$\begin{aligned}
PBS(T) &= PBS(TNOM) - TPB \cdot (T - TNOM) \\
PBSWS(T) &= PBSWS(TNOM) - TPBSW \cdot (T - TNOM) \\
PBSWGS(T) &= PBSWGS(TNOM) - TPBSWG \cdot (T - TNOM)
\end{aligned} \tag{8.30}$$

– Drain-side diode: The temperature dependences of zero-bias unit-length/area junction capacitances on the drain side are modeled by

$$\begin{aligned}
CJS(T) &= CJS(TNOM)[1 + TCJ \cdot (T - TNOM)] \\
CJSWS(T) &= CJSWS(TNOM) + TCJSW \cdot (T - TNOM) \\
CJSWGS(T) &= CJSWGS(TNOM)[1 + TCJSWG \cdot (T - TNOM)]
\end{aligned} \tag{8.31}$$

The temperature dependences of the built-in potentials on the drain side are modeled by

$$\begin{aligned}
PBD(T) &= PBD(TNOM) - TPB \cdot (T - TNOM) \\
PBSWD(T) &= PBSWD(TNOM) - TPBSW \cdot (T - TNOM) \\
PBSWGD(T) &= PBSWGD(TNOM) - TPBSWG \cdot (T - TNOM)
\end{aligned} \tag{8.32}$$

- **trap-assisted tunneling (TAT) and recombination current**

$$
\begin{aligned}
J_{tsswgs}(T) &= J_{tsswgs}(TNOM) \cdot \left(\sqrt{\frac{JTWEFF}{W_{effcj}}} + 1\right) \\
&\cdot \quad exp\left[\frac{-E_g(TNOM)}{k_b T} \cdot X_{tsswgs} \cdot \left(1 - \frac{T}{TNOM}\right)\right] \qquad (8.33) \\
J_{tssws}(T) &= J_{tssws}(TNOM) \cdot exp\left[\frac{-E_g(TNOM)}{k_b T} \cdot X_{tssws} \cdot \left(1 - \frac{T}{TNOM}\right)\right] \\
J_{tss}(T) &= J_{tss}(TNOM) \cdot exp\left[\frac{-E_g(TNOM)}{k_b T} \cdot X_{tss} \cdot \left(1 - \frac{T}{TNOM}\right)\right] \\
J_{tsswgd}(T) &= J_{tsswgd}(TNOM) \cdot \left(\sqrt{\frac{JTWEFF}{W_{effcj}}} + 1\right) \\
&\cdot \quad exp\left[\frac{-E_g(TNOM)}{k_b T} \cdot X_{tsswgd} \cdot \left(1 - \frac{T}{TNOM}\right)\right] \qquad (8.34) \\
J_{tsd}(T) &= J_{tsd}(TNOM) \cdot exp\left[\frac{-E_g(TNOM)}{k_b T} \cdot X_{tsd} \cdot \left(1 - \frac{T}{TNOM}\right)\right] \\
NJTSSWG(T) &= NJTSSWG(TNOM) \cdot \left[1 + TNJTSSWG(\frac{T}{TNOM} - 1)\right] \\
NJTSSW(T) &= NJTSSW(TNOM) \cdot \left[1 + TNJTSSW(\frac{T}{TNOM} - 1)\right] \\
NJTS(T) &= NJTS(TNOM) \cdot \left[1 + TNJTS(\frac{T}{TNOM} - 1)\right] \\
NJTSSWGD(T) &= NJTSSWGD(TNOM) \cdot \left[1 + TNJTSSWGD(\frac{T}{TNOM} - 1)\right] \\
NJTSSWD(T) &= NJTSSWD(TNOM) \cdot \left[1 + TNJTSSWD(\frac{T}{TNOM} - 1)\right] \\
NJTSSWD(T) &= NJTSSWD(TNOM) \cdot \left[1 + TNJTSSWD(\frac{T}{TNOM} - 1)\right] \\
NJTSD(T) &= NJTSD(TNOM) \cdot \left[1 + TNJTSD(\frac{T}{TNOM} - 1)\right]
\end{aligned}
\qquad (8.35)
$$

8.1.8 Temperature Dependences of E_g and n_i

- Energy-band gap of channel (E_g): The temperature dependence of E_g is modeled by

$$Eg0 = BG0SUB - \frac{TBGASUB \times Tnom^2}{Tnom + TBGBSUB} \tag{8.36}$$

$$E_g = BG0SUB - \frac{TBGASUB \times T^2}{T + TBGBSUB} \tag{8.37}$$

- Intrinsic carrier concentration of non-silicon channel (n_i)

$$n_i = NI0SUB \times \left(\frac{T}{Tnom}\right)^{(3/2)} \times exp\left(\frac{Eg}{2\frac{kTnom}{q}} - \frac{Eg}{2\frac{kT}{q}}\right) \tag{8.38}$$

9 Stress effect Model Development

9.1 Stress Effect Model

CMOS feature size aggressively scaling makes shallow trench isolation(STI) very popular active area isolation process in advanced technologies. Recent years, strain channel materials have been employed to achieve high device performance. The mechanical stress effect induced by these process causes MOSFET performance function of the active area size(OD: oxide definition) and the location of the device in the active area. And the necessity of new models to describe the layout dependence of MOS parameters due to stress effect becomes very urgent in advance CMOS technologies. Influence of stress on mobility has been well known since the 0.13um technology. The stress influence on saturation velocity is also experimentally demonstrated. Stress-induced enhancement or suppression of dopant diffusion during the processing is reported. Since the doping profile may be changed due to different STI sizes and stress, the threshold voltage shift and changes of other second-order effects, such as DIBL and body effect, were shown in process integration. BSIM-SOI 100.x.x considers the influence of stress on mobility, velocity saturation, threshold voltage, body effect, and DIBL effect.

9.1.1 Stress Effect Model Development

Experimental analysis show that there exist at least two different mechanisms within the influence of stress effect on device characteristics. The first one is mobility-related and is induced by the band structure modification. The second one is Vth-related as a result of doping profile variation. Both of them follow the same 1/LOD trend but reveal different L and W scaling. We have derived a phenomenological model based on these findings by modifying some parameters in the BSIM model. Note that the following equations have no impact on the iteration time because there are no voltage-controlled components in them.

Mobility-related Equations: This model introduces the first mechanism by adjusting the U0 and Vsat according to different W, L and OD shapes. Define mobility relative change due to stress effect as :

$$\rho_{\mu_{eff}} = \Delta\mu_{eff}/\mu_{effo} = (\mu_{eff} - \mu_{effo})/\mu_{effo} = \frac{\mu_{eff}}{\mu_{effo}} - 1 \tag{9.1}$$

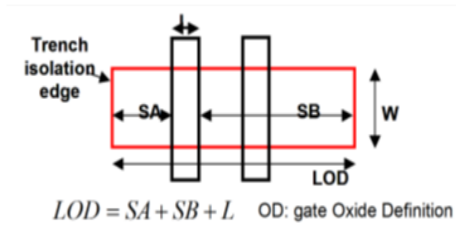

Figure 9: the typical layout of a MOSFET

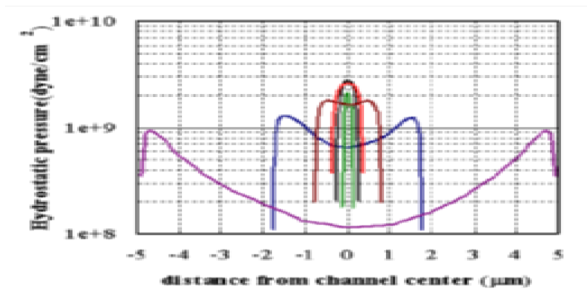

Figure 10: Stress distribution within MOSFET channel using 2D simulation

So,

$$\frac{\mu_{eff}}{\mu_{effo}} = 1 + \rho_{\mu_{eff}} \tag{9.2}$$

Figure 9 shows the typical layout of a MOSFET on active layout surrounded by STI isolation. SA, SB are the distances between isolation edge to Poly from one and the other side, respectively. 2D simulation shows that stress distribution can be expressed by a simple function of SA and SB. Assuming that mobility relative change is propotional to stress distribution. It can be described as function of SA, SB(LOD effect), L, W, and T dependence:

$$\rho_{\mu_{eff}} = \frac{KU0}{Kstress_u0} \cdot \left(Inv\_sa + Inv\_sb\right) \tag{9.3}$$

where:

$$
\begin{aligned}
Inv\_sa &= \frac{1}{SA + 0.5 \cdot L_{drawn}} && (9.4)\\
Inv\_sb &= \frac{1}{SB + 0.5 \cdot L_{drawn}} && (9.5)\\
Kstress\_u0 &= \left(1 + \frac{LKU0}{(L_{drawn}+XL)^{LLODKU0}}\right.\\
&\quad + \frac{WKU0}{(W_{drawn}+XW+WLOD)^{WLODKU0}}\\
&\quad \left. + \frac{PKU0}{(L_{drawn}+XL)^{LLODKU0} \cdot (W_{drawn}+XW+WLOD)^{WLODKU0}}\right)\\
&\quad \times \left(1 + TKU0 \cdot \left(\frac{Temperature}{TNOM} - 1\right)\right) && (9.6)
\end{aligned}
$$

So that:

$$
\begin{aligned}
\mu_{eff} &= \frac{1+\rho_{\mu_{eff}}(SA,SB)}{1+\rho_{\mu_{eff}}(SA_{ref},SB_{ref})}\mu_{effo} && (9.7)\\
\nu_{sattemp} &= \frac{1+KVSAT\cdot\rho_{\mu_{eff}}(SA,SB)}{1+KVSAT\cdot\rho_{\mu_{eff}}(SA_{ref},SB_{ref})}\nu_{sattempo} && (9.8)
\end{aligned}
$$

and SA_{ref}, SB_{ref} are reference distances between OD edge to poly from one and the other side.

Vth-related Equations: Vth0 (threshold voltage without stress effect), K2 and ETA0 are modified to cover the doping profile change in the devices with different LOD. They use the same 1/LOD formulas as shown in earlier sections, but different equations for W and L scaling:

$$
\begin{aligned}
VTH0 &= VTH0_{original} + \frac{KVTH0}{Kstress\_vth0} \cdot \left(Inv\_sa + Inv\_sb - Inv\_sa_{ref} - Inv\_sb_{ref}\right)\\
K2 &= K2_{original} + \frac{STK2}{Kstress\_vth0^{LODK2}} \cdot \left(Inv\_sa + Inv\_sb - Inv\_sa_{ref} - Inv\_sb_{ref}\right)\\
ETA0 &= ETA0_{original} + \frac{STETA0}{Kstress\_vth0^{LODETA0}} \cdot \left(Inv\_sa + Inv\_sb - Inv\_sa_{ref} - Inv\_sb_{ref}\right)
\end{aligned}
\tag{9.9}
$$

where:

$$Inv\_sa_{ref} = \frac{1}{SA_{ref} + 0.5 \cdot L_{drawn}} \tag{9.10}$$

$$Inv\_sbref = \frac{1}{SB_{ref} + 0.5 \cdot L_{drawn}} \tag{9.11}$$

$$\begin{aligned} Kstress\_vth0 = & \left(1 + \frac{LKVTH0}{(L_{drawn} + XL)^{LLODKVTH}}\right. \\ & + \frac{WKVTH0}{(W_{drawn} + XW + WLOD)^{WLODKVTH}} \\ & \left. + \frac{PKVTH0}{(L_{drawn} + XL)^{LLODKVTH} \cdot (W_{drawn} + XW + WLOD)^{WLODKVTH}}\right) \end{aligned} \tag{9.12}$$

Multiple Finger Device: For multiple finger device, the total LOD effect is the average of LOD effect to every finger. That is(see Figure 11) for the layout for multiple finger device):

$$Inv\_sa = \frac{1}{NF} \sum_{i=0}^{NF-1} \frac{1}{SA + 0.5 \cdot L_{drawn} + i \cdot (SD + L_{drawn})} \tag{9.13}$$

$$Inv\_sb = \frac{1}{NF} \sum_{i=0}^{NF-1} \frac{1}{SB + 0.5 \cdot L_{drawn} + i \cdot (SD + L_{drawn})} \tag{9.14}$$

$$\tag{9.15}$$

9.1.2 Effective SA and SB for Irregular LOD

General MOSFET has an irregular shape of active area shown in Figure 12 To fully describe the shape of OD region will require additional instance parameters. However, this will result in too many parameters in the net lists and would massively increase the read-in time and degrade the readability of parameters. One way to overcome this difficulty is the concept of effective SA and SB similar to [24]. Stress effect model as

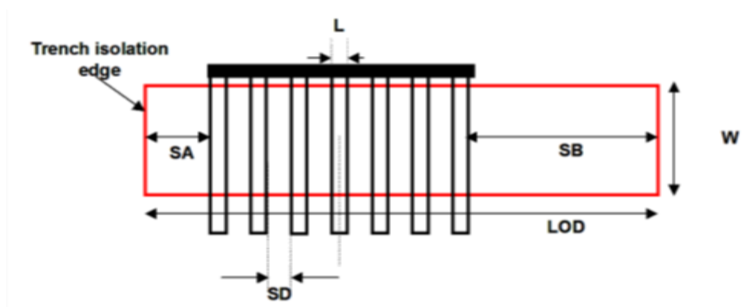

Figure 11: Layout of multiple finger MOSFET

described earlier allows an accurate and efficient layout extraction of effective SA and SB while keeping fully compatibility of the LOD model. They are expressed as:

$$\frac{1}{SA_{eff} + 0.5 \cdot L_{drawn}} = \sum_{i=1}^{n} \frac{sw_i}{W_{drawn}} \cdot \frac{1}{sa_i + 0.5 \cdot L_{drawn}} \tag{9.16}$$

$$\frac{1}{SB_{eff} + 0.5 \cdot L_{drawn}} = \sum_{i=1}^{n} \frac{sw_i}{W_{drawn}} \cdot \frac{1}{sb_i + 0.5 \cdot L_{drawn}} \tag{9.17}$$

(9.18)

Stress Model for EDGEFET: Same stress model is implemented for the EDGEFET. New parameters are **SAEDGE, SBEDGE, KVTH0EDGE, STK2EDGE, STETA0EDGE.**

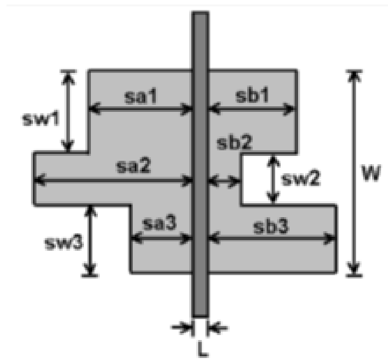

Figure 12: A typical layout of MOS devices with more instance parameters (swi, sai and sbi) in addition to the traditional L and W

10 Well Proximity Effect Model

10.1 Well Proximity Effect Model

Retrograde well profiles have several key advantages for highly scaled bulk complementary metal oxide semiconductor(CMOS) technology. With the advent of high-energy implanters and reduced thermal cycle processing, it has become possible to provide a relatively heavily doped deep nwell and pwell without affecting the critical device-related doping at the surface. The deep well implants provide a low resistance path and suppress parasitic bipolar gain for latchup protection, and can also improve soft error rate and noise isolation. A deep buried layer is also key to forming triple-well structures for isolated-well NMOSFETs. However, deep buried layers can affect devices located near the mask edge. Some of the ions scattered out of the edge of the photoresist are implanted in the silicon surface near the mask edge, altering the threshold voltage of those devices [25]. It is observed a threshold voltage shifts of up to 100 mV in a deep boron retrograde pwell, a deep phosphorus retrograde nwell, and also a triple-well implementation with a deep phosphorus isolation layer below the pwell over a lateral distance on the order of a micrometer [25]. This effect is called well proximity effect. BSIM-SOI 100.x.x considers the influence of well proximity effect on threshold voltage, mobility, and body effect. This well proximity effect model is developed by the Compact Model Council [26]. Experimental analysis [25] shows that well proximity effect is strong function of distance of FET from mask edge, and electrical quantities influenced by it

follow the same geometrical trend. A phenomenological model based on these findings has been developed by modifying some parameters in the BSIM model. Note that the following equations have no impact on the iteration time because there are no voltage controlled components in them. Well proximity affects threshold voltage, mobility, and the body effect of the device. The effect of the well proximity can be described through the following equations :
If SCA, SCB, SCC are given then WPE is given by (10.1), otherwise it is given by (10.2)

$$\begin{aligned}
Vth0 &= Vth0_{org} + KVTH0WE \cdot (SCA + WEB \cdot SCB + WEC \cdot SCC) \\
K2 &= K2_{org} + K2WE \cdot (SCA + WEB \cdot SCB + WEC \cdot SCC) \\
\mu_{eff} &= \mu_{eff,org} \cdot (1 + KU0WE \cdot (SCA + WEB \cdot SCB + WEB \cdot SCC))
\end{aligned} \tag{10.1}$$

$$\begin{aligned}
& Wdrn = W/NF \\
& T1 = SC + Wdrn \\
& T2 = 1.0/SCREF \\
& T3 = \exp(-10.0 \cdot SC \cdot T2) \\
& T4 = \exp(-10.0 \cdot T1 \cdot T2) \\
& T5 = \exp(-20.0 \cdot SC \cdot T2) \\
& T6 = \exp(-20.0 \cdot T1 \cdot T2) \\
& LOCAL_{SCA} = \frac{SCREF \cdot SCREF}{SC \cdot T1} \\
& LOCAL_{SCB} = \frac{(0.1 \cdot SC + 0.01 \cdot SCREF) \cdot T3 - (0.1 \cdot T1 + 0.01 \cdot SCREF) \cdot T4}{Wdrn} \\
& LOCAL_{SCC} = \frac{(0.05 \cdot SC + 0.0025 \cdot SCREF) \cdot T5 - (0.05 \cdot T1 + 0.0025 \cdot SCREF) \cdot T6}{Wdrn} \\
& Vth0 = Vth0_{org} + KVTH0WE \cdot (LOCAL_{SCA} + WEB \cdot LOCAL_{SCB} + WEC \cdot LOCAL_{SCC}) \\
& K2 = K2_{org} + K2WE \cdot (LOCAL_{SCA} + WEB \cdot LOCAL_{SCB} + WEC \cdot LOCAL_{SCC}) \\
& \mu_{eff} = \mu_{eff,org} \cdot (1 + KU0WE \cdot (LOCAL_{SCA} + WEB \cdot LOCAL_{SCB} + WEB \cdot LOCAL_{SCC}))
\end{aligned} \tag{10.2}$$

where SCA, SCB, SCC are instance parameters that represent the integral of the first/second/third distribution function for scattered well dopant. The guidelines for calculating the instance parameters SCA, SCB, SCC have been developed by the Compact Model Council which can be found at the CMC website [26].

Well proximity Model for EDGEFET: Same well proximity model is implemented for the EDGE- FET. New parameters are KVTH0EDGEWE, LKVTH0EDGEWE, WKVTH0EDGEWE, PKVTH0EDGWE, K2EDGEWE, LK2EDGEWE, WK2EDGEWE, PK2EDGEWE.

11 C-V Model

Inversion Charge for SOIMOD=1: Total inversion charge (including velocity saturation, CLM and poly depletion) can be expressed explicitly in terms of surface potential as follows,

Using eq. (5.470) and (5.475) the position dependence of surface potential can be obtained as :

$$x = x_m + \frac{L}{\Delta_\psi}\left[\psi_s - \psi_m - \frac{(\psi_s - \psi_m)^2}{2H}\right] \tag{11.1}$$

where, xm is average potential,

$$x_m = \frac{L}{2}\cdot\left(1 + \frac{\Delta\psi}{4\cdot H}\right) \tag{11.2}$$

$$\alpha_{DD}{}' = \alpha_{DD}\cdot\left[1 + \frac{Z_{sat}}{2}\cdot\left(\frac{D_{mob}\cdot D_{\Delta L}}{D_{vsat}}\right)^2\right] \tag{11.3}$$

$$H = \left[V_t - \frac{Q_{im}}{\alpha_{DD}{}'}\right]\cdot\left(\frac{D_{mob}\cdot D_{\Delta L}}{D_{vsat}}\right) \tag{11.4}$$

Solving for ψ_s,

$$\psi_s = \psi_m + H\cdot\left[1 - \sqrt{1 - \frac{2\Delta\psi}{L\cdot H}\cdot(x - x_m)}\right] \tag{11.5}$$

using eq. (11.5) the total gate charge is given by,

$$Q_G = \frac{1}{L}\cdot\int_0^L (vgfb - \psi_s)\,dx \tag{11.6}$$

$$= \frac{1}{L}\cdot\int_0^L (vgfb - \psi_s)\,\frac{dx}{d\psi_s}\cdot\psi_s \tag{11.7}$$

close form Q_G can be deduced as,

$$Q_G = vgfb - \psi_m + \frac{\Delta\psi^2}{12\cdot H} \tag{11.8}$$

Using Ward-Dutton partitioning,

$$Q_D = \frac{1}{L} \cdot \int_0^L \cdot \frac{x}{L} \cdot Q_i dx \tag{11.9}$$

and, hence,

$$Q_D = \frac{q_{im}}{2} + \frac{\alpha_{DD} \cdot \Delta\psi}{12} \cdot \left(1 - \frac{\Delta\psi}{2H} - \frac{\Delta\psi^2}{20H^2}\right) \tag{11.10}$$

Source charge is given by,

$$Q_S = Q_I - Q_D \tag{11.11}$$

where,

$$Q_I = q_{im} + \frac{\alpha_{DD} \cdot \Delta\psi^2}{12 \cdot H} \tag{11.12}$$

Bulk charge is given by,

$$Q_B = Q_G - Q_i \tag{11.13}$$

Note: SOIMOD=1 do no have multiple CVMOD.

Inversion Charge for SOIMOD=0: Total inversion charge (excluding velocity saturation, CLM and poly depletion) can be expressed explicitly in terms of normalized charge densities at source and drain sides as follows,

$$Q_I = W \cdot \int_0^L Q_i \ dx \tag{11.14}$$

$$= -WL \cdot C_{ox} \cdot V_t \int_0^1 2.n_q.q \ d\xi \tag{11.15}$$

$$-\frac{Q_I}{WL \cdot C_{ox} V_t} = q_I = 2.n_q \cdot \int_0^1 q \ d\xi \tag{11.16}$$

Here $\xi = \frac{x}{L}$. Inversion charge density is normalized to $-2.n_q.C_{ox}.V_t$ and voltages to V_t. From (5.484),

$$I_{ds} = -2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2 \cdot (2q+1)\frac{dq}{d\xi} \tag{11.17}$$

Normalizing current with $2 \cdot n_q \cdot \mu_{eff} \cdot \frac{W_{eff}}{L_{eff}} \cdot C_{ox} \cdot nV_t^2$,

$$i = -(2q+1)\frac{dq}{d\xi} \tag{11.18}$$

which gives $d\xi = -\frac{(2q+1)}{i}dq = -\frac{(2q+1)}{(q_s-q_d)(1+q_s+q_d)}$. Substituting in (11.16)

$$q_I = -\frac{2n_q}{(q_s-q_d)(1+q_s+q_d)}\int_{q_s}^{q_d} q(2q+1)\ dq \tag{11.19}$$

$$= -\frac{2n_q}{(q_s-q_d)(1+q_s+q_d)}\left[\frac{2}{3}({q_d}^3-{q_s}^3)+\frac{1}{2}({q_d}^2-{q_s}^2)\right] \tag{11.20}$$

on rearranging,

$$q_I = n_q \cdot \left[q_s+q_d+\frac{1}{3}\cdot\frac{(q_s-q_d)^2}{1+q_s+q_d}\right] \tag{11.21}$$

Bulk Charge: Bulk charge density is given as

$$Q_b = -C_{ox}\cdot(V_G - V_{FB} - \psi_S) - Q_i \tag{11.22}$$

$$\tag{11.23}$$

using charge linearization

$$Q_b = -C_{ox}\cdot(V_G - V_{FB} - \psi_P) - Q_i\left(1-\frac{1}{n_q}\right) \tag{11.24}$$

Total bulk charge,

$$Q_B = W\cdot\int_0^L Q_b\ dx \tag{11.25}$$

$$= -WL\cdot C_{ox}\cdot\left[V_G - V_{FB} - \psi_P + \left(1-\frac{1}{n_q}\right)\cdot\int_0^1\frac{Q_i}{C_{ox}}\ d\xi\right] \tag{11.26}$$

Normalizing the bulk charge to $-W.L.C_{ox}.V_t$,

$$q_B = v_g - v_{fb} - \psi_p - 2(n_q - 1) \cdot \int_0^1 q \ d\xi \tag{11.27}$$

We know that $i_{ds} = -(2q+1)\frac{dq}{d\xi}$ with i_{ds} given by (5.493). Thus $d\xi = -\frac{-(2q+1)}{i_{ds}}dq$,

$$q_B = v_g - v_{fb} - \psi_p + \frac{2(n_q - 1)}{i_{ds}} \cdot \int_{q_s}^{q_d} q(2q+1) \ dq \tag{11.28}$$

$$= v_g - v_{fb} - \psi_p + \frac{2(n_q - 1)}{(q_s - q_d)(1 + q_s + q_d)} \int_{q_s}^{q_d} \left[\frac{2q^3}{3} + \frac{q^2}{2}\right] \tag{11.29}$$

$$= v_g - v_{fb} - \psi_p + \frac{2(n_q - 1)}{(q_s - q_d)(1 + q_s + q_d)} \left[\frac{2}{3} \cdot (q_d - q_s)(q_d^2 + q_s^2 + q_s.q_d) + \frac{1}{2} \cdot (q_d - q_s)(q_d + q_s)\right] \tag{11.30}$$

which on rearrangement becomes,

$$q_B = v_g - v_{fb} - \psi_p - (n_q - 1)\left[q_s + q_d + \frac{1}{3} \cdot \frac{(q_s - q_d)^2}{1 + q_s + q_d}\right] \tag{11.31}$$

Flexibility of Tuning Cgg in Accumulation
'$PO\_psip((vgfbCV + DELVFBACC * inv\_Vt), gam, 0, phib\_n, psipACC)$

$$q_B = v_g - v_{fb} - psipACC - (n_q - 1)\left[q_s + q_d + \frac{1}{3} \cdot \frac{(q_s - q_d)^2}{1 + q_s + q_d}\right] \tag{11.32}$$

And old ψ_p will be used for calculating qs and qdeff

Bulk charge with poly depletion effect :

$$q_B = A + B + \frac{1}{3} . \frac{\Delta q^2}{C^3} \cdot \left[\frac{4}{8} \cdot \left(C^2 + P.Q\right) \cdot \frac{1}{1 + q_s + q_d} + \frac{2}{\gamma_g^2}\right] - n_q . \left[q_s + q_d + \frac{1}{3} \cdot \frac{(q_s - q_d)^2}{1 + q_s + q_d}\right] \tag{11.33}$$

where

$$P = \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}} \tag{11.34}$$

$$Q = \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_d}{\gamma_g^2}} \tag{11.35}$$

$$A = \frac{v_g - v_{fb} - \psi_p + 2.q_s}{1 + 2.\sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}}} \tag{11.36}$$

$$B = \frac{v_g - v_{fb} - \psi_p + 2.q_d}{1 + 2.\sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}}} \tag{11.37}$$

$$C = \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}} + \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_d}{\gamma_g^2}} \tag{11.38}$$

Bulk charge with poly depletion effect, CLM and Velocity Saturation effects :

$$d_{qgeff} = (v_g - v_{fb} - \psi_p + 2.q_d)\left[\frac{2\sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_d}{\gamma_g^2}} - 1}{2\sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_d}{\gamma_g^2}} + 1}\right] \tag{11.39}$$

$$q_{beff} = \Bigg[(v_g - v_{fb} - \psi_p) - 2(n_q - 1)q_d + d_{qgeff}\Bigg] \tag{11.40}$$

$$\begin{aligned} q_B = \frac{1}{MDL}\Bigg[A + B + \frac{1}{3}.\frac{\Delta q^2}{C^3} \cdot \left(\frac{4}{8} \cdot (C^2 + P.Q) \cdot \frac{1}{1 + q_s + q_d} + \frac{2}{\gamma_g^2}\right) - \\ n_q.\left(q_s + q_d + \frac{1}{3} \cdot \frac{(q_s - q_d)^2}{1 + q_s + q_d}\right)\Bigg] + (MDL - 1)q_{beff} \end{aligned} \tag{11.41}$$

where

$$P = \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}} \tag{11.42}$$

$$Q = \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_d}{\gamma_g^2}} \tag{11.43}$$

$$A = \frac{v_g - v_{fb} - \psi_p + 2.q_s}{1 + 2.\sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}}} \tag{11.44}$$

$$B = \frac{v_g - v_{fb} - \psi_p + 2.q_d}{1 + 2.\sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}}} \tag{11.45}$$

$$C = \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_s}{\gamma_g^2}} + \sqrt{\frac{1}{4} + \frac{v_g - v_{fb} - \psi_p + 2.q_d}{\gamma_g^2}} \tag{11.46}$$

Source and Drain Charges

$$Q_s = \frac{n_q}{3}\left[2 \cdot q_s + q_{deff} + \frac{1}{2} \cdot \left(1 + \frac{4}{5} \cdot q_s + \frac{6}{5} \cdot q_{deff}\right)\left(\frac{q_s - q_{deff}}{1 + q_s + q_{deff}}\right)^2\right] \tag{11.47}$$

$$Q_d = \frac{n_q}{3}\left[q_s + 2 \cdot q_{deff} + \frac{1}{2} \cdot \left(1 + \frac{6}{5} \cdot q_s + \frac{4}{5} \cdot q_{deff}\right)\left(\frac{q_s - q_{deff}}{1 + q_s + q_{deff}}\right)^2\right] \tag{11.48}$$

Source and Drain Charges with CLM and Velocity Saturation Model

$$Q_i = \frac{n_q}{MDL}\left[(q_s + q_{deff}) + \frac{1}{3}(q_s - q_{deff})^2 \frac{DVSAT}{MDL \cdot (1 + q_s + q_{deff})} + 2n_q(MDL - 1)q_{deff}\right] \tag{11.49}$$

$$Q_d = \frac{n_q}{3 \cdot MDL^2}\left[q_s + 2 \cdot q_{deff} + \frac{1}{2} \cdot \left(\frac{1}{5} \cdot (q_s - q_{deff}) + \frac{MDL}{DVSAT} \cdot (1 + q_s + q_{deff})\right)\left(\frac{q_s - q_{deff}}{1 + q_s + q_{deff}}\right)^2 \left(\frac{DVSAT}{MDL}\right)^2\right] + n_q\left[MDL - \frac{1}{MDL}\right]$$

$$Q_s = Q_i - Q_d \tag{11.50}$$

$$\tag{11.51}$$

Flexibility of tuning Cgg in strong inversion for $V_{ds} \neq 0$ (AbulkCV implementation)

$$T1 = \frac{Leff}{Leff + sqrt(XJ_i * Xdep)} \tag{11.52}$$

$$AbulkCV = 1 + \frac{A0CV * T1 - AGSCV * q_s * V_T * T1}{1 + KETACV * V_{bsx}} \tag{11.53}$$

$$VdssatCV = VdssatCV/AbulkCV \tag{11.54}$$

If A0CV and AGSCV=0, then ABULKCV=ABULK, to maintain backward compatibility

$$T7 = pow(Vds/VdssatCV, 1/Delta\_t) \tag{11.55}$$

$$T8 = pow(1 + T7, -Delta\_t) \tag{11.56}$$

$$Vdseff = Vds * T8 \tag{11.57}$$

$$Vdeff = (Vdseff + Vs) * inv\_Vt \tag{11.58}$$

Calculate qdeff using new Vdeff and then recalculate Qi

$$Q_i = \frac{n_q}{MDL}\left[(q_s + q_{deff}) + \frac{1}{3}(q_s - q_{deff})^2 \frac{AbulkCV * DVSAT}{MDL \cdot (1 + q_s + q_{deff})} + 2n_q(MDL - 1)q_{deff}\right] \tag{11.59}$$

Quantum Mechanical Effect (same for both the SOIMOD)

$$X_{DC}^{inv} = \frac{ADOS \cdot (1.9 \cdot 10^{-9})}{1 + \left[\frac{Q_i + ETAQM \cdot Q_B}{QM0}\right]^{0.7*BDOS}} \tag{11.60}$$

$$C_{ox}^{inv} = \frac{3.9 \cdot \epsilon_0}{TOXP \cdot \frac{3.9}{EPSROX} + \frac{X_{DC}^{inv}}{\epsilon_{ratio}}} \tag{11.61}$$

Intrinsic Charge expressions:

$$WLCOXVt_{inv} = NF \cdot Wact \cdot Lact \cdot C_{ox}^{inv} \cdot nVt \tag{11.62}$$

$$QBi = -NF \cdot Wact \cdot Lact \cdot \left(\frac{\epsilon_0 \cdot EPSROX}{TOXP}\right) \cdot nVt \cdot Qb \tag{11.63}$$

$$QSi = -WLCOXVt_{inv} \cdot Qs \tag{11.64}$$

$$QDi = -WLCOXVt_{inv} \cdot Qd \tag{11.65}$$

$$QGi = QSi + QDi + QBi \tag{11.66}$$

Bias-dependent overlap capacitance model (same for both the SOIMOD)
An accurate overlap capacitance model is essential. This is especially true for the drain side where the effect of the capacitance is amplified by the transistor gain. The overlap capacitance changes with gate to source and gate to drain biases. In LDD MOSFETs a substantial portion of the LDD region can be depleted, both in the vertical and lateral directions. This can lead to a large reduction of the overlap capacitance. This LDD region can be in accumulation or depletion. We use a single equation for both regions by using such smoothing parameters as $V_{gs,overlap}$ and $V_{gd,overlap}$ for the source and drain side, respectively. Unlike the case with the intrinsic capacitance, the overlap capacitances are reciprocal. In other words, $C_{gs,overlap} = C_{sg,overlap}$ and $C_{gd,overlap} = C_{dg,overlap}$.

The bias-dependent overlap capacitance model in BSIM-SOI 100.0.0 is adopted from BSIM4/BSIM-BULK 107.1.0. The overlap charge is given by:

$$V_{gs,overlap} = \frac{1}{2}\left[V_{gs} - V_{fbsd} + \delta_1 - \sqrt{(V_{gs} - V_{fbsd} + \delta_1)^2 + 4\delta_1}\right] \tag{11.67}$$

$$T6 = \frac{V_{gs,overlap}}{[1 + (\frac{-V_{gs,overlap}}{CKAPPAS1})]^{1/CKAPPAS2}} \tag{11.68}$$

$$\begin{aligned}\frac{Q_{gs,ov}}{NF \cdot W_{effCV}} &= CGSO \cdot V_{gs} + \\ &CGSL \cdot \left[V_{gs} - V_{fbsd} - V_{gs,overlap} - \frac{CKAPPAS}{2}\left(\sqrt{1 - \frac{4 * T6}{CKAPPAS}} - 1\right)\right]\end{aligned} \tag{11.69}$$

$$V_{gd,overlap} = \frac{1}{2}\left[V_{gd} - V_{fbsd} + \delta_1 - \sqrt{(V_{gd} - V_{fbsd} + \delta_1)^2 + 4\delta_1}\right] \tag{11.70}$$

$$T6 = \frac{V_{gd,overlap}}{[1 + (\frac{-V_{gd,overlap}}{CKAPPAD1})]^{1/CKAPPAD2}} \tag{11.71}$$

$$\frac{Q_{gd,ov}}{NF \cdot W_{effCV}} = CGDO \cdot V_{gd} + CGDL \cdot \left[V_{gd} - V_{fbsd} - V_{gd,overlap} - \frac{CKAPPAD}{2}\left(\sqrt{1 - \frac{4 * T6}{CKAPPAD}} - 1\right)\right] \tag{11.72}$$

$$\delta_1 = 0.02V \tag{11.73}$$

Outer Fringing Capacitance (same for both the SOIMOD)
The fringing capacitance consists of a bias-independent outer fringing capacitance and a bias-dependent inner fringing capacitance. Only the bias-independent outer fringing capacitance is modeled. If CF is not given then outer fringe capacitance is calculated as

$$CF = \frac{2 \cdot EPSROX \cdot \epsilon_0}{\pi} \cdot ln[CFRCOEFF \cdot (1 + \frac{0.4e-6}{TOX})] \tag{11.74}$$

12 List of Parameters

12.1 Instance Parameters

Name	**Unit**	**Default**	**Min**	**Max**	**Description**
L	m	1e-5	-	-	Length
W	m	1e-5	-	-	Total width including fingers
NF	-	1	1	inf	Number of fingers
NRS	-	1	-	-	Number of squares in source
NRD	-	1	-	-	Number of squares in drain
VFBSDOFF	V	0	-	-	Flatband voltage offset parameter
MINZ	-	0	0	1	Minimize either drain or source
RGATEMOD	-	0	0	3	Gate resistance model selector
GEOMOD	-	0	0	10	Geometry-dependent parasitics model
RGEOMOD	-	0	0	8	Geometry-dependent source/drain resistance, 0: RSH-based, 1: Holistic
SA	m	0	-	-	Distance between OD edge from poly from one side
SB	m	0	-	-	Distance between OD edge from poly from other side
SD	m	0	-	-	Distance between neighboring fingers
SCA	-	0	-inf	inf	Integral of the first distribution function for scattered well dopants
SCB	-	0	-inf	inf	Integral of second distribution function for scattered well dopants
SCC	-	0	-inf	inf	Integral of third distribution function for scattered well dopants
SC	m	0	-inf	inf	Distance to a single well edge; if ¡= 0.0, turn off WPE
AS	m^2	0	-	-	Source-to-Body junction area
AD	m^2	0	-	-	Drain-to-Body junction area
PS	m	0	-	-	Source-to-Body junction perimeter
PD	m	0	-	-	Drain-to-Body junction perimeter

12.2 Both Model and Instance Parameters

Name	Unit	Default	Min	Max	Description
XGW	m	0	-	-	Distance from gate contact center to device edge
NGCON	-	1	1	2	Number of gate contacts
DTEMP	K	0	-	-	Offset of device temperature
MULU0	$m^2/(V*s)$	1	-	-	Multiplication factor for low field mobility
DELVTO	V	0	-	-	Zero bias threshold voltage variation
IDS0MULT	-	1	-	-	Variability in drain current for miscellaneous reasons
EDGEFET	-	0	0	1	0: Edge FET Model OFF, 1: Edge FET Model ON
SSLMOD	-	0	0	1	Sub-surface leakage drain current, 0: Turn off 1: Turn on

12.3 Model Selectors/Controllers

Note: Binnable parameters are marked in italics and with $^{(b)}$ subscripts.

Name	Unit	Default	Min	Max	Description
SOIMOD	-	0	0	1	0: PDSOI mode, 1: DDSOI mode
TYPE	-	‘ntype	-	-	N-type = 1, P-type = -1
CVMOD	-	0	0	1	0: Consistent I-V/C-V, 1: Different I-V/C-V
COVMOD	-	0	0	1	0: Use bias-independent overlap capacitances, 1: Use bias-dependent overlap capacitances
RDSMOD	-	0	0	2	0: Internal bias dependent and external bias independent S/D resistance model, 1: External S/D resistance model, 2: Internal S/D resistance model
WPEMOD	-	0	0	1	Model flag

Name	Unit	Default	Min	Max	Description
ASYMMOD	-	0	0	1	0: Asymmetry model turned off - forward mode parameters used, 1: Asymmetry model turned on
GIDLMOD	-	0	0	1	0: Turn off GIDL current, 1: Turn on GIDL current
IGCMOD	-	0	0	1	0: Turn off Igc, Igs and Igd, 1: Turn on Igc, Igs and Igd
IGBMOD	-	0	0	1	0: Turn off Igb, 1: Turn on Igb
TNOIMOD	-	0	0	1	Thermal noise model selector
TNODEOUT	-	0	0	1	Flag indicating external temperature node
SHMOD	-	0	0	1	0: Self heating model OFF, 1: Self heating model ON
MOBSCALE	-	0	0	1	Mobility scaling model, 0: Old Model, 1: New Model
BODYMOD	-	0	0	2	Flag for body contact mode 0: Floating body, 1: linear 2: non-linear (bias-dependent RB)
IIIMOD	-	0	0	2	Flag for III selector
MODAGBCP2	-	0	0	1	0: AGBCP2 OFF, 1: AGBCP2 ON
PDEMOD	-	0	0	1	0: PDE OFF, 1: PDE ON - for DD model
FBODY1	-	0	0	1	0: Extrinsic S/D substrate charge ON, 1: PExtrinsic S/D substrate charge OFF
BINUNIT	-	1	-	-	Unit of L and W for Binning, 1: micro-meter, 0: default

12.4 Basic Model Parameters

Name	Unit	Default	Min	Max	Description
DLBIN	-	0	-	-	Length reduction parameter for binning
DWBIN	-	0	-	-	Width reduction parameter for binning
LLONG	m	1e-5	-	-	L of extracted long channel device
LMLT	-	1	-	-	Length shrinking parameter
WMLT	-	1	-	-	Width shrinking parameter

Name	Unit	Default	Min	Max	Description
XL	m	0	-	-	L offset for channel length due to mask/etch effect
WWIDE	m	1e-5	-	-	W of extracted wide channel device
XW	m	0	-	-	W offset for channel width due to mask/etch effect
LINT	m	0	-	-	Delta L for I-V
LL	$m^{(1+LLN)}$	0	-	-	Length reduction parameter
$LW^{(b)}$	$m^{(1+LWN)}$	0	-	-	Length reduction parameter [Scaling parameters: LWL]
LLN	-	1	-	-	Length reduction parameter
LWN	-	1	-	-	Length reduction parameter
WINT	m	0	-	-	Delta W for I-V
$WL^{(b)}$	$m^{(1+WLN)}$	0	-	-	Width reduction parameter
$WW^{(b)}$	$m^{(1+WWN)}$	0	-	-	Width reduction parameter [Scaling parameters: WWL]
WLN	-	1	-	-	Width reduction parameter
WWN	-	1	-	-	Width reduction parameter
DLC	m	0	-	-	Delta L for C-V
LLC	$m^{(1+LLN)}$	0	-	-	Length reduction parameter
LWC	$m^{(1+LWN)}$	0	-	-	Length reduction parameter
LWLC	$m^{(1+LWN+LLN)}$	0	-	-	Length reduction parameter
DWC	m	0	-	-	Delta W for C-V
$WLC^{(b)}$	$m^{(1+WLN)}$	0	-	-	Width reduction parameter
WWC	$m^{(1+WWN)}$	0	-	-	Width reduction parameter
TSI	m	4e-8	0.0	inf	Silicon-on-insulator thickness
TBOX	m	2e-7	0.0	inf	Back gate oxide thickness in meters
TOXE	m	3e-9	0.0	inf	Effective gate dielectric thickness relative to SiO2
TOXP	m	TOXE	0.0	inf	Physical gate dielectric thickness. If not given, TOXP is calculated from TOXE and DTOX
DTOX	m	0	-	-	Difference between effective dielectric thickness

Name	Unit	Default	Min	Max	Description
$NDEP^{(b)}$	$1/m^3$	1e+24	-	-	Channel doping concentration for I-V [Scaling parameters: NDEPL1, NDEPLEXP1, NDEPL2, NDEPLEXP2, NDEPW, NDEPWEXP, NDEPWL, NDEPWLEXP]
$NDEPCV^{(b)}$	$1/m^3$	NDEP	-	-	Channel doping concentration for C-V [Scaling parameters: NDEPCVL1, NDEPCVLEXP1, NDEPCVL2, NDEPCVLEXP2, NDEPCVW, NDEPCVWEXP, NDEPCVWL, NDEPCVWLEXP]
$NGATE^{(b)}$	$1/m^3$	5e+25	-	-	Gate doping concentration
NI0SUB	$1/m^3$	1e+16	-	-	Intrinsic carrier concentration of the substrate at 300.15K
BG0SUB	eV	1	0.0	inf	Bandgap of substrate at 300.15K
EPSRSUB	-	1e+1	0.0	inf	Relative dielectric constant of the channel material
EPSROX	-	4	0.0	inf	Relative dielectric constant of the gate dielectric
$XJ^{(b)}$	m	1e-7	-	-	S/D junction depth
$VFB^{(b)}$	V	-5e-1	-	-	Flatband voltage [Scaling parameters: VFBL, VFBLEXP, VFBW, VFBWEXP, VFBWL, VFBWLEXP]
$VFBB^{(b)}$	V	0	-	-	Flatband voltage for back gate
$VFBCV^{(b)}$	V	VFB	-	-	Flatband voltage for C-V [Scaling parameters: VFBCVL, VFBCVLEXP, VFBCVW, VFBCVWEXP, VFBCVWL, VFBCVWLEXP]
DELVFBACC	-	0	-	-	VFB shift in the accumulation region, valid for CVMOD = 1 only
VFBAGBCP2	V	VFB	-	-	Flatband voltage for AGBCP2 C-V
NDEPAGBCP2	$1/m^3$	NDEP	0	inf	Channel doping concentration for AGBCP2
$NSD^{(b)}$	$1/m^3$	1e+26	-	-	S/D doping concentration

Name	Unit	Default	Min	Max	Description
$DVTP0^{(b)}$	m	0	-	-	DITS
$DVTP1^{(b)}$	$1/V$	0	-	-	DITS
$DVTP2^{(b)}$	$m*V$	0	-	-	DITS
$DVTP3^{(b)}$	-	0	-	-	DITS
$DVTP4^{(b)}$	$1/V$	0	-	-	DITS
$DVTP5^{(b)}$	V	0	-	-	DITS
$DVBD0^{(b)}$	-	0	-	-	coupling from Vd to Vbs for improved dVbi model
$DVBD1^{(b)}$	-	0	-	-	coupling from Vd to Vbs for improved dVbi model
$VSCE^{(b)}$	-	0	-	-	coupling from Vd to Vbs for improved dVbi model
$CDSBS1^{(b)}$	-	1	-	-	coupling from Vd to Vbs for improved dVbi model
$CDSBS^{(b)}$	-	0	-	-	coupling from Vd to Vbs for improved dVbi model
$PHIN^{(b)}$	V	4e-2	-	-	Non-uniform vertical doping effect on surface potential
$ETA0^{(b)}$	-	8e-2	-	-	DIBL coefficient
$ETA0R^{(b)}$	-	ETA0	-	-	DIBL coefficient
DSUB	-	1	-	-	Length scaling exponent for DIBL
$ETAB^{(b)}$	$1/V$	-7e-2	-	-	Body bias coefficient for subthreshold DIBL effect [Scaling parameters: ETABEXP]
$K1^{(b)}$	$V^0.5$	0	-	-	First-order body-bias Vth shift due to vertical non-uniform doping [Scaling parameters: K1L, K1LEXP, K1W, K1WEXP, K1WL, K1WLEXP]
$K2^{(b)}$	V	0	-	-	Vth shift due to vertical non-uniform doping [Scaling parameters: K2L, K2LEXP, K2W, K2WEXP, K2WL, K2WLEXP]
ADOS	-	0	-	-	Quantum mechanical effect pre-factor switch in inversion

Name	Unit	Default	Min	Max	Description
BDOS	-	1	-	-	Charge centroid parameter - slope of C-V curve under QME in inversion
QM0	-	1e-3	-	-	Charge centroid parameter - starting point for QME in inversion
ETAQM	-	5e-1	-	-	Bulk charge coefficient for charge centroid in inversion
$CIT^{(b)}$	F/m^2	0	-	-	Parameter for interface traps
$NFACTOR^{(b)}$	-	0	-	-	Subthreshold slope factor [Scaling parameters: NFACTORL, NFACTORLEXP, NFACTORW, NFACTORWEXP, NFACTORWL, NFACTORWLEXP]
$ASCL^{(b)}$	-	0	-	-	Parameter for back-gate dependent scale length
$BSCL^{(b)}$	-	0	-	-	Parameter for back-gate dependent scale length
$DVT1^{(b)}$	-	1	-	-	SCE coefficient
$CDSCD^{(b)}$	$F/m^2/V$	0	-	-	Drain bias sensitivity of subthreshold slope [Scaling parameters: CDSCDL, CDSCDLEXP]
$CDSC^{(b)}$	$F/m^2/V$	1e-9	-	-	Coupling capacitance between S/D and channel
CSECSED	$F/(m^2 * V^2)$	0	-	-	Substrate-bias sensitivity of CDSCD
CBCBD	$F/(m^2 * V^2)$	0	-	-	Body bias sensitivity of CDSCD
$CSECSE0^{(b)}$	$F/(m^2 * V)$	0	-	-	Substrate-bias sensitivity of SS for long channel [Scaling parameters: CSECSE0P]
$CSECSE^{(b)}$	$F/(m^2 * V)$	0	-	-	Substrate-bias sensitivity of SS for long channel [Scaling parameters: CSECSEP]
$CBCB^{(b)}$	$F/(m^2 * V)$	0	-	-	Substrate-bias sensitivity of SS for long channel [Scaling parameters: CBCBP]
$CBCB0^{(b)}$	$F/(m^2 * V)$	0	-	-	Body-bias sensitivity of SS for long channel [Scaling parameters: CBCB0P]
$CDSCDR^{(b)}$	$F/m^2/V$	CDSCD	-	-	Drain bias sensitivity of subthreshold slope

Name	Unit	Default	Min	Max	Description
$CDSCB^{(b)}$	$F/m^2/V$	0	-	-	Body-bias sensitivity of subthreshold slope [Scaling parameters: CDSCBL, CDSCBLEXP]
VBSA	V	0	-	-	vbsa offset voltage
$VSAT^{(b)}$	m/s	1e+5	-	-	Saturation velocity [Scaling parameters: VSATL, VSATLEXP, VSATW, VSATWEXP, VSATWL, VSATWLEXP]
$VSATR^{(b)}$	m/s	VSAT	-	-	Saturation velocity
$DELTA^{(b)}$	-	1e-1	-	-	Smoothing function factor for Vdsat [Scaling parameters: DELTAL, DELTALEXP]
$VSATCV^{(b)}$	m/s	VSAT	-	-	VSAT parameter for C-V [Scaling parameters: VSATCVL, VSATCVLEXP, VSATCVW, VSATCVWEXP, VSATCVWL, VSATCVWLEXP]
$THESAT^{(b)}$	-	3e-1	-	-	Saturation velocity dependent parameter
$LPE1^{(b)}$	-	0	-	-	Equivalent length of pocket region at zero bias
UP1	-	0	-inf	inf	Mobility channel length coefficient
LP1	m	1e-8	-	-	Mobility channel length exponential coefficient
UP2	-	0	-inf	inf	Mobility channel length coefficient
LP2	m	1e-8	-	-	Mobility channel length exponential coefficient
$U0^{(b)}$	$m^2/V/s$	7e-2	-	-	Low Field mobility. [Scaling parameters: U0L, U0LEXP]
$U0R^{(b)}$	$m^2/V/s$	U0	-	-	Reverse-mode Low Field mobility.
ETAMOB	-	1	-	-	Effective field parameter (should be kept close to 1)
$UA^{(b)}$	$(m/V)^{EU}$	1e-3	-	-	Mobility reduction coefficient [Scaling parameters: UAL, UALEXP, UAW, UAWEXP, UAWL, UAWLEXP]
$UAR^{(b)}$	$(m/V)^{EU}$	UA	-	-	Reverse-mode mobility reduction coefficient

Name	Unit	Default	Min	Max	Description
$EU^{(b)}$	-	2	-	-	Mobility reduction exponent [Scaling parameters: EUL, EULEXP, EUW, EUWEXP, EUWL, EUWLEXP]
$UD^{(b)}$	-	1e-3	-	-	Coulomb scattering parameter [Scaling parameters: UDL, UDLEXP]
$UDR^{(b)}$	-	UD	-	-	Reverse-mode Coulomb scattering parameter
$UCS^{(b)}$	-	2	-	-	Coulomb scattering parameter
$UCSR^{(b)}$	-	UCS	-	-	Reverse-mode Coulomb scattering parameter
$UC^{(b)}$	$(m/V)^{EU}/V$	0	-	-	Mobility reduction with body bias [Scaling parameters: UCL, UCLEXP, UCW, UCWEXP, UCWL, UCWLEXP]
$UCR^{(b)}$	$(m/V)^{EU}/V$	UC	-	-	Reverse-mode mobility reduction with body bias
$PCLM^{(b)}$	-	3e-3	-	-	CLM pre-factor [Scaling parameters: PCLML, PCLMLEXP]
$PCLMR^{(b)}$	-	PCLM	-	-	Reverse-mode CLM pre-factor
PCLMG	V	0	-	-	CLM pre-factor gate voltage dependence
$PCLMCV^{(b)}$	-	PCLM	-	-	CLM parameter for C-V [Scaling parameters: PCLMCVL, PCLMCVLEXP]
$PSCBE1^{(b)}$	V/m	4e+8	-	-	Substrate current body-effect coefficient
$PSCBE2^{(b)}$	m/V	1e-8	-	-	Substrate current body-effect coefficient
$PDITS^{(b)}$	$1/V$	0	-	-	Coefficient for drain-induced Vth shift [Scaling parameters: PDITSL]
$PDITSD^{(b)}$	$1/V$	0	-	-	Vds dependence of drain-induced Vth shift
RSH	$ohm/square$	0	-	-	Source-drain sheet resistances
$PRWG^{(b)}$	$1/V$	1	-	-	Gate bias dependence of S/D extension resistances
$PRWB^{(b)}$	$1/V$	0	-	-	Body bias dependence of resistances [Scaling parameters: PRWBL, PRWBLEXP]
$WR^{(b)}$	-	1	-	-	W dependence parameter of S/D extension resistances

Name	Unit	Default	Min	Max	Description
$RSWMIN^{(b)}$	$ohm * um^{WR}$	0	-	-	Source resistance per unit width at high Vgs (RDSMOD = 1)
$RSW^{(b)}$	$ohm * um^{WR}$	1e+1	-	-	Zero bias source resistance (RDSMOD = 1) [Scaling parameters: RSWL, RSWLEXP]
$RDWMIN^{(b)}$	$ohm * um^{WR}$	RSWMIN	-	-	Drain resistance per unit width at high Vgs (RDSMOD = 1)
$RDW^{(b)}$	$ohm * um^{WR}$	RSW	-	-	Zero bias drain resistance (RDSMOD = 1) [Scaling parameters: RDWL, RDWLEXP]
$RDSWMIN^{(b)}$	$ohm * um^{WR}$	0	-	-	S/D Resistance per unit width at high Vgs (RDSMOD = 0 and RDSMOD = 2)
$RDSW^{(b)}$	$ohm * um^{WR}$	2e+1	-	-	Zero bias resistance (RDSMOD = 0 and RDSMOD = 2) [Scaling parameters: RDSWL, RDSWLEXP]
$PSAT^{(b)}$	-	1	-	-	Gmsat variation with gate bias [Scaling parameters: PSATL, PSATLEXP]
$PSATB^{(b)}$	$1/V$	0	-	-	Body bias effect on Idsat
$PSATR^{(b)}$	-	PSAT	-	-	Reverse-mode Gmsat variation with gate bias
PSATX	-	1	-	-	Fine tuning of PTWG effect
$PTWG^{(b)}$	-	0	-	-	Idsat variation with gate bias [Scaling parameters: PTWGL, PTWGLEXP]
VP	-	5e-2	-	-	parameter for CLM dependence on VSAT, DD Model
ALP	-	1e-2	-	-	parameter for CLM dependence on VSAT, DD Model
$PTWGR^{(b)}$	-	PTWG	-	-	Reverse-mode Idsat variation with gate bias
$KSATIV^{(b)}$	-	1	-	-	Parameter for Vdsat
$A1^{(b)}$	$1/V^2$	0	-	-	Non-saturation effect parameter for strong inversion region
$A11^{(b)}$	-	0	-	-	Temperature dependence of A1

Name	Unit	Default	Min	Max	Description
$A2^{(b)}$	$1/V$	0	-	-	Non-saturation effect parameter for moderate inversion region
$A21^{(b)}$	-	0	-	-	Temperature dependence of A2
$PDIBLC^{(b)}$	-	0	-	-	Parameter for DIBL effect on Rout [Scaling parameters: PDIBLCL, PDIBLCLEXP]
$PDIBLCR^{(b)}$	-	PDIBLC	-	-	Reverse-mode parameter for DIBL effect on Rout
$PDIBLCB^{(b)}$	$1/V$	0	-	-	Parameter for DIBL effect on Rout
$PVAG^{(b)}$	-	1	-	-	Vg dependence of early voltage
$FPROUT^{(b)}$	$V/m^{0.5}$	0	-	-	Gds degradation factor due to pocket implants [Scaling parameters: FPROUTL, FPROUTLEXP]
BJTOFF	-	0	0	1	BJT on/off flag
$VABJT^{(b)}$	V	1e+1	-	-	Early voltage for bipolar current
$AELY^{(b)}$	V/m	0	-	-	Channel length dependency of early voltage for bipolar current
$AHLI^{(b)}$	-	0	-	-	High level injection parameter for bipolar current
$AHLID^{(b)}$	-	AHLI	-	-	High level injection parameter for bipolar current
$XBJT^{(b)}$	-	1	-	-	Temperature coefficient for Isbjt
$NDIODE^{(b)}$	-	1	-	-	Diode non-ideality factor
$ISBJT^{(b)}$	A/m^2	0	-	-	BJT injection saturation current
$IDBJT^{(b)}$	A/m^2	ISBJT	-	-	BJT injection saturation current
$NBJT^{(b)}$	-	1	-	-	Power coefficient of channel length dependency for bipolar current
LLBJT0	m^2	0	-	-	Length dependence of lbjt0
WLBJT0	m^2	0	-	-	Width dependence of lbjt0
PLBJT0	m^3	0	-	-	Cross-term dependence of lbjt0
$LBJT0^{(b)}$	m	2e-7	-	-	Reference channel length for bipolar current
LN	m	2e-6	0	inf	Electron/hole diffusion length

Name	Unit	Default	Min	Max	Description
$VDSATII0^{(b)}$	-	9e-1	-	-	Nominal drain saturation voltage at threshold for impact ionization current
TII	-	0	-	-	Temperature dependent parameter for impact ionization
$ALPHA0^{(b)}$	m/V	0	-	-	substrate current model parameter [Scaling parameters: ALPHA0L, ALPHA0LEXP]
$BETA0^{(b)}$	$1/V$	0	-	-	First Vds dependent parameter of impact ionization current
$BETA1^{(b)}$	-	0	-	-	Second Vds dependent parameter of impact ionization current
$BETA2^{(b)}$	V	1e-1	-	-	Third Vds dependent parameter of impact ionization current
$LII^{(b)}$	-	0	-	-	Channel length dependent parameter at threshold for impact ionization current
$SII0^{(b)}$	-	5e-1	-	-	First Vgs dependent parameter for impact ionization current
$SII1^{(b)}$	-	1e-1	-	-	Second Vgs dependent parameter for impact ionization current
$SII2^{(b)}$	-	0	-	-	Third Vgs dependent parameter for impact ionization current
$SIID^{(b)}$	-	0	-	-	Vds dependent parameter of drain saturation voltage for impact ionization current
$ESATII^{(b)}$	V/m	1e+7	-	-	Saturation electric field for impact ionization
IIMOD2CLAMP1	V	1e-1	-	-	Clamp1 of SII1 * Vg term in IIMOD = 2
IIMOD2CLAMP2	V	1e-1	-	-	Clamp2 of SII0 * Vg term in IIMOD = 2
IIMOD2CLAMP3	V	1e-1	-	-	Clamp3 of ratio term in IIMOD = 2
$FBJTII^{(b)}$	-	0	-	-	Fraction of bipolar current affecting the impact ionization
$EBJTII^{(b)}$	-	0	-	-	Impact ionization parameter for BJT part
$CBJTII^{(b)}$	m	0	-	-	Length scaling parameter for II BJT part
$ABJTII^{(b)}$	$1/V$	0	-	-	Exponent factor for avalanche current

Name	Unit	Default	Min	Max	Description
$VBCI^{(b)}$	V	0	-	-	Internal B-C built-in potential
TVBCI	-	0	-	-	Temperature coefficient for VBCI
$MBJTII^{(b)}$	-	4e-1	-	-	Internal B-C grading coefficient
VECB	-	3e-2	0.0	inf	Vaux parameter for conduction-band electron tunneling
$ALPHAGB1^{(b)}$	$1/V$	3e-1	-	-	First Vox dependent parameter for gate current in inversion
$ALPHAGB1\_T^{(b)}$	-	0	-	-	Temperature coefficient of ALPHAGB1
$BETAGB1^{(b)}$	$1/V^2$	3e-2	-	-	Second Vox dependent parameter for gate current in inversion
$ALPHAGB2^{(b)}$	$1/V$	4e-1	-	-	First Vox dependent parameter for gate current in accumulation
$ALPHAGB2\_T^{(b)}$	-	0	-	-	Temperature coefficient of ALPHAGB2
$BETAGB2^{(b)}$	$1/V^2$	5e-2	-	-	Second Vox dependent parameter for gate current in accumulation
VGB2	V	2e+1	-	-	Third Vox dependent parameter for gate current in accumulation
VGB1	V	3e+2	-	-	Third Vox dependent parameter for gate current in inversion
AGB1	-	4e-7	-	-	'A' for Igb1 Tunneling current model
BGB1	-	-3e+10	-	-	'B' for Igb1 Tunneling current model
AGB2	-	5e-7	-	-	'A' for Igb2 Tunneling current model
BGB2	-	-2e+10	-	-	'B' for Igb2 Tunneling current model
AGBC2N	-	3e-7	-	-	NMOS 'A' for tunneling current model
AGBC2P	-	5e-7	-	-	PMOS 'A' for tunneling current model
BGBC2N	-	1e+12	-	-	NMOS 'B' for tunneling current model
BGBC2P	-	7e+11	-	-	PMOS 'B' for tunneling current model
$EIGBINV^{(b)}$	V	1	-	-	Parameter for the Si bandgap for Igbinv
$AIGC^{(b)}$	$(F*s^2/g)^{0.5}/m$	((TYPE == 'ntype) ? 1.36e-2 : 9.8e-3)	-	-	Parameter for Igc [Scaling parameters: AIGCL, AIGCW]

Name	Unit	Default	Min	Max	Description
$BIGC^{(b)}$	$(F * s^2/g)^{0.5}/m/V$	((TYPE == 'ntype') ? 1.71e-3 : 7.59e-4)	-	-	Parameter for Igc
$CIGC^{(b)}$	$1/V$	((TYPE == 'ntype') ? 0.075 : 0.03)	-	-	Parameter for Igc
$AIGS^{(b)}$	$(F*s^2/g)^{0.5}/m$	((TYPE == 'ntype') ? 1.36e-2 : 9.8e-3)	-	-	Parameter for Igs [Scaling parameters: AIGSL, AIGSW]
$AIGS1^{(b)}$	-	0	-	-	Temperature coefficient of AIGD
$BIGS^{(b)}$	$(F * s^2/g)^{0.5}/m/V$	((TYPE == 'ntype') ? 1.71e-3 : 7.59e-4)	-	-	Parameter for Igs
$CIGS^{(b)}$	$1/V$	((TYPE == 'ntype') ? 0.075 : 0.03)	-	-	Parameter for Igs
$AIGD^{(b)}$	$(F*s^2/g)^{0.5}/m$	((TYPE == 'ntype') ? 1.36e-2 : 9.8e-3)	-	-	Parameter for Igd [Scaling parameters: AIGDL, AIGDW]
$AIGD1^{(b)}$	-	0	-	-	Temperature coefficient of AIGD

Name	Unit	Default	Min	Max	Description
$BIGD^{(b)}$	$(F * s^2/g)^{0.5}/m/V$	((TYPE == 'ntype) ? 1.71e-3 : 7.59e-4)	-	-	Parameter for Igd
$CIGD^{(b)}$	$1/V$	((TYPE == 'ntype) ? 0.075 : 0.03)	-	-	Parameter for Igd
$DLCIG^{(b)}$	m	LINT	-	-	Delta L for Ig model
$DLCIGD^{(b)}$	m	DLCIG	-	-	Delta L for Ig model
$POXEDGE^{(b)}$	-	1	-	-	Factor for the gate edge Tox
$NTOX^{(b)}$	-	1	-	-	Exponent for Tox ratio
TOXREF	m	3e-9	-	-	Target tox value
$PIGCD^{(b)}$	-	1	-50.0	50.0	Igc, S/D partition parameter [Scaling parameters: PIGCDL]
$AIGC1^{(b)}$	-	0	-	-	Temperature coefficient of AIGC
$AIGBCP2^{(b)}$	$1/V^2$	4e-2	-	-	First Vgp dependent parameter for gate current in accumulation in AGBCP2 region
$AIGBCP2\_T^{(b)}$	-	0	-	-	Temperature coefficient of AIGBCP2
$BIGBCP2^{(b)}$	$1/V^2$	5e-3	-	-	Second Vgp dependent parameter for gate current in accumulation in AGBCP2 region
$CIGBCP2^{(b)}$	$1/V^2$	7e-3	-	-	Third Vgp dependent parameter for gate current in accumulation in AGBCP2 region
$AGIDL^{(b)}$	V/m	0	-	-	Pre-exponential coefficient for GIDL [Scaling parameters: AGIDLL, AGIDLW]
$BGIDL^{(b)}$	V/m	2e+9	-	-	Exponential coefficient for GIDL
$BGIDL1^{(b)}$	V/m	0	-	-	Temperature coefficient of BGIDL
$CGIDL^{(b)}$	V/m	5e-1	-	-	Exponential coefficient for GIDL

Name	Unit	Default	Min	Max	Description
$EGIDL^{(b)}$	V	8e-1	-	-	Band bending parameter for GIDL
$AGISL^{(b)}$	V/m	AGIDL	-	-	Pre-exponential coefficient for GISL [Scaling parameters: AGISLL, AGISLW]
$BGISL^{(b)}$	V/m	BGIDL	-	-	Exponential coefficient for GISL
$BGISL1^{(b)}$	V/m	BGIDL1	-	-	Temperature coefficient of BGISL
$CGISL^{(b)}$	V/m	CGIDL	-	-	Exponential coefficient for GISL
$EGISL^{(b)}$	V	EGIDL	-	-	Band bending parameter for GISL
$RGIDL^{(b)}$	-	1	-	-	GIDL vg parameter
$KGIDL^{(b)}$	V	0	-	-	GIDL vb parameter
$FGIDL^{(b)}$	-	0	-	-	GIDL vb parameter
$RGISL^{(b)}$	-	RGIDL	-	-	GISL vg parameter
$KGISL^{(b)}$	V	KGIDL	-	-	GISL vb parameter
$FGISL^{(b)}$	-	FGIDL	-	-	GISL vb parameter
$CF^{(b)}$	F/m	0	-	-	Outer fringe capacitance
CFRCOEFF	F/m	1	1.0	inf	Coefficient for outer fringe capacitance
CGSO	F/m	0	-	-	Gate-to-source overlap capacitance
CGDO	F/m	0	-	-	Gate-to-drain overlap capacitance
CGBO	F/m	0	-	-	Gate-to-body overlap capacitance
$CGSL^{(b)}$	F/m	0	-	-	Overlap capacitance between gate and lightly-doped source region
$CGDL^{(b)}$	F/m	0	-	-	Overlap capacitance between gate and lightly-doped drain region
$CKAPPAS^{(b)}$	V	6e-1	-	-	Coefficient of bias-dependent overlap capacitance for the source side
$CKAPPAD^{(b)}$	V	6e-1	-	-	Coefficient of bias-dependent overlap capacitance for the drain side
CKAPPAD1	-	1e+6	-	-	Parameter for tuning CGD
CKAPPAD2	-	1	-	-	Parameter for tuning CGD
CKAPPAS1	-	1e+6	-	-	Parameter for tuning CGD
CKAPPAS2	-	1	-	-	Parameter for tuning CGD
XRCRG1	-	1e+1	-	-	1st fitting parameter the bias-dependent Rg

Name	Unit	Default	Min	Max	Description
XRCRG2	-	1	-	-	2nd fitting parameter the bias-dependent Rg
SSL0	A/m	4e+2	-	-	Temperature- and doping-independent parameter for sub-surface leakage drain current
SSL1	$1/m$	3e+8	-	-	Temperature- and doping-independent parameter for gate length for sub-surface leakage drain current
SSL2	-	2e-1	-	-	Fitting parameter for sub-surface leakage drain current: barrier height
SSL3	V	3e-1	-	-	Fitting parameter for sub-surface leakage drain current: gate voltage effect
SSL4	$1/V$	1	-	-	Fitting parameter for sub-surface leakage drain current: gate voltage effect
SSL5	$1/V$	0	-	-	Fitting parameter for sub-surface leakage drain current: gate voltage effect
SSLEXP1	-	5e-1	-	-	Fitting exponent for SSL doping effect
SSLEXP2	-	1	-	-	Fitting exponent for SSL temperature
AVDSX	-	2e+1	5.0	100.0	Smoothing parameter in Vdsx in Vbsx
ABULK	-	1	1.0	2.0	For flexibility of tuning Cgg in strong inversion
A0	-	0	-	-	Coefficient of depletion width dependence in AbulkIV for Id-Vd flexibility
AGS	-	0	-	-	Coefficient of Source Charge dependence in AbulkIV for Id-Vd flexibility
AGS1	-	1	-	-	qs nonlinear term for Id-Vd flexibility
KETA	-	0	-	-	Coefficient of back-bias dependence in AbulkIV for Id-Vd flexibility
A0CV	-	A0	-	-	Coefficient of depletion width dependence in AbulkCV for Capacitance flexibility
AGSCV	-	AGS	-	-	Coefficient of Source Charge dependence in AbulkCV for Capacitance flexibility

Name	Unit	Default	Min	Max	Description
KETACV	-	KETA	-	-	Coefficient of back-bias dependence in AbulkCV for Capacitance flexibility
$C0^{(b)}$	V	0	-	-	Lateral NUD1 voltage parameter
$C01^{(b)}$	$1/K$	0	-	-	Temperature dependence of lateral NUD1 voltage parameter
$C0SI^{(b)}$	V	1	-	-	Correction factor for strong inversion used in Mnud1. After binning it should be within (0 : inf)
$C0SI1^{(b)}$	$1/K$	0	-	-	Temperature dependence of C0SI1
$C0SISAT^{(b)}$	V	0	-	-	Correction factor for strong inversion used in Mnud1
$C0SISAT1^{(b)}$	$1/K$	0	-	-	Temperature dependence of C0SISAT1
IGCLAMP	-	1	0	1	Model flag for Ig clamping
$K0^{(b)}$	-	0	-	-	Non-saturation effect parameter for strong inversion region
$K01^{(b)}$	$1/K$	0	-	-	Temperature coefficient for K0
$M0^{(b)}$	-	1	-	-	offset of non-saturation effect parameter for strong inversion region
$M01^{(b)}$	$1/K$	0	-	-	Temperature coefficient for M0
minr	Ohm	1e-3	-	-	minr is the value below which the simulator expects elimination of source/drain resistance, and it will improve simulation efficiency without significantly altering the results.

12.5 Junction Diode Parameters

Name	Unit	Default	Min	Max	Description
CJS	F/m^2	5e-4	-	-	Unit area source-side junction capacitance at zero bias
CJD	F/m^2	CJS	-	-	Unit area drain-side junction capacitance at zero bias

Name	**Unit**	**Default**	**Min**	**Max**	**Description**
CJSWS	F/m	5e-10	-	-	Unit length source-side side-wall junction capacitance at zero bias
CJSWD	F/m	CJSWS	-	-	Unit length drain-side side-wall junction capacitance at zero bias
CJSWGS	F/m	0	-	-	Unit length source-side gate side-wall junction capacitance at zero bias
CJSWGD	F/m	CJSWGS	-	-	Unit length drain-side gate side-wall junction capacitance at zero bias
PBS	V	1	-	-	Source-side bulk junction built-in potential
PBD	V	PBS	-	-	Drain-side bulk junction built-in potential
PBSWS	V	1	-	-	Built-in potential for Source-side side-wall junction capacitance
PBSWD	V	PBSWS	-	-	Built-in potential for Drain-side side-wall junction capacitance
PBSWGS	V	PBSWS	-	-	Built-in potential for Source-side gate side-wall junction capacitance
PBSWGD	V	PBSWGS	-	-	Built-in potential for Drain-side gate side-wall junction capacitance
MJS	-	5e-1	-	-	Source bottom junction capacitance grading coefficient
MJD	-	MJS	-	-	Drain bottom junction capacitance grading coefficient
MJSWS	-	3e-1	-	-	Source side-wall junction capacitance grading coefficient
MJSWD	-	MJSWS	-	-	Drain side-wall junction capacitance grading coefficient
MJSWGS	-	MJSWS	-	-	Source-side gate side-wall junction capacitance grading coefficient
MJSWGD	-	MJSWGS	-	-	Drain-side gate side-wall junction capacitance grading coefficient
TT	s	1e-12	-	-	Diffusion capacitance transit time coefficient

Name	**Unit**	**Default**	**Min**	**Max**	**Description**
LDIF0	-	1	-	-	Channel-length dependency coefficient of diffusion cap
$NDIF^{(b)}$	-	-1	-	-	Power coefficient of channel length dependency for diffusion capacitance
VTM00	V	3e-2	0.0	1	Hard coded 25 degC thermal voltage
PERMOD	-	1	0	1	Whether PS/PD (when given) include gate-edge perimeter
DWJ	m	DWC	-	-	Delta W for S/D junctions
$XDIF^{(b)}$	-	XBJT	-	-	Temperature coefficient for Isdif
$ISDIF^{(b)}$	A/m^2	1e-7	-	-	Body to source/drain injection saturation current
$IDDIF^{(b)}$	A/m^2	ISDIF	-	-	Body to source/drain injection saturation current
$NRECF0^{(b)}$	-	2	-	-	Recombination non-ideality factor at forward bias
$NRECR0^{(b)}$	-	1e+1	-	-	Recombination non-ideality factor at reversed bias
$XREC^{(b)}$	-	1	-	-	Temperature coefficient for Isrec
$ISREC^{(b)}$	A/m^2	1e-5	-	-	Recombination in depletion saturation current
$IDREC^{(b)}$	A/m^2	ISREC	-	-	Recombination in depletion saturation current
$NTRECF^{(b)}$	-	0	-	-	Temperature coefficient for Nrecf
$NTRECR^{(b)}$	-	0	-	-	Temperature coefficient for Nrecr
$ISTUN^{(b)}$	A/m^2	1e-8	-	-	Reverse tunneling saturation current
$IDTUN^{(b)}$	A/m^2	ISTUN	-	-	Reverse tunneling saturation current
$XTUN^{(b)}$	-	0	-	-	Temperature coefficient for Istun
$XTUND^{(b)}$	-	XTUN	-	-	Temperature coefficient for Idtun
$NTUN^{(b)}$	-	1e+1	-	-	Reverse tunneling non-ideality factor
$NTUND^{(b)}$	-	NTUN	-	-	Reverse tunneling non-ideality factor
$VTUN0^{(b)}$	V	0	-	-	Voltage dependent parameter for tunneling current

Name	Unit	Default	Min	Max	Description
$VTUN0D^{(b)}$	V	VTUN0	-	-	Voltage dependent parameter for tunneling current
$VREC0^{(b)}$	V	0	-	-	Voltage dependent parameter for recombination current
$VREC0D^{(b)}$	V	VREC0	-	-	Voltage dependent parameter for recombination current

12.6 Layout-Dependent Parasitic Model Parameters

Name	Unit	Default	Min	Max	Description
DMCG	m	0	-	-	Distance of mid-contact to gate edge
DMCI	m	DMCG	-	-	Distance of mid-contact to isolation
DMDG	m	0	-	-	Distance of mid-diffusion to gate edge
DMCGT	m	0	-	-	Distance of mid-contact to gate edge in test
XGL	m	0	-inf	$(L * LMLT + XL)$	Variation in Ldrawn
RSHG	ohm	1e-1	-	-	Gate sheet resistance

12.7 EdgeFET Parameters

Name	Unit	Default	Min	Max	Description
WEDGE	m	1e-8	1.0e-9	inf	Edge FET width
$DGAMMAEDGE^{(b)}$	-	0	-inf	inf	Different in body-bias coefficient between Edge-FET and Main-FET [Scaling parameters: DGAMMAEDGEL, DGAMMAEDGELEXP]
DVTEDGE	-	0	-inf	inf	Vth shift for Edge FET
$NDEPEDGE^{(b)}$	$1/m^3$	1e+24	-	-	Channel doping concentration for EDGEFET

Name	Unit	Default	Min	Max	Description
$NFACTOREDGE^{(b)}$	-	0	-	-	NFACTOR for Edge FET
$CITEDGE^{(b)}$	F/m^2	0	-	-	CIT for Edge FET
$CDSCEDGE^{(b)}$	$F/m^2/V$	1e-9	-	-	CDSC for edge FET
$CDSCDEDGE^{(b)}$	$F/m^2/V$	1e-9	-	-	CDSCD for edge FET
$CDSCDEDGER^{(b)}$	$F/m^2/V$	1e-9	-	-	CDSCDR for edge FET
$CSECSEEDGE^{(b)}$	$F/(m^2 * V)$	0	-	-	CSECSE for edge FET
CSECSEPEDGE	$F/(m^2 * V)$	0	-	-	CSECSEP for edge FET
CSECSE0EDGE	$F/(m^2 * V)$	0	-	-	CSECSE0 for edge FET
CSECSE0PEDGE	$F/(m^2 * V^2)$	0	-	-	CSECSE0P for edge FET
CSECSEDEDGE	$F/(m^2 * V^2)$	0	-	-	CSECSED for edge FET
CBCB0EDGE	$F/(m^2 * V)$	0	-	-	CBCB0 for edge FET
CBCB0PEDGE	$F/(m^2 * V^2)$	0	-	-	CBCB0P for edge FET
$CDSCBEDGE^{(b)}$	$F/m^2/V$	0	-	-	CDSCB for edge FET
CBCBPEDGE	$F/(m^2 * V)$	0	-	-	CBCBP for edge FET
$CBCBEDGE^{(b)}$	$F/(m^2 * V)$	0	-	-	CBCB for edge FET
CBCBDEDGE	$F/(m^2 * V^2)$	0	-	-	CBCBD for edge FET
$K1EDGE^{(b)}$	$V^{0.5}$	0	-	-	K1 for edge FET
K1LEDGE	-	0	-	-	length dependence coefficient of K1EDGE
K1LEXPEDGE	-	1	-	-	Length dependence exponent coefficient of K1EDGE
K1WEDGE	-	0	-	-	Width dependence coefficient of K1EDGE
K1WEXPEDGE	-	1	-	-	Width dependence exponent coefficient of K1EDGE
K1WLEDGE	-	0	-	-	Width-length dependence coefficient of K1EDGE
K1WLEXPEDGE	-	1	-	-	Width-length dependence exponent coefficient of K1EDGE
$ETA0EDGE^{(b)}$	-	8e-2	-	-	DIBL parameter for edge FET
$ETABEDGE^{(b)}$	$1/V$	-7e-2	-	-	ETAB for edge FET
$KT1EDGE^{(b)}$	V	-1e-1	-	-	Temperature dependence parameter of threshold voltage for edge FET

Name	Unit	Default	Min	Max	Description
$KT1LEDGE^{(b)}$	$V*m$	0	-	-	Temperature dependence parameter of threshold voltage for edge FET
$KT2EDGE^{(b)}$	-	2e-2	-	-	Temperature dependence parameter of threshold voltage for edge FET
$KT1EXPEDGE^{(b)}$	-	1	-	-	Temperature dependence parameter of threshold voltage for edge device
$TNFACTOREDGE^{(b)}$	-	0	-	-	Temperature dependence parameter of subthreshold slope factor for edge
$TETA0EDGE^{(b)}$	-	0	-	-	Temperature dependence parameter of DIBL parameter for edge FET
$DVTP0EDGE^{(b)}$	m	0	-	-	DVTP0 for edge FET
$DVTP1EDGE^{(b)}$	$1/V$	0	-	-	DVTP1 for edge FET
$DVTP2EDGE^{(b)}$	$m*V$	0	-	-	DVTP2 for edge FET
$DVTP3EDGE^{(b)}$	-	0	-	-	DVTP3 for edge FET
$DVTP4EDGE^{(b)}$	$1/V$	0	-	-	DVTP4 for edge FET
$DVTP5EDGE^{(b)}$	V	0	-	-	DVTP5 for edge FET
DVT0EDGE	-	2	-	-	First coefficient of SCE effect on Vth for Edge FET
DVT1EDGE	-	5e-1	-	-	Second coefficient of SCE effect on Vth for Edge FET
DVT2EDGE	$1/V$	0	-	-	Body-bias coefficient for SCE effect for Edge FET
$K2EDGE^{(b)}$	V	0	-	-	K2 for edge FET
K2LEDGE	m^{K2LEXP}	0	-	-	Length dependence coefficient of K2EDGE
K2LEXPEDGE	-	1	-	-	Length dependence exponent coefficient of K2EDGE
K2WEDGE	m^{K2WEXP}	0	-	-	Width dependence coefficient of K2EDGE
K2WEXPEDGE	-	1	-	-	Width dependence exponent coefficient of K2EDGE
K2WLEDGE	$m^{(2*K2WLEXP)}$	0	-	-	Width-length dependence coefficient of K2EDGE

Name	Unit	Default	Min	Max	Description
K2WLEXPEDGE	-	1	-	-	Width-length dependence exponent coefficient of K2EDGE
$KVTH0EDGE^{(b)}$	$V * m$	0	-	-	Threshold Shift parameter for stress effect
$KVTH0EDGEWE^{(b)}$	-	0	-	-	Threshold Shift parameter due to Well proximity in EDGEFET
$K2EDGEWE^{(b)}$	-	0	-	-	Vth shift due to Vertical Non-uniform doping due to Well proximity in EDGE-FET
$STK2EDGE^{(b)}$	m	0	-	-	K2 shift factor related to Vth change
$STETA0EDGE^{(b)}$	m	0	-	-	ETA0 shift related to Vth0 change

12.8 Noise Parameters

Name	Unit	Default	Min	Max	Description
EF	-	1	0.0	2.0	Flicker noise frequency exponent
EM	V/m	4e+7	-	-	Saturation field
NOIA	$s^{(1-EF)}/(eV)^1/m^3$	6e+40	-	-	Flicker noise parameter A
NOIB	$s^{(1-EF)}/(eV)^1/m$	3e+25	-	-	Flicker noise parameter B
NOIC	$s^{(1-EF)} * m/(eV)^1$	9e+8	-	-	Flicker noise parameter C
LINTNOI	m	0	-	-	Length reduction parameter offset
NOIA1	-	0	-	-	Flicker noise fitting parameter in strong inversion
NOIAX	-	1	-	-	Flicker noise fitting parameter in strong inversion for high VDS
NTNOI	-	1	-	-	Noise factor for short-channel devices for TNOIMOD = 0 only
RNOIA	-	6e-1	-	-	Noise parameter for TNOIMOD = 1
RNOIB	-	5e-1	-	-	Noise parameter for TNOIMOD = 1
RNOIC	-	4e-1	-	-	Noise parameter for TNOIMOD = 1
TNOIA	-	2	-inf	inf	Noise parameter for TNOIMOD = 1
TNOIB	-	4	-inf	inf	Noise parameter for TNOIMOD = 1

Name	Unit	Default	Min	Max	Description
TNOIC	-	0	-inf	inf	Noise correlation coefficient for TNOIMOD = 1
LP	m	1e-5	-	-	Length scaling parameter for thermal noise
RNOIK	-	0	-	-	Exponential coefficient for enhanced correlated thermal noise
TNOIK	$1/m$	0	-inf	inf	Empirical parameter for Leff trend of Sid at low Ids
TNOIK2	$1/m$	1e-1	-	-	Empirical parameter for sensitivity of RNOIK
NEDGE	-	1	-	-	Flicker noise parameter for edge fet transistor
NOIA1_EDGE	-	0	-	-	Flicker noise fitting parameter in strong inversion for edge fet transistor
NOIAX_EDGE	-	1	-	-	Flicker noise fitting parameter in strong inversion for edge fet transistor
FNOIMOD	-	0	0	1	Flicker noise model selector
LH	m	1e-8	0.0	L	Length of halo transistor
NOIA2	$s^{(1-EF)}/(eV)^1/m^3$	NOIA	-	-	Flicker noise parameter A for Halo
HNDEP	$1/m^3$	NDEP	-	-	Halo doping concentration for I-V

12.9 Additional Model Parameters for Body-Contact Parasitics

Name	Unit	Default	Min	Max	Description
RBODY	$ohm/square$	0	-	-	Intrinsic body contact sheet resistance
FRBODY	-	1	-	-	layout dependent body-resistance coefficient
RBSH	$ohm/square$	0	-	-	Extrinsic body contact sheet resistance
NRB	-	1	-	-	Number of squares in body
RHALO	ohm/m	1e+15	-	-	body halo sheet resistance
$UB^{(b)}$	$m^2/V/s$	7e-2	-	-	Mobility of majority carriers in body.
$UBTE^{(b)}$	-	1	-	-	Mobility temperature exponent

Name	Unit	Default	Min	Max	Description
$NEFF^{(b)}$	$1/m^3$	5e+24	-	-	Effective substrate doping
NSEG	-	1	0	inf	Number segments for width partitioning
RBODYAGBCP2	ohm	1e-3	-	-	Body resistance for AGBCP2 FET
NBC	-	0	-	-	Number of body contact isolation edge 1 = TBODY, 2 = HBODY
DWBC	m	0	-	-	Width offset for body contact isolation edge
PDBCP	m	0	-	-	Perimeter length for bc parasitics at drain side
PSBCP	m	0	-	-	Perimeter length for bc parasitics at source side
AGBCP	m^2	0	-	-	Gate to body overlap area for bc parasitics
AGBCP2	m^2	2e-12	-	-	Parasitic Gate to body overlap area for bc parasitics /* v4.1 improvement on BC
AGBCPD	m^2	AGBCP	-	-	Gate to body overlap area for bc parasitics in DC
AEBCP	m^2	0	-	-	Substrate to body overlap area for bc parasitics
EGGBCP2	eV	1	-	-	Bandgap in Agbcp2 region

12.10 Temperature Dependence and Self-Heating Parameters

Name	Unit	Default	Min	Max	Description
TNOM	$degC$	3e+1	-	-	Temperature at which the model was extracted
TBGASUB	eV/K	5e-4	-	-	Bandgap temperature coefficient
TBGBSUB	K	6e+2	-	-	Bandgap temperature coefficient
TNFACTOR	-	0	-	-	Temperature exponent for NFACTOR
$UTE^{(b)}$	-	-2	-	-	Mobility temperature exponent [Scaling parameters: UTEL]
$UA1^{(b)}$	m/V	1e-3	-	-	Temperature coefficient for UA [Scaling parameters: UA1L]

Name	Unit	Default	Min	Max	Description
$UC1^{(b)}$	$1.0/K$	6e-11	-	-	Temperature coefficient for UC
$UD1^{(b)}$	$1/m^2$	0	-	-	Temperature coefficient for UD [Scaling parameters: UD1L]
$EU1^{(b)}$	-	0	-	-	Temperature coefficient for EU
$UCSTE^{(b)}$	-	-5e-3	-	-	Temperature coefficient for UCS
TETA0	-	0	-	-	Temperature coefficient for ETA0
$PRT^{(b)}$	-	0	-	-	Temperature coefficient for resistance
$AT^{(b)}$	m/s	-2e-3	-	-	Temperature coefficient for saturation velocity [Scaling parameters: ATL]
TDELTA	$1/K$	0	-	-	Temperature coefficient for DELTA
$PTWGT^{(b)}$	$1/K$	0	-	-	Temperature coefficient for PTWG [Scaling parameters: PTWGTL]
$KT1^{(b)}$	V	-1e-1	-	-	Temperature coefficient for Vth [Scaling parameters: KT1EXP, KT1L]
$KT2^{(b)}$	-	2e-2	-	-	Temperature coefficient for Vth
$IIT^{(b)}$	-	0	-	-	Temperature coefficient for BETA0
$IGT^{(b)}$	-	2	-	-	Gate current temperature dependence
TCJ	$1/K$	0	-	-	Temperature coefficient for CJS/CJD
TCJSW	$1/K$	0	-	-	Temperature coefficient for CJSWS/CJSWD
TCJSWG	$1/K$	0	-	-	Temperature coefficient for CJSWGS/CJSWGD
TPB	V/K	0	-	-	Temperature coefficient for PBS/PBD
TPBSW	V/K	0	-	-	Temperature coefficient for PBSWS/PBSWD
TPBSWG	V/K	0	-	-	Temperature coefficient for PBSWGS/PBSWGD
RTH0	$m * K/W$	0	0.0	inf	Thermal resistance
CTH0	$s*W/(m*K)$	1e-5	0.0	inf	Thermal capacitance
WTH0	m	0	-	-	Width dependence coefficient for Rth and Cth

12.11 Stress Releated and Well-Proximity Model Parameters

Name	Unit	Default	Min	Max	Description
SAREF	m	1e-6	-	-	Reference distance between OD edge from Poly from one side
SBREF	m	1e-6	-	-	Reference distance between OD edge from Poly from other side
WLOD	m	0	-	-	Width parameter for stress effect
$KU0^{(b)}$	m	0	-	-	Mobility degradation/enhancement parameter for stress effect
KVSAT	m	0	-	-	Saturation velocity degradation/enhancement parameter for stress effect
TKU0	-	0	-	-	Temperature coefficient for KU0
LLODKU0	-	0	-	-	Length parameter for U0 stress effect
WLODKU0	-	0	-	-	Width parameter for U0 stress effect
$KVTH0^{(b)}$	$V * m$	0	-	-	Threshold shift parameter for stress effect
LLODVTH	-	0	-	-	Length parameter for Vth stress effect
WLODVTH	-	0	-	-	Width parameter for Vth stress effect
STK2	m	0	-	-	K2 shift factor related to Vth change
LODK2	-	0	-	-	K2 shift modification factor for stress effect
STETA0	m	0	-	-	ETA0 shift related to Vth0 change
LODETA0	-	0	-	-	ETA0 modification factor for stress effect
WEB	-	0	-	-	Coefficient for SCB (¿ 0)
WEC	-	0	-	-	Coefficient for SCC (¿ 0)
$KVTH0WE^{(b)}$	-	0	-	-	Threshold shift factor for well proximity effect
$K2WE^{(b)}$	-	0	-	-	K2 shift factor for well proximity effect
$KU0WE^{(b)}$	-	0	-	-	Mobility degradation factor for well proximity effect
SCREF	m	1e-6	0.0	inf	Reference distance to calculate SCA,SCB and SCC (¡ 0)

13 Appendix A : About Optional Nodes, usage of TNODEOUT

There are two optional nodes: B and T nodes. B node is used for body contact devices.

Let us consider the case when TNODEOUT is not set. If user specifies four nodes, this element is a 4-terminal device, i.e., floating body. If user specifies five nodes, the fifth node represents the external body contact node (B). There is a body resistance between internal body node and B node.

If TNODEOUT flag is set, the last node is interpreted as the temperature node. In this case, if user specifies five nodes, it is a floating body case, and the fifth terminal is a temperature node. If user specifies six nodes, it is a body-contacted case with fifth terminal as a body contact and sixth terminal as a temperature node. The temperature node is useful for thermal coupling simulation.

The following table shows the detailed possible cases:

Node Activation & Assignment Matrix:

CASE	**BSIMSOI Nodes as seen from user POV**	**Nodes supplied on instance call**	**Node-B**	**Node-T**	**TNODE OUT**	**SHMOD**	**Warning Message**
1	D G S E	4	F	F	0	0	None
2	D G S E B	5	T	F	0	0	None
3	D G S E B T	6	T	T	0	0	None
4	D G S E	4	F	F	1	0	None
5	D G S E T	5	T	F	1	0	None
6	D G S E B T	6	T	T	1	0	None
7	D G S E	4	F	F	0	1	None
8	D G S E B	5	T	F	0	1	None
9	D G S E B T	6	T	T	0	1	None
10	D G S E	4	F	F	1	1	**w1**
11	D G S E T	5	T	F	1	1	**w2**
12	D G S E B T	6	T	T	1	1	None

Warning Messages:

w1: Warning: TNODEOUT = 1 requesting an external temperature node, but a fifth node for temperature is not supplied.

w2: Warning: A fifth node has been specified with TNODEOUT = 1. The fifth node will be treated as the temperature node 'T'.

Implementation Notes:

1) $port_connected(NODE) only provides information about whether a node is present in the node list of the instance in the netlist. $port_connected does not provide any information about whether or where NODE is actually connected in the circuit.

2) TNODEOUT is an integer instance parameter (default 0).

3) When TNODEOUT=1, the right-most node is always interpreted as the temperature node T.

4) When SHMOD=0 and TNODEOUT=1 there are no self-heating messages because self-heating is inactive.

5) When SHMOD=1 and TNODEOUT=1 and number of nodes=5, switch SH network to use the 5th node (case 11)

6) When SHMOD=1 and TNODEOUT=1 and number of nodes=6, switch SH network to use the 6th node (case 12).

14 Appendix B : Smoothing Function

14.1 Polynomial Smoothing

The polynomial smoothing is used for a smooth transition between boundaries, maintaining exact values at all the corner points. Consider the function

$$f(x) = \quad x \quad \textit{if } x > \frac{\Delta x}{2} \tag{14.1}$$

$$= \quad k \quad \textit{if } x < \frac{-\Delta x}{2} \tag{14.2}$$

where k is some constant. The function is undefined for the region $\frac{-\Delta x}{2} < x < \frac{\Delta x}{2}$. If this region is approximated by a polynomial function, the complete function and even derivatives can be made continuous. Now consider the more generalized case

$$f(x) = \quad x \quad \textit{if } x > x_1 \tag{14.3}$$

$$= \quad k \quad \textit{if } x < x_2 \tag{14.4}$$

To express (14.4) in the form of (14.2), x is linearly transformed into z. Defining

$$x_0 = \frac{x_1 + x_2}{2} \tag{14.5}$$

$$\Delta x = x_1 - x_2 \tag{14.6}$$

then the boundary points becomes

$$x_1 = x_0 + \frac{\Delta x}{2} \tag{14.7}$$

$$x_2 = x_0 - \frac{\Delta x}{2} \tag{14.8}$$

Let $z = \frac{x - x_0}{\Delta x}$. Thus the above boundary points in z domain becomes,

$$z_1 = \frac{x_1 - x_0}{\Delta x} = \frac{1}{2} \tag{14.9}$$

$$z_2 = \frac{x_2 - x_0}{\Delta x} = -\frac{1}{2} \tag{14.10}$$

so that the function becomes

$$f(z) = z.\Delta x - x_0 \quad if\ z > \frac{1}{2} \tag{14.11}$$

$$= k \quad if\ z < -\frac{1}{2} \tag{14.12}$$

the region $-\frac{1}{2} \leq z \leq \frac{1}{2}$ is modeled by the polynomial function whose order depends on the number of boundary conditions. For example, to have continuous derivatives upto third order, we need seventh order polynomial as there are 8 boundary conditions.

$$f(z) = a.z^7 + b.z^6 + c.z^5 + d.z^4 + e.z^3 + f.z^2 + g.z + 1 \tag{14.13}$$

Then boundary conditions can be applied to derivatives to determine the polynomial coefficients. For the case of continuous third order derivatives, we found that

$$f(x) = x_0 + \Delta x.\left[\frac{5}{64} + \frac{z}{2} + z^2.\left[\frac{15}{16} - z^2.\left(\frac{5}{4} - z^2\right)\right]\right] \tag{14.14}$$

while for continuous second order derivative

$$f(x) = x_0 + \Delta x.\left[\frac{3}{32} + \frac{z}{2} + z^2.\left[\frac{3}{4} - z^2.\left(\frac{3}{4} - \frac{z^2}{2}\right)\right]\right] \tag{14.15}$$

with $z = \frac{x-x_0}{\Delta x}$. Figure 13 illustrate the concept of polynomial smoothing.

An Example : Let the function be given as

$$f(x) = x \quad if\ x > -90 \tag{14.16}$$

$$= -100 \quad if\ x < -110 \tag{14.17}$$

with the condition that third derivative to exist. From (14.6)

$$x_0 = -100 \tag{14.18}$$

$$\Delta x = 20 \tag{14.19}$$

$$z = \frac{x + 100}{20} \tag{14.20}$$

and function becomes,

$$f(z) = 20.z + 100 \quad if\ z > \frac{1}{2} \tag{14.21}$$

$$= -100 \quad if\ z < -\frac{1}{2} \tag{14.22}$$

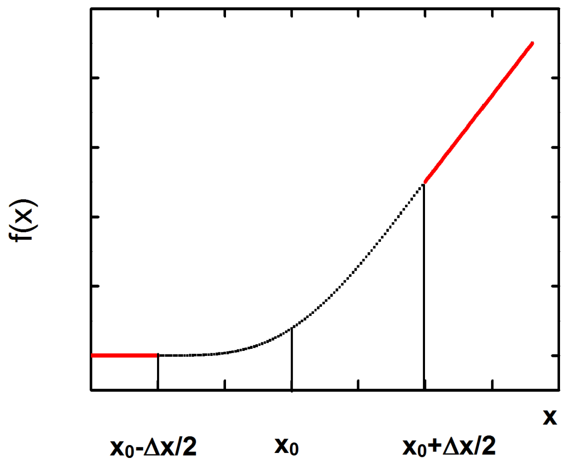

Figure 13: Illustration of Polynomial Smoothing

Boundary Conditions

$$f(\frac{1}{2}) = -90, f(-\frac{1}{2}) = -100 \tag{14.23}$$

$$f^{'}(\frac{1}{2}) = 20, f^{'}(-\frac{1}{2}) = 0 \tag{14.24}$$

$$f^{''}(\frac{1}{2}) = 0, f^{''}(-\frac{1}{2}) = 0 \tag{14.25}$$

$$f^{'''}(\frac{1}{2}) = 0, f^{'''}(-\frac{1}{2}) = 0 \tag{14.26}$$

We have 8 boundary conditions. So let

$$f(z) = a.z^7 + b.z^6 + c.z^5 + d.z^4 + e.z^3 + f.z^2 + g.z + 1 \tag{14.27}$$

Now we have 8 equations and 8 unknowns and hence all the coefficients can be derived. By substituting (14.23-14.26) in (14.27) we get

$$a = 0,\ b = 20,\ c = 0$$

$$d = -25,\ e = 0,\ f = \frac{75}{4} \tag{14.28}$$

$$g = 10,\ h = -\frac{6300}{64} \tag{14.29}$$

Thus

$$f(z) = 20.z^6 - 25.z^4 + \frac{75}{4}.z^2 + 10.z - \frac{6300}{64} \tag{14.30}$$

Figure:14 shows the above function. As can be seen that due to polynomial nature, the

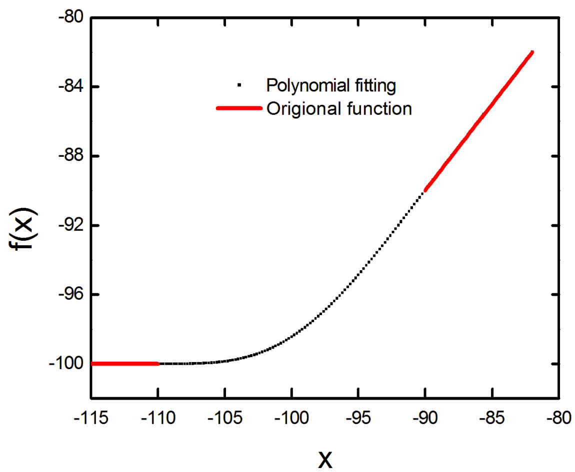

Figure 14: Polynomial Smoothing Function

approximated function undergoes smooth transitions around the boundary points.

14.2 Some Useful Functions

functions used in SOIMOD=1:

1. sigma0(d,f,tau,eta,z) (14.31)

$\mathrm{nu} = \mathrm{d} + \mathrm{f}$

$\mathrm{mutau} = \mathrm{nu} \cdot \mathrm{nu} + \mathrm{tau} \cdot \left(\frac{\mathrm{f}^2}{2} - \mathrm{d}\right)$

$\mathrm{z} = \mathrm{eta} + \frac{\mathrm{d} \cdot \mathrm{nu} \cdot \mathrm{tau}}{(\mathrm{mutau} + \frac{\mathrm{nu}}{\mathrm{mutau}} \cdot \mathrm{tau}^2 \cdot \mathrm{f}(\mathrm{f}^2 \cdot \frac{1}{3} - \mathrm{d}))}$

2. sigma1(d,e,f,tau,eta,z) (14.32)

$$\mathrm{nu} = \mathrm{d} + \mathrm{f}$$

$$\mathrm{mutau} = \mathrm{nu}^2 + \mathrm{tau} \cdot \left(\frac{\mathrm{f}^2}{2} - \mathrm{d} \cdot \mathrm{e}\right)$$

$$\mathrm{z} = \mathrm{eta} + \frac{\mathrm{d} \cdot \mathrm{nu} \cdot \mathrm{tau}}{(\mathrm{mutau} + \frac{\mathrm{nu}}{\mathrm{mutau}} \cdot \mathrm{tau}^2 \cdot \mathrm{f}(\mathrm{f}^2 \cdot \frac{1}{3} - \mathrm{d} \cdot \mathrm{e}))}$$

PE3 3rd order polynomial expansion of exp()

3. PE3(k)

$$\mathrm{PE3(k)} = 1.0 + \mathrm{k} \cdot \left(1.0 + 0.5 \cdot \left(\mathrm{k} \cdot (1.0 + \frac{\mathrm{k}}{3})\right)\right)$$

15 Appendix C : Release Notes

1. **BSIMSOI 100.1.1**

a. Removal of additional VAMPyRE warnings.

b. Updating the technical manual.

c. Removal of unused "tempc1" variable from code.

d. Removal of the kink in the Ids vs. Vg at different Ve for CDSBS1=0.1.

e. Correcting TNODEOUT dependency of AGBCP2 region gate to body tunneling current.

f. Removal of redundant assignment of Vgfbb, vgfb1.

g. Adding gmin between bi2 and si/di nodes.

h. Correcting dvth_dibl definition for SOIMOD=1.

i. Correcting overflow issue related to mobility model.

j. Correcting the definition of BODYMOD mod-selectors and addition of warnings.

k. Correcting the length and width dependencies of VFB_i, VFBCV_i, VSATCV_i.

l. Removal of floating-point errors.

m. Correcting the reference node voltage definition for calculation of gate current AGBCP2 FET.

n. Correcting the floating body definition in 5 terminal devices with TNODEOUT=1.

o. Additional QA tests included for parameter coverage.

p. Removal of VAMPyRE warnings/errors.

q. Correcting the definition of VFBB and updating the Technical Manual.

3. **BSIMSOI 100.1.0:**

a. Determination of last node as thermal node—TNODEOUT instance parameter.

b. Independent body resistance for body-contact parasitic AGBCP2 FET from main FET.

c. EDGEFET enablement for SOIMOD=1.

d. Removal of redundant nodes sbulk and dbulk.

e. Removal of unwanted kinks in g_{ds} vs V_d for SOIMOD = 0.

f. Addition of noise QA tests with all test results above simulator tolerance.

g. Removal of redundant calculations of alpha1.

h. Initialization of Mscbe = 1 for X7_s <= 0.

i. Removal of the blip in C_{gg}-V_{gs} for different V_e values.

j. Removal of bias-dependent strobe conditions, as applicable.

k. Introduce PMOS QA tests for SOIMOD=1.

l. Removal of VAMPyRE errors: superfluous assignments, and updating binning parameter units.

m. Correcting the NF scaling in leakage currents in both SOIMOD=0 and SOIMOD=1.

n. Correcting the default value and unit of NSUB parameter.

o. Correction regarding description of FBODY1 parameter in Technical Manual.

p. Removal of floating point errors (FPEs).

4. **BSIMSOI 100.0.1:**

a. Removal of floating point errors (FPEs).

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